Mathematical Equations for Singularity Analysis of Hybrid Serial-Cable Driven Planar Robot

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Contents

1 Robot Configuration and Parameters **6**
1.1 System Description . . . 6
1.2 Specific Robot Configuration . . . 7
1.2.1 Serial Chain . . . 7
1.2.2 Cable System . . . 7
1.2.3 Force System . . . 7
1.2.4 Cable Force Vectors . . . 7
1.3 Robot Parameters . . . 8
1.3.1 Geometric Parameters . . . 8
1.3.2 Cable Geometry . . . 8
1.4 Coordinate Systems . . . 8
1.4.1 Transformation Matrices . . . 8
1.4.2 Denavit-Hartenberg Parameters . . . 14

2 Forward Kinematics **17**
2.1 Serial Chain Kinematics . . . 17
2.1.1 Joint 2 Position (Elbow) . . . 17
2.1.2 End-Effector Position . . . 17
2.1.3 Transformation Matrices . . . 17
2.2 Cable Kinematics . . . 19
2.2.1 Cable Length Equations . . . 19
2.2.2 Cable Direction Vectors . . . 20
2.2.3 Cable Angles . . . 20
2.3 Complete Forward Kinematics . . . 21
2.3.1 Primary Positioning (Serial Chain) . . . 21
2.3.2 Cable Length Constraints . . . 21
2.3.3 Hybrid System Formulation . . . 21
2.3.4 Constraint Equations . . . 21

3 Jacobian Matrices **23**
3.1 Velocity Kinematics . . . 23
3.2 Serial Joint Jacobian . . . 23
3.3 Cable-Constraint Jacobian . . . 24
3.3.1 Constraint Jacobian with respect to Joint Variables . . . 24
3.3.2 Constraint Jacobian with respect to Cable Lengths . . . 24

4 Constraint Analysis **25**
4.1 Constraint Jacobian . . . 25
4.2 Reduced Jacobian . . . 25

5 Singularity Analysis **26**
5.1 Classification of Singularities . . . 26
5.2 Singularity Types . . . 26
5.2.1 Type I: Serial Chain Singularities . . . 26
5.2.2 Type II: Cable Constraint Singularities . . . 26
5.2.3 Type III: Combined Singularities . . . 27
5.2.4 Type IV: Constraint Singularities . . . 27
5.2.5 Type V: Actuation Singularities . . . 27
5.3 Mathematical Characterization . . . 27
5.3.1 Screw Theory Approach . . . 27
5.3.2 Grassmann-Plücker Coordinates . . . 27

6 Cable Tension Constraints and Negative Tension Analysis **27**
6.1 Negative Cable Tension Problem . . . 28
6.2 Solving for Cable Tensions . . . 28
6.2.1 Force Equilibrium at End-Effector . . . 28
6.2.2 Matrix Formulation (Recommended for Implementation) . . . 29
6.2.3 Direct Solution via Matrix Inversion . . . 29
6.2.4 Explicit Cable Tension Formulas . . . 29
6.2.5 Numerical Example . . . 29
6.2.6 Stability and Computational Considerations . . . 30
6.2.7 Algorithm Details and Implementation Notes . . . 32
6.2.8 Python Implementation Pseudocode . . . 32
6.3 Conditions for Negative Tensions . . . 33
6.4 Mathematical Analysis of Negative Tension Regions . . . 34
6.5 Singularity Connection . . . 34
6.5.1 Motor Capability Loss . . . 34
6.5.2 Infinite Tension Conditions . . . 35
6.6 Workspace Analysis . . . 35
6.6.1 Workspace Definitions . . . 35
6.6.2 Load-Dependent Workspace . . . 35
6.7 Design and Control Implications . . . 36
6.7.1 Design Optimization . . . 36
6.7.2 Control Strategies . . . 36
6.8 Practical Implementation Considerations . . . 37
6.8.1 Sensor Requirements . . . 37
6.8.2 Safety Considerations . . . 38
6.8.3 Computational Requirements . . . 38
6.9 Summary . . . 38
6.10 Singularity Conditions . . . 39
6.10.1 Determinant-Based Analysis . . . 39
6.10.2 SVD-Based Analysis . . . 39
6.11 Manipulability Measures . . . 39
6.11.1 Yoshikawa's Manipulability . . . 39
6.11.2 Condition Number . . . 39

7 Advanced Singularity Analysis **39**
7.1 Geometric Interpretation . . . 39
7.1.1 Instantaneous Center of Rotation . . . 39
7.1.2 Velocity Space Analysis . . . 40
7.2 Bifurcation Analysis . . . 40
7.2.1 Fold Singularities . . . 40
7.2.2 Cusp Singularities . . . 40
7.3 Dynamic Singularities . . . 40
7.3.1 Acceleration Analysis . . . 40

8 Workspace Analysis **40**
8.1 Workspace Classification . . . 40
8.1.1 Primary Workspace . . . 40
8.1.2 Secondary Workspace . . . 41
8.1.3 Dexterous Workspace . . . 41
8.2 Workspace Boundary . . . 41
8.3 Singular Workspace . . . 41

9 Computational Methods **41**
9.1 Efficient Jacobian Computation . . . 41
9.1.1 Analytical Differentiation . . . 41
9.1.2 Recursive Computation . . . 42
9.2 Singularity Detection Algorithms . . . 42
9.2.1 Multi-Method Approach . . . 42
9.2.2 Adaptive Tolerance Selection . . . 42
9.3 Workspace Generation Algorithms . . . 43
9.3.1 Discretization Method . . . 43
9.4 Optimization-Based Methods . . . 43
9.4.1 Singularity-Free Path Planning . . . 43
9.4.2 Singularity Avoidance . . . 43
9.5 Workspace Discretization . . . 43

10 Performance Analysis **44**
10.1 Computational Complexity . . . 44
10.1.1 Jacobian Computation . . . 44
10.1.2 Singularity Detection . . . 44
10.1.3 Workspace Generation . . . 44
10.2 Accuracy Analysis . . . 44
10.2.1 Numerical Sensitivity . . . 44
10.2.2 Condition Number Scaling . . . 44

11 Experimental Validation **44**
11.1 Hardware Considerations . . . 44
11.1.1 Sensor Requirements . . . 44
11.1.2 Actuator Limitations . . . 45
11.2 Validation Protocols . . . 45
11.2.1 Static Singularity Verification . . . 45
11.2.2 Dynamic Response Testing . . . 45
11.3 Benchmark Problems . . . 45
11.3.1 Analytical Test Cases . . . 45

12 Validation and Testing **45**
12.1 Systematic Test Framework . . . 45
12.1.1 Unit Tests . . . 45
12.2 Test Cases . . . 46
12.2.1 Known Singular Configurations . . . 46
12.2.2 Boundary Conditions . . . 46
12.3 Numerical Tolerances . . . 46

13 Practical Implementation Considerations **46**
13.1 Real-Time Implementation . . . 46
13.1.1 Computational Optimization . . . 46
13.1.2 Memory Management . . . 46
13.2 Error Propagation Analysis . . . 47
13.2.1 Parameter Uncertainties . . . 47
13.2.2 Measurement Noise . . . 47
13.3 Safety Considerations . . . 47
13.3.1 Singularity Avoidance Zones . . . 47
13.3.2 Emergency Procedures . . . 47

14 Software Architecture **47**
14.1 Class Hierarchy . . . 47
14.2 Configuration Management . . . 48
14.3 Implementation Notes for atan2 . . . 48

15 References and Further Reading **49**
15.1 Primary References . . . 49
15.2 Supplementary References . . . 49
15.3 Cable-Driven Robot Specific . . . 49

16 Dynamic Analysis using Euler-Lagrange Formulation **50**
16.1 Euler-Lagrange Method Foundation . . . 50
16.2 Kinetic Energy . . . 50
16.3 Potential Energy . . . 51
16.4 7.4 Lagrangian . . . 51
16.5 Partial Derivatives for Euler-Lagrange Equation . . . 51
16.5.1 Derivative with respect to velocities . . . 51
16.5.2 Derivatives with respect to positions . . . 51
16.6 Equations of Motion . . . 52
16.7 Matrix Form of Equations of Motion . . . 52
16.8 Component Matrices . . . 52
16.8.1 Mass Matrix . . . 52
16.8.2 Coriolis and Centrifugal Matrix . . . 52
16.8.3 Gravitational Torque Vector . . . 53
16.9 Cable Tension Coupling . . . 53
16.10 Simplified Equations of Motion . . . 53
16.11 Inverse Dynamics . . . 53
16.12 Singularity Effects on Dynamics . . . 53
16.12.1 Singularity Configuration . . . 53
16.12.2 Near-Singularity Behavior . . . 54
16.12.3 Acceleration Amplification . . . 54
16.13 Numerical Example: Dynamic Simulation . . . 54
16.14 7.14 Connection to Singularity Analysis . . . 55

16.15 Advanced Topics . . . 56
16.15.1 Damping and Friction . . . 56
16.15.2 Constraint Forces . . . 56
16.15.3 Elastic Cables . . . 56
16.15.4 Nonlinear Control . . . 56
16.16 Cable-Driven Robot Simplification (Pure Cable Actuation) . . . 56
16.16.1 Simplified Dynamics with tau = 0 . . . 56
16.16.2 Explicit Form for 2-DOF Cable-Driven Arm . . . 57
16.16.3 Force Balance Equations . . . 57
16.16.4 Physical Interpretation . . . 57
16.16.5 Comparison: Motor vs. Cable Actuation . . . 58
16.16.6 Numerical Form of Force Balance (Equation 6) . . . 58
16.16.7 Singularity Effects on Force Balance . . . 58
16.16.8 Control Strategy for Cable-Driven Systems . . . 59
16.16.9 Summary: Cable-Driven Dynamics Simplification . . . 59
16.16.10 Wrench Analysis and Force Balance at End-Effector . . . 59
16.17 Summary: Euler-Lagrange Dynamics . . . 77

A Notation and Symbol Reference **78**
A.1 Robot Geometry and Configuration . . . 78
A.2 Cable System Parameters . . . 78
A.3 Coordinate Frames and Positions . . . 80
A.4 Cable Anchor Points and Geometry . . . 80
A.5 Denavit-Hartenberg Parameters . . . 80
A.6 Transformation Matrices . . . 80
A.7 Kinematics and Dynamics . . . 80
A.8 Jacobian Matrices . . . 80
A.9 Constraint Equations . . . 80
A.10 Forces and Unit Vectors . . . 82
A.11 Mathematical Functions and Operators . . . 82
A.12 Singularity Analysis Notation . . . 82
A.13 Important Relationships and Formulas . . . 82
A.14 Workspace Definitions . . . 82
A.15 Notation Conventions . . . 82
A.16 Critical Distinctions . . . 82

B Derivation Details **83**

C Code Implementation Notes **83**

1 Robot Configuration and Parameters

1.1 System Description

The hybrid serial-cable driven planar robot combines a serial manipulator with cable actuation. The system comprises:

- A 2-DOF serial kinematic chain (links connected by revolute joints)
- Two cables both attached to the end-effector point M
- Fixed cable anchor points at positions $(-L, 0)$ and $(L, 0)$
- Cable actuation providing additional forces and constraints at point M
- Hybrid actuation: joint motors + cable tensions

Key Characteristics:

- The serial chain provides the primary positioning mechanism
- Cables apply forces directly to the end-effector point M
- Cable lengths vary as the serial chain moves
- Cable tensions can assist or modify the motion capabilities
- System redundancy allows for singularity avoidance

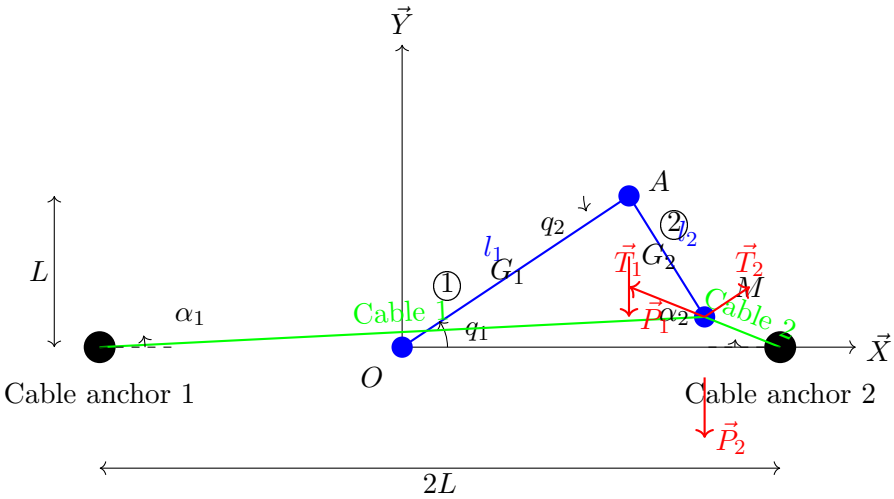

Figure 1: Schematic of the hybrid serial-cable driven planar robot. Cable angles α_1 and α_2 are computed using atan2 for robust quadrant handling.

The hybrid configuration provides:

1. Extended workspace through serial chain reach
2. Enhanced stiffness and precision via cable pre-tensioning
3. Singularity avoidance through redundant actuation
4. Improved dynamic performance with distributed actuation

1.2 Specific Robot Configuration

Based on the schematic in Figure 1, the robot has the following characteristics:

1.2.1 Serial Chain

- Two revolute joints with angles q_1 and q_2
- Link lengths: l_1 and l_2
- Joint centers at O (base), A (elbow), and M (end-effector)
- Centers of mass at G_1 and G_2

1.2.2 Cable System

- Two cables attached to the end-effector point M
- Fixed cable anchor points positioned at $(-L, 0)$ and $(L, 0)$
- Both cables connect from their respective anchor points to point M
- Cable tensions T_1 and T_2 act directly on the end-effector
- Cable angles α_1 and α_2 measured from horizontal to cable directions

1.2.3 Force System

- Gravitational forces P_1 and P_2 acting at centers of mass G_1 and G_2
- Cable tensions T_1 and T_2 both acting at end-effector point M
- Cable forces act along the directions from M to respective anchor points
- Joint torques from serial actuators (if present)
- Net force equilibrium at point M from both cable tensions

1.2.4 Cable Force Vectors

The cable tension forces acting on the end-effector are:

$$\boldsymbol{F}_1 = T_1 \boldsymbol{u}_1 = T_1 \frac{\boldsymbol{A}_1 - \boldsymbol{p}_M}{\rho_1} = -T_1 \frac{\boldsymbol{p}_M - \boldsymbol{A}_1}{\rho_1} \tag{1}$$

$$\boldsymbol{F}_2 = T_2 \boldsymbol{u}_2 = T_2 \frac{\boldsymbol{A}_2 - \boldsymbol{p}_M}{\rho_2} = -T_2 \frac{\boldsymbol{p}_M - \boldsymbol{A}_2}{\rho_2} \tag{2}$$

The total cable force on the end-effector is:

$$\boldsymbol{F}_{cable} = \boldsymbol{F}_1 + \boldsymbol{F}_2 \tag{3}$$

$$= -T_1 \frac{1}{\rho_1} \begin{bmatrix} x_M + L \\ y_M \end{bmatrix} - T_2 \frac{1}{\rho_2} \begin{bmatrix} x_M - L \\ y_M \end{bmatrix} \tag{4}$$

1.3 Robot Parameters

For the specific configuration shown in Figure 1:

$$n_j = 2 \quad \text{(number of serial joints)} \tag{5}$$
$$n_c = 2 \quad \text{(number of cables)} \tag{6}$$
$$n_{dof} = 2 \quad \text{(planar positioning: } x, y\text{)} \tag{7}$$
$$l_1, l_2 = \text{link lengths of serial chain} \tag{8}$$
$$L = \text{half-span of cable anchor points} \tag{9}$$
$$m_1, m_2 = \text{masses of links 1 and 2} \tag{10}$$
$$I_1, I_2 = \text{moments of inertia about centers of mass} \tag{11}$$

1.3.1 Geometric Parameters

$$\text{Base frame origin:} \quad O = (0, 0) \tag{12}$$
$$\text{Cable anchor points:} \quad A_1 = (-L, 0), \quad A_2 = (L, 0) \tag{13}$$
$$\text{Joint 1 position:} \quad \text{at origin } O \tag{14}$$
$$\text{Joint 2 position:} \quad A = (l_1 \cos q_1, l_1 \sin q_1) \tag{15}$$
$$\text{End-effector position:} \quad M = A + (l_2 \cos(q_1 + q_2), l_2 \sin(q_1 + q_2)) \tag{16}$$

1.3.2 Cable Geometry

$$\text{Cable 1 length:} \quad \rho_1 = \|M - A_1\| \tag{17}$$
$$\text{Cable 2 length:} \quad \rho_2 = \|M - A_2\| \tag{18}$$
$$\text{Cable 1 angle:} \quad \alpha_1 = \text{atan2}(y_M, x_M + L) \tag{19}$$
$$\text{Cable 2 angle:} \quad \alpha_2 = \text{atan2}(y_M, x_M - L) \tag{20}$$

1.4 Coordinate Systems

The robot coordinate systems are defined as follows:

$$\{O\} : \text{base frame (fixed world frame)} \tag{21}$$
$$\{E\} : \text{end-effector frame (moving frame)} \tag{22}$$
$$\{C_i\} : \text{cable attachment frame } i, \quad i = 1, 2, \ldots, n_c \tag{23}$$
$$\{J_j\} : \text{joint frame } j, \quad j = 1, 2, \ldots, n_j \tag{24}$$

1.4.1 Transformation Matrices

Homogeneous transformation matrices provide a unified framework for representing both rotation and translation in robotics. For the hybrid serial-cable driven planar robot, these transformations are essential for relating different coordinate frames.

General Homogeneous Transformation: The homogeneous transformation from frame $\{i\}$ to frame $\{j\}$ is:

$$\boldsymbol{T}_j^i = \begin{bmatrix} \boldsymbol{R}_j^i & \boldsymbol{p}_j^i \\ \boldsymbol{0}^T & 1 \end{bmatrix} \tag{25}$$

Where:

- $\boldsymbol{R}_j^i \in SO(2)$: rotation matrix from frame $\{i\}$ to frame $\{j\}$

- $\boldsymbol{p}_j^i \in \mathbb{R}^2$: position vector of origin of frame $\{j\}$ expressed in frame $\{i\}$
- $\mathbf{0}^T = [0 \quad 0]$: zero row vector

2D Planar Transformation: For planar motion, the transformation matrix becomes:

$$\boldsymbol{T}_j^i = \begin{bmatrix} \cos\theta_{ij} & -\sin\theta_{ij} & x_{ij} \\ \sin\theta_{ij} & \cos\theta_{ij} & y_{ij} \\ 0 & 0 & 1 \end{bmatrix} \tag{26}$$

Where:

$$\theta_{ij} : \text{rotation angle from frame } \{i\} \text{ to frame } \{j\} \tag{27}$$

$$x_{ij}, y_{ij} : \text{translation components from frame } \{i\} \text{ to frame } \{j\} \tag{28}$$

Properties of Transformation Matrices:

1. **Composition:** $\boldsymbol{T}_k^i = \boldsymbol{T}_j^i \boldsymbol{T}_k^j$
2. **Inverse:** $(\boldsymbol{T}_j^i)^{-1} = \boldsymbol{T}_i^j = \begin{bmatrix} (\boldsymbol{R}_j^i)^T & -(\boldsymbol{R}_j^i)^T \boldsymbol{p}_j^i \\ \mathbf{0}^T & 1 \end{bmatrix}$
3. **Identity:** $\boldsymbol{T}_i^i = \boldsymbol{I}_3$
4. **Determinant:** $\det(\boldsymbol{T}_j^i) = 1$

Derivation Methodology: Step-by-Step Approach
To systematically derive transformation matrices, follow this structured approach:
General Derivation Steps:

1. **Frame Assignment**: Clearly define coordinate frames with their origins and orientations
2. **Geometric Analysis**: Identify translation and rotation relationships between frames
3. **Parameter Extraction**: Extract angle θ and translation components (x, y)
4. **Matrix Construction**: Build the transformation matrix using the standard form
5. **Verification**: Check results using geometric reasoning or known points

Use Case Exercise: Mobile Robot Navigation
Problem Statement: A mobile robot starts at the origin of a world frame $\{W\}$. The robot first translates 3 units along the positive x-axis, then rotates 45° counterclockwise, and finally translates 2 units along its new local x-axis direction. Find the complete transformation matrix from the world frame to the robot's final frame $\{R\}$.

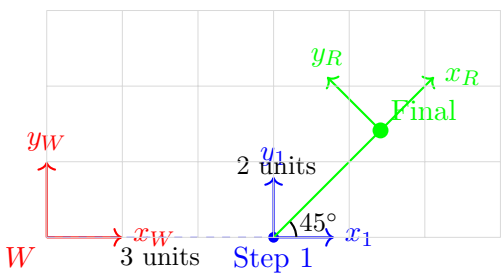

Figure 2: Mobile robot navigation exercise showing sequential transformations: translation, rotation, and final translation.

Step-by-Step Solution:
Step 1: Frame Assignment and Analysis

- World frame $\{W\}$: Fixed reference frame at origin
- Intermediate frame $\{I\}$: After first translation (3,0) with same orientation as $\{W\}$
- Final robot frame $\{R\}$: After rotation and second translation

Step 2: Break Down the Motion

Sub-transformation 1: World to Intermediate ($\boldsymbol{T}_I^W$) - Pure translation by 3 units along x_W - No rotation ($\theta = 0°$)

$$\boldsymbol{T}_I^W = \begin{bmatrix} \cos(0°) & -\sin(0°) & 3 \\ \sin(0°) & \cos(0°) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{29}$$

Sub-transformation 2: Intermediate to Robot ($\boldsymbol{T}_R^I$) - Rotation by 45° counterclockwise - Translation by 2 units along rotated x-axis - Combined rotation and translation:

$$\theta = 45° = \frac{\pi}{4} \text{ rad} \tag{30}$$

$$x_{trans} = 2\cos(45°) = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2} \tag{31}$$

$$y_{trans} = 2\sin(45°) = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2} \tag{32}$$

$$\boldsymbol{T}_R^I = \begin{bmatrix} \cos(45°) & -\sin(45°) & \sqrt{2} \\ \sin(45°) & \cos(45°) & \sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \sqrt{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} \tag{33}$$

Step 3: Compose Complete Transformation

$$\boldsymbol{T}_R^W = \boldsymbol{T}_I^W \boldsymbol{T}_R^I \tag{34}$$

$$= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \sqrt{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} \tag{35}$$

Step 4: Matrix Multiplication

Rotation part (upper-left 2×2):

$$R_{11} = 1 \cdot \frac{\sqrt{2}}{2} + 0 \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2} \tag{36}$$

$$R_{12} = 1 \cdot (-\frac{\sqrt{2}}{2}) + 0 \cdot \frac{\sqrt{2}}{2} = -\frac{\sqrt{2}}{2} \tag{37}$$

$$R_{21} = 0 \cdot \frac{\sqrt{2}}{2} + 1 \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2} \tag{38}$$

$$R_{22} = 0 \cdot (-\frac{\sqrt{2}}{2}) + 1 \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2} \tag{39}$$

Translation part:

$$T_x = 1 \cdot \sqrt{2} + 0 \cdot \sqrt{2} + 3 \cdot 1 = \sqrt{2} + 3 \tag{40}$$

$$T_y = 0 \cdot \sqrt{2} + 1 \cdot \sqrt{2} + 0 \cdot 1 = \sqrt{2} \tag{41}$$

Step 5: Final Result

$$\boldsymbol{T}_R^W = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 3+\sqrt{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \sqrt{2} \\ 0 & 0 & 1 \end{bmatrix} \tag{42}$$

Step 6: Numerical Verification Using $\sqrt{2} \approx 1.414$:

$$\boldsymbol{T}_R^W \approx \begin{bmatrix} 0.707 & -0.707 & 4.414 \\ 0.707 & 0.707 & 1.414 \\ 0 & 0 & 1 \end{bmatrix} \tag{43}$$

Step 7: Geometric Verification
The robot's final position should be:

$$\begin{aligned} \text{Position} &= (3,0) + 2(\cos(45^\circ), \sin(45^\circ)) && (44) \\ &= (3,0) + (1.414, 1.414) && (45) \\ &= (4.414, 1.414) && (46) \end{aligned}$$

This matches our transformation matrix result: $(T_x, T_y) = (4.414, 1.414)$ ✓
Step 8: Alternative Verification Using Point Transformation
Transform the origin of frame $\{R\}$ to world coordinates:

$$\begin{bmatrix} x_W \\ y_W \\ 1 \end{bmatrix} = \boldsymbol{T}_R^W \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4.414 \\ 1.414 \\ 1 \end{bmatrix} \tag{47}$$

This confirms our final position calculation.
Key Learning Points:

1. **Sequential transformations**: Complex motions break down into simpler components
2. **Matrix composition**: Order matters in matrix multiplication
3. **Translation after rotation**: The translation vector must account for the rotated coordinate system
4. **Verification importance**: Always check results using multiple methods
5. **Geometric intuition**: Mathematical results should match physical expectations

Illustrative Example: Simple 2D Pendulum
To demonstrate transformation matrix concepts, consider a simple 2D pendulum system shown in Figure 3:
System Description:

- Base frame $\{0\}$: Fixed coordinate system at the pivot point
- Pendulum frame $\{1\}$: Moving coordinate system attached to the mass
- Length: L (distance from pivot to mass)
- Angle: θ (measured from vertical, positive clockwise)

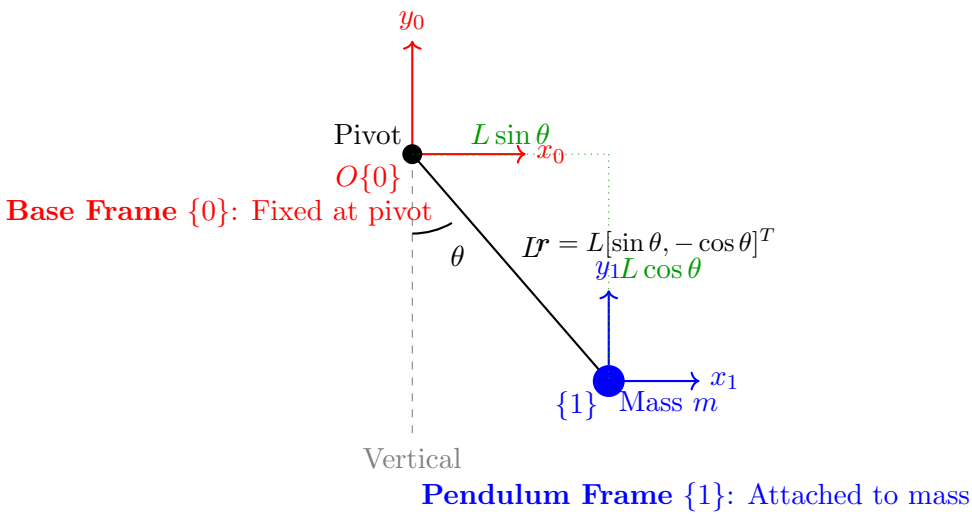

Figure 3: Simple 2D pendulum system demonstrating coordinate frame transformations. The base frame $\{0\}$ is fixed at the pivot point, while the pendulum frame $\{1\}$ moves with the mass. The position vector components and transformation angle θ are clearly illustrated.

Transformation Matrix Derivation:
The transformation from base frame $\{0\}$ to pendulum frame $\{1\}$ is:

$$\boldsymbol{T}_1^0 = \begin{bmatrix} \cos\theta & -\sin\theta & L\sin\theta \\ \sin\theta & \cos\theta & -L\cos\theta \\ 0 & 0 & 1 \end{bmatrix} \tag{48}$$

Step-by-Step Derivation:
1) **Position of pendulum mass in base frame:**

$$x_m = L\sin\theta \tag{49}$$
$$y_m = -L\cos\theta \tag{50}$$

2) **Orientation of pendulum frame:** The pendulum frame is rotated by angle θ relative to the base frame.
3) **Complete transformation:**

$$\begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = \boldsymbol{T}_1^0 \begin{bmatrix} x_0 \\ y_0 \\ 1 \end{bmatrix} \tag{51}$$
$$= \begin{bmatrix} \cos\theta & -\sin\theta & L\sin\theta \\ \sin\theta & \cos\theta & -L\cos\theta \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ 1 \end{bmatrix} \tag{52}$$

Numerical Example:
For $\theta = 30°$ and $L = 1$ meter:

$$\boldsymbol{T}_1^0 = \begin{bmatrix} \cos(30°) & -\sin(30°) & 1\cdot\sin(30°) \\ \sin(30°) & \cos(30°) & -1\cdot\cos(30°) \\ 0 & 0 & 1 \end{bmatrix} \tag{53}$$
$$= \begin{bmatrix} 0.866 & -0.5 & 0.5 \\ 0.5 & 0.866 & -0.866 \\ 0 & 0 & 1 \end{bmatrix} \tag{54}$$

Verification: The origin of frame {1} (pendulum mass position) in frame {0}:

$$\begin{bmatrix} x_m \\ y_m \\ 1 \end{bmatrix} = \boldsymbol{T}_1^0 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.866 \\ 1 \end{bmatrix} \tag{55}$$

This matches our geometric calculation: $x_m = L\sin(30°) = 0.5$, $y_m = -L\cos(30°) = -0.866$.

Connection to Robot Analysis:

This pendulum example demonstrates the same principles used in our robot:

1. **Frame assignment:** Each moving part gets its own coordinate frame
2. **DH-like parameterization:** Angle θ and length L define the transformation
3. **Forward kinematics:** Transformation matrix gives end-effector position
4. **Singularity concepts:** When $\theta = 0°$ or $180°$, the pendulum loses a degree of freedom

The robot's two-link structure can be viewed as two connected pendulums, making this a perfect stepping stone to understanding the more complex system.

Specific Transformations for the Robot:

1. Base to Joint 1 Transformation: Since joint 1 is located at the base origin with rotation q_1:

$$\boldsymbol{T}_1^0 = \begin{bmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{56}$$

2. Joint 1 to Link 1 End Transformation: Translation along link 1 of length l_1:

$$\boldsymbol{T}_{1e}^1 = \begin{bmatrix} 1 & 0 & l_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{57}$$

3. Link 1 End to Joint 2 Transformation: Rotation by angle q_2 at joint 2:

$$\boldsymbol{T}_2^{1e} = \begin{bmatrix} \cos q_2 & -\sin q_2 & 0 \\ \sin q_2 & \cos q_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{58}$$

4. Joint 2 to End-Effector Transformation: Translation along link 2 of length l_2:

$$\boldsymbol{T}_M^2 = \begin{bmatrix} 1 & 0 & l_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{59}$$

Complete Chain Transformation: The transformation from base to end-effector is:

$$\boldsymbol{T}_M^0 = \boldsymbol{T}_1^0 \boldsymbol{T}_{1e}^1 \boldsymbol{T}_2^{1e} \boldsymbol{T}_M^2 \tag{60}$$

$$= \begin{bmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & l_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos q_2 & -\sin q_2 & 0 \\ \sin q_2 & \cos q_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & l_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{61}$$

Simplified End-Effector Transformation: After matrix multiplication, the final transformation becomes:

$$\boldsymbol{T}_M^0 = \begin{bmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & l_1\cos q_1 + l_2\cos(q_1+q_2) \\ \sin(q_1+q_2) & \cos(q_1+q_2) & l_1\sin q_1 + l_2\sin(q_1+q_2) \\ 0 & 0 & 1 \end{bmatrix} \tag{62}$$

Cable Anchor Transformations: For cable analysis, we need transformations from the base frame to cable anchor points:

Left Cable Anchor:

$$\boldsymbol{T}_{A1}^0 = \begin{bmatrix} 1 & 0 & -L \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{63}$$

Right Cable Anchor:

$$\boldsymbol{T}_{A2}^0 = \begin{bmatrix} 1 & 0 & L \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{64}$$

Point Transformation: To transform a point $\boldsymbol{p}_i$ from frame $\{i\}$ to frame $\{j\}$:

$$\boldsymbol{p}_j = \boldsymbol{T}_j^i \begin{bmatrix} \boldsymbol{p}_i \\ 1 \end{bmatrix} \tag{65}$$

$$\begin{bmatrix} x_j \\ y_j \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x_{trans} \\ \sin\theta & \cos\theta & y_{trans} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} \tag{66}$$

Vector Transformation: To transform a vector $\boldsymbol{v}_i$ (direction only):

$$\boldsymbol{v}_j = \boldsymbol{R}_j^i \boldsymbol{v}_i \tag{67}$$

$$\begin{bmatrix} v_{x,j} \\ v_{y,j} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} v_{x,i} \\ v_{y,i} \end{bmatrix} \tag{68}$$

Applications in Singularity Analysis: Transformation matrices are crucial for:

1. **Jacobian Computation:** Relating joint velocities to end-effector velocities
2. **Cable Vector Calculation:** Determining cable directions in different frames
3. **Force Transformation:** Converting forces between coordinate frames
4. **Workspace Analysis:** Mapping reachable positions and orientations

1.4.2 Denavit-Hartenberg Parameters

The Denavit-Hartenberg (DH) convention provides a systematic method for describing the kinematic structure of serial manipulators. For the 2-DOF planar serial chain, we establish the DH parameters that define the relationship between consecutive coordinate frames.

DH Convention Rules: The DH convention defines four parameters for each link:

- a_i: Link length - distance from z_{i-1} to z_i along x_i
- α_i: Link twist - angle from z_{i-1} to z_i about x_i
- d_i: Link offset - distance from x_{i-1} to x_i along z_{i-1}

- θ_i: Joint angle - angle from x_{i-1} to x_i about z_{i-1}

Frame Assignment for 2-DOF Planar Robot:
Frame 0 (Base Frame):

- Origin at joint 1 center
- z_0-axis: perpendicular to the plane (pointing out of page)
- x_0-axis: along the initial direction of link 1
- y_0-axis: completes the right-hand coordinate system

Frame 1 (After Joint 1):

- Origin at joint 2 center (end of link 1)
- z_1-axis: parallel to z_0 (planar motion)
- x_1-axis: along the direction of link 2
- y_1-axis: completes the right-hand coordinate system

Frame 2 (End-Effector Frame):

- Origin at end-effector point M
- z_2-axis: parallel to z_0 and z_1
- x_2-axis: along the final link direction
- y_2-axis: completes the right-hand coordinate system

DH Parameters for the 2-DOF Planar Robot:

Link	a_i	α_i	d_i	θ_i	Description
1	l_1	0	0	q_1^*	First revolute joint
2	l_2	0	0	q_2^*	Second revolute joint

Table 1: Denavit-Hartenberg parameters for 2-DOF planar serial chain

Where q_i^* indicates that $\theta_i = q_i$ (joint variable).
Detailed Parameter Justification:
Link 1 Parameters:

$$a_1 = l_1 \quad \text{(length of link 1)} \tag{69}$$
$$\alpha_{DH,1} = 0 \quad \text{(planar motion, no twist)} \tag{70}$$
$$d_1 = 0 \quad \text{(revolute joint, no offset)} \tag{71}$$
$$\theta_1 = q_1 \quad \text{(joint variable)} \tag{72}$$

Link 2 Parameters:

$$a_2 = l_2 \quad \text{(length of link 2)} \tag{73}$$
$$\alpha_{DH,2} = 0 \quad \text{(planar motion, no twist)} \tag{74}$$
$$d_2 = 0 \quad \text{(revolute joint, no offset)} \tag{75}$$
$$\theta_2 = q_2 \quad \text{(joint variable)} \tag{76}$$

DH Transformation Matrix: The standard DH transformation matrix from frame $i-1$ to frame i is:

$$\boldsymbol{T}_i^{i-1} = \mathrm{Rot}_z(\theta_i) \cdot \mathrm{Trans}_z(d_i) \cdot \mathrm{Trans}_x(a_i) \cdot \mathrm{Rot}_x(\alpha_i) \tag{77}$$

$$= \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{78}$$

Simplified 2D Planar Case: For planar motion ($\alpha_i = 0$, $d_i = 0$), the transformation simplifies to:

$$\boldsymbol{T}_i^{i-1} = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i & a_i\sin\theta_i \\ 0 & 0 & 1 \end{bmatrix} \tag{79}$$

Individual Link Transformations:
Base to Frame 1 (Joint 1):

$$\boldsymbol{T}_1^0 = \begin{bmatrix} \cos q_1 & -\sin q_1 & l_1\cos q_1 \\ \sin q_1 & \cos q_1 & l_1\sin q_1 \\ 0 & 0 & 1 \end{bmatrix} \tag{80}$$

Frame 1 to Frame 2 (Joint 2):

$$\boldsymbol{T}_2^1 = \begin{bmatrix} \cos q_2 & -\sin q_2 & l_2\cos q_2 \\ \sin q_2 & \cos q_2 & l_2\sin q_2 \\ 0 & 0 & 1 \end{bmatrix} \tag{81}$$

Forward Kinematics via DH Parameters: The complete transformation from base to end-effector:

$$\boldsymbol{T}_2^0 = \boldsymbol{T}_1^0\boldsymbol{T}_2^1 \tag{82}$$

$$= \begin{bmatrix} \cos q_1 & -\sin q_1 & l_1\cos q_1 \\ \sin q_1 & \cos q_1 & l_1\sin q_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos q_2 & -\sin q_2 & l_2\cos q_2 \\ \sin q_2 & \cos q_2 & l_2\sin q_2 \\ 0 & 0 & 1 \end{bmatrix} \tag{83}$$

After multiplication:

$$\boldsymbol{T}_2^0 = \begin{bmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & l_1\cos q_1 + l_2\cos(q_1+q_2) \\ \sin(q_1+q_2) & \cos(q_1+q_2) & l_1\sin q_1 + l_2\sin(q_1+q_2) \\ 0 & 0 & 1 \end{bmatrix} \tag{84}$$

End-Effector Position: From the transformation matrix, the end-effector position is:

$$\boldsymbol{p}_M = \begin{bmatrix} x_M \\ y_M \end{bmatrix} = \begin{bmatrix} l_1\cos q_1 + l_2\cos(q_1+q_2) \\ l_1\sin q_1 + l_2\sin(q_1+q_2) \end{bmatrix} \tag{85}$$

Validation of DH Parameters: The DH parameters can be validated by checking:

1. **Workspace boundary:** Maximum reach $= l_1 + l_2$, minimum reach $= |l_1 - l_2|$
2. **Joint limits:** Typical range $q_i \in [-\pi, \pi]$ for full rotation
3. **Singularity conditions:** When $q_2 = 0$ or $q_2 = \pi$ (extended/folded)

4. **Frame consistency:** Right-hand rule preserved through all transformations

Alternative DH Conventions: Note that some literature uses modified DH parameters. The relationship between standard and modified DH is:

$$\boldsymbol{T}_{i,modified}^{i-1} = \mathrm{Rot}_x(\alpha_{i-1}) \cdot \mathrm{Trans}_x(a_{i-1}) \cdot \mathrm{Rot}_z(\theta_i) \cdot \mathrm{Trans}_z(d_i) \tag{86}$$

For our planar robot, both conventions yield the same result due to $\alpha_i = 0$ and $d_i = 0$.
Applications in Singularity Analysis: DH parameters are fundamental for:

1. **Jacobian derivation:** Systematic computation of velocity relationships
2. **Workspace analysis:** Determining reachable positions and orientations
3. **Inverse kinematics:** Solving for joint angles given end-effector pose
4. **Dynamic modeling:** Computing inertia matrices and Coriolis terms

2 Forward Kinematics

2.1 Serial Chain Kinematics

For the 2-DOF serial chain, the forward kinematics are:

2.1.1 Joint 2 Position (Elbow)

$$\boldsymbol{p}_A = \begin{bmatrix} x_A \\ y_A \end{bmatrix} = \begin{bmatrix} l_1 \cos q_1 \\ l_1 \sin q_1 \end{bmatrix} \tag{87}$$

2.1.2 End-Effector Position

$$\boldsymbol{p}_M = \begin{bmatrix} x_M \\ y_M \end{bmatrix} \tag{88}$$

$$= \boldsymbol{p}_A + l_2 \begin{bmatrix} \cos(q_1 + q_2) \\ \sin(q_1 + q_2) \end{bmatrix} \tag{89}$$

$$= \begin{bmatrix} l_1 \cos q_1 + l_2 \cos(q_1 + q_2) \\ l_1 \sin q_1 + l_2 \sin(q_1 + q_2) \end{bmatrix} \tag{90}$$

2.1.3 Transformation Matrices

Derivation of Transformation Matrices:
The transformation matrices are derived systematically using the Denavit-Hartenberg parameters established in the previous section. We'll show the step-by-step derivation for each transformation.
Step 1: Base to Frame 1 Transformation ($\boldsymbol{T}_1^0$)
Starting from the DH parameters for link 1: - $a_1 = l_1$, $\alpha_{DH,1} = 0$, $d_1 = 0$, $\theta_1 = q_1$
The general DH transformation matrix is:

$$\boldsymbol{T}_i^{i-1} = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{91}$$

Substituting the parameters for link 1 ($\theta_1 = q_1$, $\alpha_1 = 0$, $d_1 = 0$, $a_1 = l_1$):

$$\cos\alpha_{DH,1} = \cos(0) = 1 \tag{92}$$
$$\sin\alpha_{DH,1} = \sin(0) = 0 \tag{93}$$

Therefore:

$$\boldsymbol{T}_1^0 = \begin{bmatrix} \cos q_1 & -\sin q_1 \cdot 1 & \sin q_1 \cdot 0 & l_1 \cos q_1 \\ \sin q_1 & \cos q_1 \cdot 1 & -\cos q_1 \cdot 0 & l_1 \sin q_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{94}$$
$$= \begin{bmatrix} \cos q_1 & -\sin q_1 & 0 & l_1 \cos q_1 \\ \sin q_1 & \cos q_1 & 0 & l_1 \sin q_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{95}$$

For 2D planar motion, we reduce this to:

$$\boldsymbol{T}_1^0 = \begin{bmatrix} \cos q_1 & -\sin q_1 & l_1 \cos q_1 \\ \sin q_1 & \cos q_1 & l_1 \sin q_1 \\ 0 & 0 & 1 \end{bmatrix} \tag{96}$$

Physical Interpretation: This transformation represents:

- **Rotation by** q_1: The 2×2 rotation submatrix rotates frame 0 by angle q_1
- **Translation by** l_1: The translation vector moves along the rotated x_1-axis by distance l_1

Step 2: Frame 1 to Frame 2 Transformation ($\boldsymbol{T}_2^1$)
For link 2 with DH parameters: - $a_2 = l_2$, $\alpha_{DH,2} = 0$, $d_2 = 0$, $\theta_2 = q_2$
Following the same process:

$$\boldsymbol{T}_2^1 = \begin{bmatrix} \cos q_2 & -\sin q_2 & l_2 \cos q_2 \\ \sin q_2 & \cos q_2 & l_2 \sin q_2 \\ 0 & 0 & 1 \end{bmatrix} \tag{97}$$

Step 3: Complete Chain Transformation ($\boldsymbol{T}_2^0$)
The complete transformation from base to end-effector is obtained by matrix multiplication:

$$\boldsymbol{T}_2^0 = \boldsymbol{T}_1^0 \boldsymbol{T}_2^1 \tag{98}$$
$$= \begin{bmatrix} \cos q_1 & -\sin q_1 & l_1 \cos q_1 \\ \sin q_1 & \cos q_1 & l_1 \sin q_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos q_2 & -\sin q_2 & l_2 \cos q_2 \\ \sin q_2 & \cos q_2 & l_2 \sin q_2 \\ 0 & 0 & 1 \end{bmatrix} \tag{99}$$

Detailed Matrix Multiplication:
Let's compute each element of the result matrix $\boldsymbol{T}_2^0$:
Rotation Matrix Elements (upper-left 2×2):

$$R_{11} = \cos q_1 \cos q_2 - (-\sin q_1)(\sin q_2) = \cos q_1 \cos q_2 + \sin q_1 \sin q_2 = \cos(q_1 + q_2) \tag{100}$$
$$R_{12} = \cos q_1(-\sin q_2) - (-\sin q_1)(\cos q_2) = -\cos q_1 \sin q_2 + \sin q_1 \cos q_2 = -\sin(q_1 + q_2) \tag{101}$$
$$R_{21} = \sin q_1 \cos q_2 - \cos q_1 \sin q_2 = \sin q_1 \cos q_2 - \cos q_1 \sin q_2 = \sin(q_1 + q_2) \tag{102}$$
$$R_{22} = \sin q_1(-\sin q_2) - \cos q_1(-\cos q_2) = -\sin q_1 \sin q_2 + \cos q_1 \cos q_2 = \cos(q_1 + q_2) \tag{103}$$

Translation Vector Elements:

$$T_x = \cos q_1(l_2 \cos q_2) + (-\sin q_1)(l_2 \sin q_2) + l_1 \cos q_1 \tag{104}$$
$$= l_2(\cos q_1 \cos q_2 - \sin q_1 \sin q_2) + l_1 \cos q_1 \tag{105}$$
$$= l_2 \cos(q_1 + q_2) + l_1 \cos q_1 \tag{106}$$
$$T_y = \sin q_1(l_2 \cos q_2) + \cos q_1(l_2 \sin q_2) + l_1 \sin q_1 \tag{107}$$
$$= l_2(\sin q_1 \cos q_2 + \cos q_1 \sin q_2) + l_1 \sin q_1 \tag{108}$$
$$= l_2 \sin(q_1 + q_2) + l_1 \sin q_1 \tag{109}$$

Final Result:

$$\boldsymbol{T}_2^0 = \begin{bmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & l_1 \cos q_1 + l_2 \cos(q_1+q_2) \\ \sin(q_1+q_2) & \cos(q_1+q_2) & l_1 \sin q_1 + l_2 \sin(q_1+q_2) \\ 0 & 0 & 1 \end{bmatrix} \tag{110}$$

Verification Using Trigonometric Identities:
The derivation uses the angle addition formulas:

$$\cos(A+B) = \cos A \cos B - \sin A \sin B \tag{111}$$
$$\sin(A+B) = \sin A \cos B + \cos A \sin B \tag{112}$$

Where $A = q_1$ and $B = q_2$.
Alternative Derivation - Geometric Approach:
We can also derive this geometrically:

1. **Joint 1 contribution:** Point moves from origin to $(l_1 \cos q_1, l_1 \sin q_1)$
2. **Joint 2 contribution:** From joint 1 position, move distance l_2 in direction $(q_1 + q_2)$
3. **Final position:** Sum of both contributions gives the end-effector position

Summary of Individual Transformations:

$$\boldsymbol{T}_1^0 = \begin{bmatrix} \cos q_1 & -\sin q_1 & l_1 \cos q_1 \\ \sin q_1 & \cos q_1 & l_1 \sin q_1 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{(Base to Joint 2)} \tag{113}$$
$$\boldsymbol{T}_2^1 = \begin{bmatrix} \cos q_2 & -\sin q_2 & l_2 \cos q_2 \\ \sin q_2 & \cos q_2 & l_2 \sin q_2 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{(Joint 2 to End-Effector)} \tag{114}$$
$$\boldsymbol{T}_2^0 = \boldsymbol{T}_1^0 \boldsymbol{T}_2^1 \quad \text{(Complete Chain)} \tag{115}$$

2.2 Cable Kinematics

Both cables are attached to the end-effector point M and extend to fixed anchor points:

2.2.1 Cable Length Equations

Cable 1 connects anchor point $A_1 = (-L, 0)$ to end-effector M:

$$\rho_1 = \|\boldsymbol{p}_M - \boldsymbol{A}_1\| \tag{116}$$
$$= \sqrt{(x_M + L)^2 + y_M^2} \tag{117}$$
$$= \sqrt{(l_1 \cos q_1 + l_2 \cos(q_1+q_2) + L)^2 + (l_1 \sin q_1 + l_2 \sin(q_1+q_2))^2} \tag{118}$$

Cable 2 connects anchor point $A_2 = (L, 0)$ to end-effector M:

$$\rho_2 = \|\boldsymbol{p}_M - \boldsymbol{A}_2\| \tag{119}$$
$$= \sqrt{(x_M - L)^2 + y_M^2} \tag{120}$$
$$= \sqrt{(l_1 \cos q_1 + l_2 \cos(q_1 + q_2) - L)^2 + (l_1 \sin q_1 + l_2 \sin(q_1 + q_2))^2} \tag{121}$$

2.2.2 Cable Direction Vectors

The unit vectors along the cables (from anchor points toward end-effector) are:

$$\boldsymbol{u}_1 = \frac{\boldsymbol{p}_M - \boldsymbol{A}_1}{\rho_1} = \frac{1}{\rho_1} \begin{bmatrix} x_M + L \\ y_M \end{bmatrix} \tag{122}$$
$$\boldsymbol{u}_2 = \frac{\boldsymbol{p}_M - \boldsymbol{A}_2}{\rho_2} = \frac{1}{\rho_2} \begin{bmatrix} x_M - L \\ y_M \end{bmatrix} \tag{123}$$

Alternatively, using the cable angles:

$$\boldsymbol{u}_1 = \begin{bmatrix} \cos \alpha_1 \\ \sin \alpha_1 \end{bmatrix} = \begin{bmatrix} \cos(\text{atan2}(y_M, x_M + L)) \\ \sin(\text{atan2}(y_M, x_M + L)) \end{bmatrix} \tag{124}$$
$$\boldsymbol{u}_2 = \begin{bmatrix} \cos \alpha_2 \\ \sin \alpha_2 \end{bmatrix} = \begin{bmatrix} \cos(\text{atan2}(y_M, x_M - L)) \\ \sin(\text{atan2}(y_M, x_M - L)) \end{bmatrix} \tag{125}$$

Note that:

$$\cos(\text{atan2}(y, x)) = \frac{x}{\sqrt{x^2 + y^2}} \tag{126}$$
$$\sin(\text{atan2}(y, x)) = \frac{y}{\sqrt{x^2 + y^2}} \tag{127}$$

2.2.3 Cable Angles

Using the two-argument arctangent function for proper quadrant handling:

$$\alpha_1 = \text{atan2}(y_M, x_M + L) \tag{128}$$
$$\alpha_2 = \text{atan2}(y_M, x_M - L) \tag{129}$$

Critical for this robot configuration:

- When $x_M = -L$: Cable 1 is vertical, $\text{atan2}(y_M, 0)$ handles this correctly
- When $x_M = +L$: Cable 2 is vertical, $\text{atan2}(y_M, 0)$ handles this correctly
- When $y_M = 0$: Both cables horizontal, $\text{atan2}(0, x_M \pm L)$ returns 0 or π
- Workspace boundaries: Continuous angle computation throughout reachable space

Where $\text{atan2}(y, x)$ returns the angle $\theta \in (-\pi, \pi]$ such that:

$$x = r \cos \theta \tag{130}$$
$$y = r \sin \theta \tag{131}$$

This formulation:

- Handles all four quadrants correctly
- Avoids division by zero when $x_M = -L$ or $x_M = L$
- Provides continuous angle values as the end-effector moves
- Returns angles in the range $(-\pi, \pi]$

2.3 Complete Forward Kinematics

In this hybrid system, the end-effector position is determined by both the serial chain configuration and cable length constraints:

2.3.1 Primary Positioning (Serial Chain)

The serial chain provides the primary positioning capability:

$$\boldsymbol{p}_{M,serial} = \begin{bmatrix} x_M \\ y_M \end{bmatrix} \tag{132}$$

$$= \begin{bmatrix} l_1 \cos q_1 + l_2 \cos(q_1 + q_2) \\ l_1 \sin q_1 + l_2 \sin(q_1 + q_2) \end{bmatrix} \tag{133}$$

2.3.2 Cable Length Constraints

The cables impose constraints on the achievable positions:

$$\rho_1 = f_1(\boldsymbol{q}) = \sqrt{(x_M(\boldsymbol{q}) + L)^2 + y_M(\boldsymbol{q})^2} \tag{134}$$

$$\rho_2 = f_2(\boldsymbol{q}) = \sqrt{(x_M(\boldsymbol{q}) - L)^2 + y_M(\boldsymbol{q})^2} \tag{135}$$

2.3.3 Hybrid System Formulation

For a given cable length pair (ρ_1, ρ_2), the end-effector must satisfy:

$$\boldsymbol{g}(\boldsymbol{q}, \boldsymbol{\rho}) = \begin{bmatrix} g_1(\boldsymbol{q}, \rho_1) \\ g_2(\boldsymbol{q}, \rho_2) \end{bmatrix} = \boldsymbol{0} \tag{136}$$

Where:

$$g_1(\boldsymbol{q}, \rho_1) = (x_M + L)^2 + y_M^2 - \rho_1^2 = 0 \tag{137}$$

$$g_2(\boldsymbol{q}, \rho_2) = (x_M - L)^2 + y_M^2 - \rho_2^2 = 0 \tag{138}$$

2.3.4 Constraint Equations

Important Clarification: Nature of Cable Constraints

In a cable-driven robot, the cable lengths ρ_1 and ρ_2 are indeed **variable** - they change as the robot moves and are controlled by motors. However, the "constraints" here refer to the **geometric relationships** that must always be satisfied, not fixed length constraints.

Types of Constraints in Hybrid Serial-Cable Systems:

1. Geometric Constraints (Always Satisfied): These are the fundamental geometric relationships that define how cable lengths relate to robot configuration:

$$\rho_1^2 = (x_M + L)^2 + y_M^2 \quad \text{(Left cable geometric constraint)} \tag{139}$$

$$\rho_2^2 = (x_M - L)^2 + y_M^2 \quad \text{(Right cable geometric constraint)} \tag{140}$$

Where: - ρ_1, ρ_2: **Variable** cable lengths (controlled inputs) - x_M, y_M: End-effector position (functions of joint angles q_1, q_2) - L: Fixed distance between cable anchor points

2. Kinematic Constraint (Forward Kinematics): The end-effector position is determined by the serial joint configuration:

$$x_M = l_1 \cos q_1 + l_2 \cos(q_1 + q_2) \tag{141}$$

$$y_M = l_1 \sin q_1 + l_2 \sin(q_1 + q_2) \tag{142}$$

3. Cable Length Relationships: Substituting the kinematic equations into the geometric constraints:

$$\rho_1^2 = [l_1 \cos q_1 + l_2 \cos(q_1 + q_2) + L]^2 + [l_1 \sin q_1 + l_2 \sin(q_1 + q_2)]^2 \tag{143}$$

$$\rho_2^2 = [l_1 \cos q_1 + l_2 \cos(q_1 + q_2) - L]^2 + [l_1 \sin q_1 + l_2 \sin(q_1 + q_2)]^2 \tag{144}$$

Different Control Scenarios:

Scenario A: Joint-Space Control (Serial Joints as Inputs) - **Inputs**: Joint angles q_1, q_2 - **Determined**: End-effector position (x_M, y_M) via forward kinematics - **Calculated**: Required cable lengths ρ_1, ρ_2 via geometric constraints - **Constraint form**: $\rho_i = f(\boldsymbol{q})$ (cable lengths as functions of joint angles)

Scenario B: Cable-Space Control (Cable Lengths as Inputs) - **Inputs**: Desired cable lengths ρ_1, ρ_2 - **Constraint**: The two cable length equations must be satisfied simultaneously - **Determined**: End-effector position (x_M, y_M) via constraint intersection - **Calculated**: Required joint angles q_1, q_2 via inverse kinematics

Mathematical Formulation for Cable-Space Control:

The constraint equations become:

$$g_1(\boldsymbol{q}, \rho_1) = (x_M(\boldsymbol{q}) + L)^2 + y_M(\boldsymbol{q})^2 - \rho_1^2 = 0 \tag{145}$$

$$g_2(\boldsymbol{q}, \rho_2) = (x_M(\boldsymbol{q}) - L)^2 + y_M(\boldsymbol{q})^2 - \rho_2^2 = 0 \tag{146}$$

Where $x_M(\boldsymbol{q})$ and $y_M(\boldsymbol{q})$ are the forward kinematic functions.

Constraint Jacobian for Cable-Space Control:

$$\boldsymbol{G}_q = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{q}} = \begin{bmatrix} \frac{\partial g_1}{\partial q_1} & \frac{\partial g_1}{\partial q_2} \\ \frac{\partial g_2}{\partial q_1} & \frac{\partial g_2}{\partial q_2} \end{bmatrix} \tag{147}$$

Physical Interpretation:

- **NOT fixed constraints**: The cables can change length freely
- **Geometric relationships**: Given any robot configuration, the cable lengths are geometrically determined
- **Control redundancy**: We can control either joints OR cable lengths, but not both independently
- **Constraint satisfaction**: Any feasible motion must satisfy the geometric relationships

Example - Cable Length Calculation:

For joint configuration $q_1 = 30°$, $q_2 = 45°$ with $l_1 = l_2 = 1$ m, $L = 2$ m:

Step 1: Calculate end-effector position

$$x_M = 1 \cdot \cos(30°) + 1 \cdot \cos(75°) = 0.866 + 0.259 = 1.125 \text{ m} \tag{148}$$

$$y_M = 1 \cdot \sin(30°) + 1 \cdot \sin(75°) = 0.500 + 0.966 = 1.466 \text{ m} \tag{149}$$

Step 2: Calculate required cable lengths

$$\rho_1 = \sqrt{(1.125 + 2)^2 + (1.466)^2} = \sqrt{9.766 + 2.151} = 3.452 \text{ m} \tag{150}$$

$$\rho_2 = \sqrt{(1.125 - 2)^2 + (1.466)^2} = \sqrt{0.766 + 2.151} = 1.708 \text{ m} \tag{151}$$

Key Distinction: - The constraint equations don't fix the cable lengths - They define the **required** cable lengths for any given robot configuration - Or conversely, they define the **achievable** robot configurations for given cable lengths

3 Jacobian Matrices

3.1 Velocity Kinematics

Notation Clarification: To avoid confusion, we establish clear notation for different types of lengths:

$$l_1, l_2 : \text{Fixed link lengths of the serial arm} \tag{152}$$

$$\rho_1, \rho_2 : \text{Variable cable lengths (controlled inputs)} \tag{153}$$

$$\boldsymbol{\rho} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} \text{ (cable length vector)} \tag{154}$$

Hybrid System Velocity Kinematics: The relationship between joint velocities, cable length rates, and end-effector velocities:

$$\dot{\boldsymbol{x}} = \boldsymbol{J}(\boldsymbol{q}, \boldsymbol{\rho}) \begin{bmatrix} \dot{\boldsymbol{q}} \\ \dot{\boldsymbol{\rho}} \end{bmatrix} \tag{155}$$

$$= \begin{bmatrix} \boldsymbol{J}_q & \boldsymbol{J}_\rho \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{q}} \\ \dot{\boldsymbol{\rho}} \end{bmatrix} \tag{156}$$

Where:

- $\dot{\boldsymbol{x}} = [\dot{x}_M, \dot{y}_M]^T$: End-effector velocity
- $\dot{\boldsymbol{q}} = [\dot{q}_1, \dot{q}_2]^T$: Joint angular velocities
- $\dot{\boldsymbol{\rho}} = [\dot{\rho}_1, \dot{\rho}_2]^T$: Cable length change rates
- $\boldsymbol{J}_q$: Serial joint Jacobian matrix
- $\boldsymbol{J}_\rho$: Cable Jacobian matrix

Important Note on Cable Jacobian: In our hybrid system, both cables attach to the same end-effector point M. This means:

- The cables don't directly control end-effector motion independently
- Cable motions are **constrained** by the serial arm configuration
- The effective Jacobian depends on which actuators are active (joints vs. cables)

3.2 Serial Joint Jacobian

The Jacobian with respect to serial joints:

$$\boldsymbol{J}_q = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{q}} \tag{157}$$

$$= \begin{bmatrix} \frac{\partial x}{\partial q_1} & \frac{\partial x}{\partial q_2} & \cdots & \frac{\partial x}{\partial q_{n_j}} \\ \frac{\partial y}{\partial q_1} & \frac{\partial y}{\partial q_2} & \cdots & \frac{\partial y}{\partial q_{n_j}} \\ \frac{\partial \theta}{\partial q_1} & \frac{\partial \theta}{\partial q_2} & \cdots & \frac{\partial \theta}{\partial q_{n_j}} \end{bmatrix} \tag{158}$$

Individual elements for the 2-DOF serial chain:

$$\frac{\partial x_M}{\partial q_1} = -l_1 \sin q_1 - l_2 \sin(q_1 + q_2) \tag{159}$$

$$\frac{\partial x_M}{\partial q_2} = -l_2 \sin(q_1 + q_2) \tag{160}$$

$$\frac{\partial y_M}{\partial q_1} = l_1 \cos q_1 + l_2 \cos(q_1 + q_2) \tag{161}$$

$$\frac{\partial y_M}{\partial q_2} = l_2 \cos(q_1 + q_2) \tag{162}$$

Therefore, the serial joint Jacobian is:

$$\boldsymbol{J}_q = \begin{bmatrix} -l_1 \sin q_1 - l_2 \sin(q_1 + q_2) & -l_2 \sin(q_1 + q_2) \\ l_1 \cos q_1 + l_2 \cos(q_1 + q_2) & l_2 \cos(q_1 + q_2) \end{bmatrix} \tag{163}$$

3.3 Cable-Constraint Jacobian

Since both cables attach to the same point M, we analyze the constraint Jacobian rather than a direct cable Jacobian.

3.3.1 Constraint Jacobian with respect to Joint Variables

$$\boldsymbol{G}_q = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{q}} = \begin{bmatrix} \frac{\partial g_1}{\partial q_1} & \frac{\partial g_1}{\partial q_2} \\ \frac{\partial g_2}{\partial q_1} & \frac{\partial g_2}{\partial q_2} \end{bmatrix} \tag{164}$$

Computing the partial derivatives:

$$\frac{\partial g_1}{\partial q_1} = 2(x_M + L)\frac{\partial x_M}{\partial q_1} + 2y_M \frac{\partial y_M}{\partial q_1} \tag{165}$$

$$= 2(x_M + L)(-l_1 \sin q_1 - l_2 \sin(q_1 + q_2)) + 2y_M(l_1 \cos q_1 + l_2 \cos(q_1 + q_2)) \tag{166}$$

$$\frac{\partial g_1}{\partial q_2} = 2(x_M + L)(-l_2 \sin(q_1 + q_2)) + 2y_M(l_2 \cos(q_1 + q_2)) \tag{167}$$

$$\frac{\partial g_2}{\partial q_1} = 2(x_M - L)(-l_1 \sin q_1 - l_2 \sin(q_1 + q_2)) + 2y_M(l_1 \cos q_1 + l_2 \cos(q_1 + q_2)) \tag{168}$$

$$\frac{\partial g_2}{\partial q_2} = 2(x_M - L)(-l_2 \sin(q_1 + q_2)) + 2y_M(l_2 \cos(q_1 + q_2)) \tag{169}$$

3.3.2 Constraint Jacobian with respect to Cable Lengths

$$\boldsymbol{G}_\rho = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{\rho}} = \begin{bmatrix} -2\rho_1 & 0 \\ 0 & -2\rho_2 \end{bmatrix} \tag{170}$$

For this robot, the end-effector position is determined by the serial chain configuration, not directly by cable lengths. However, if we consider the cables as additional constraints, we need to analyze the constraint Jacobian.

The constraint equations are:

$$g_1(\boldsymbol{q}, \rho_1) = \rho_1^2 - [(x_M + L)^2 + y_M^2] = 0 \tag{171}$$

$$g_2(\boldsymbol{q}, \rho_2) = \rho_2^2 - [(x_M - L)^2 + y_M^2] = 0 \tag{172}$$

The constraint Jacobian with respect to joint variables:

$$\boldsymbol{G}_q = \begin{bmatrix} \frac{\partial g_1}{\partial q_1} & \frac{\partial g_1}{\partial q_2} \\ \frac{\partial g_2}{\partial q_1} & \frac{\partial g_2}{\partial q_2} \end{bmatrix} \tag{173}$$

$$= -2 \begin{bmatrix} (x_M + L)\frac{\partial x_M}{\partial q_1} + y_M \frac{\partial y_M}{\partial q_1} & (x_M + L)\frac{\partial x_M}{\partial q_2} + y_M \frac{\partial y_M}{\partial q_2} \\ (x_M - L)\frac{\partial x_M}{\partial q_1} + y_M \frac{\partial y_M}{\partial q_1} & (x_M - L)\frac{\partial x_M}{\partial q_2} + y_M \frac{\partial y_M}{\partial q_2} \end{bmatrix} \tag{174}$$

The constraint Jacobian with respect to cable lengths:

$$\boldsymbol{G}_\rho = \begin{bmatrix} 2\rho_1 & 0 \\ 0 & 2\rho_2 \end{bmatrix} \tag{175}$$

4 Constraint Analysis

4.1 Constraint Jacobian

The constraint Jacobian relates constraint variations to joint and cable variations:

$$\frac{d\boldsymbol{g}}{dt} = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{q}}\dot{\boldsymbol{q}} + \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{l}}\dot{\boldsymbol{l}} = \boldsymbol{0} \tag{176}$$

$$\boldsymbol{G}_q\dot{\boldsymbol{q}} + \boldsymbol{G}_l\dot{\boldsymbol{l}} = \boldsymbol{0} \tag{177}$$

Where:

$$\boldsymbol{G}_q = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{q}} = \begin{bmatrix} \frac{\partial g_1}{\partial q_1} & \cdots & \frac{\partial g_1}{\partial q_{n_j}} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_{n_c}}{\partial q_1} & \cdots & \frac{\partial g_{n_c}}{\partial q_{n_j}} \end{bmatrix} \tag{178}$$

$$\boldsymbol{G}_\rho = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{\rho}} = \begin{bmatrix} \frac{\partial g_1}{\partial \rho_1} & \cdots & \frac{\partial g_1}{\partial \rho_{n_c}} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_{n_c}}{\partial \rho_1} & \cdots & \frac{\partial g_{n_c}}{\partial \rho_{n_c}} \end{bmatrix} \tag{179}$$

NOTATION CORRECTION: Fixed inconsistency - using $\boldsymbol{\rho} = [\rho_1, \rho_2]^T$ for cable lengths consistently throughout.

4.2 Reduced Jacobian

Assuming $\boldsymbol{G}_\rho$ is invertible, we can solve for cable length velocities:

$$\dot{\boldsymbol{\rho}} = -\boldsymbol{G}_\rho^{-1}\boldsymbol{G}_q\dot{\boldsymbol{q}} \tag{180}$$

$$\dot{\boldsymbol{x}} = \boldsymbol{J}_q\dot{\boldsymbol{q}} + \boldsymbol{J}_\rho\dot{\boldsymbol{\rho}} \tag{181}$$

$$= \boldsymbol{J}_q\dot{\boldsymbol{q}} + \boldsymbol{J}_\rho(-\boldsymbol{G}_\rho^{-1}\boldsymbol{G}_q\dot{\boldsymbol{q}}) \tag{182}$$

$$= (\boldsymbol{J}_q - \boldsymbol{J}_\rho\boldsymbol{G}_\rho^{-1}\boldsymbol{G}_q)\dot{\boldsymbol{q}} \tag{183}$$

$$= \boldsymbol{J}_{red}\dot{\boldsymbol{q}} \tag{184}$$

The reduced Jacobian is:

$$\boldsymbol{J}_{red} = \boldsymbol{J}_q - \boldsymbol{J}_\rho\boldsymbol{G}_\rho^{-1}\boldsymbol{G}_q \tag{185}$$

5 Singularity Analysis

5.1 Classification of Singularities

Singularities in hybrid serial-cable robots can be classified into several categories based on their physical interpretation and mathematical characteristics.

5.2 Singularity Types

5.2.1 Type I: Serial Chain Singularities

For the 2-DOF serial chain, singularities occur when:

$$\det(\boldsymbol{J}_q) = 0 \tag{186}$$

$$\text{where } \det(\boldsymbol{J}_q) = l_2(-l_1 \sin q_1 \cos(q_1 + q_2) + l_1 \cos q_1 \sin(q_1 + q_2)) \tag{187}$$

$$= l_1 l_2 \sin((q_1 + q_2) - q_1) \tag{188}$$

$$= l_1 l_2 \sin q_2 \tag{189}$$

Therefore, serial chain singularities occur when:

$$\sin q_2 = 0 \tag{190}$$

$$\Rightarrow q_2 = 0 \text{ or } q_2 = \pi \tag{191}$$

These correspond to:

- $q_2 = 0$: Fully extended configuration
- $q_2 = \pi$: Fully folded configuration

5.2.2 Type II: Cable Constraint Singularities

Since both cables attach to the same point M, cable singularities occur when the constraint Jacobian $\boldsymbol{G}_q$ becomes singular, meaning the constraint equations become dependent.

Geometric Interpretation: Cable singularities occur when the end-effector position cannot be uniquely determined by the cable length constraints, which happens when:

1. **Collinear Configuration:** End-effector lies on the X-axis ($y_M = 0$):

$$\text{When } y_M = 0: \quad g_1 = (x_M + L)^2 - \rho_1^2 = 0 \tag{192}$$

$$g_2 = (x_M - L)^2 - \rho_2^2 = 0 \tag{193}$$

The constraint gradients become:

$$\nabla g_1 = [2(x_M + L), 0] \tag{194}$$

$$\nabla g_2 = [2(x_M - L), 0] \tag{195}$$

These are parallel (linearly dependent) when $x_M = 0$.

2. **Symmetric Configuration:** End-effector on Y-axis ($x_M = 0$):

$$\rho_1 = \rho_2 = \sqrt{L^2 + y_M^2} \tag{196}$$

In this case, both constraints provide the same information about the distance from the X-axis.

Mathematical Condition:

$$\det(\boldsymbol{G}_q) = 0 \tag{197}$$

$$\text{or equivalently:} \quad \operatorname{rank}(\boldsymbol{G}_q) < 2 \tag{198}$$

5.2.3 Type III: Combined Singularities

Occur when the complete system Jacobian loses rank:

$$\text{rank}(\boldsymbol{J}) < 3 \tag{199}$$

$$\det(\boldsymbol{J}\boldsymbol{J}^T) = 0 \quad \text{(for non-square Jacobians)} \tag{200}$$

5.2.4 Type IV: Constraint Singularities

Occur when the constraint Jacobian $\boldsymbol{G}_l$ becomes singular:

$$\det(\boldsymbol{G}_l) = 0 \tag{201}$$

$$\text{rank}(\boldsymbol{G}_l) < n_c \tag{202}$$

This represents configurations where cable length constraints become dependent.

5.2.5 Type V: Actuation Singularities

Occur when the system cannot generate forces in certain directions:

$$\text{rank}(\boldsymbol{J}^T) < 3 \tag{203}$$

5.3 Mathematical Characterization

5.3.1 Screw Theory Approach

Using screw theory, each joint and cable contributes a twist:

$$\boldsymbol{\xi}_i = \begin{bmatrix} \boldsymbol{\omega}_i \\ \boldsymbol{v}_i \end{bmatrix} \quad \text{(joint twist)} \tag{204}$$

$$\boldsymbol{\zeta}_j = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{u}_j \end{bmatrix} \quad \text{(cable twist)} \tag{205}$$

Where $\boldsymbol{u}_j$ is the unit vector along cable j.

5.3.2 Grassmann-Plücker Coordinates

For planar motion, singularities can be analyzed using Plücker coordinates:

$$\boldsymbol{L}_{ij} = \boldsymbol{\xi}_i \times \boldsymbol{\xi}_j \tag{206}$$

$$\text{Singularity:} \quad \boldsymbol{L}_{12} \cdot \boldsymbol{L}_{34} = 0 \tag{207}$$

6 Cable Tension Constraints and Negative Tension Analysis

Fundamental Physical Constraint: A critical aspect of cable-driven systems is that cables can only pull, never push. This imposes strict inequality constraints on cable tensions:

$$T_1 \geq 0 \tag{208}$$

$$T_2 \geq 0 \tag{209}$$

Physical Constraints:
$T_1 \geq 0$ (Cable 1 can only pull)
$T_2 \geq 0$ (Cable 2 can only pull)

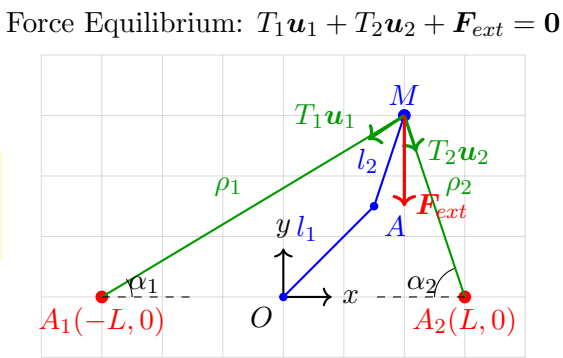

Figure 4: Cable tension force equilibrium at end-effector point M. The system must satisfy both force equilibrium and non-negative tension constraints simultaneously.

6.1 Negative Cable Tension Problem

When mathematical analysis yields negative cable tensions, the system cannot achieve the desired force equilibrium through cables alone. This represents a fundamental limitation of cable actuation.

Force Equilibrium Equations: For static equilibrium at the end-effector point M under external load $\boldsymbol{F}_{ext}$:

$$\boldsymbol{F}_1 + \boldsymbol{F}_2 + \boldsymbol{F}_{ext} = \mathbf{0} \tag{210}$$

$$T_1\boldsymbol{u}_1 + T_2\boldsymbol{u}_2 + \boldsymbol{F}_{ext} = \mathbf{0} \tag{211}$$

Expanding in component form:

$$T_1\frac{x_M + L}{\rho_1} + T_2\frac{x_M - L}{\rho_2} = -F_{ext,x} \tag{212}$$

$$T_1\frac{y_M}{\rho_1} + T_2\frac{y_M}{\rho_2} = -F_{ext,y} \tag{213}$$

6.2 Solving for Cable Tensions

The goal is to determine the cable tensions T_1 and T_2 required to maintain equilibrium at the end-effector point M under an external force $\boldsymbol{F}_{ext} = [F_{ext,x}, F_{ext,y}]^T$. This is solved using static equilibrium conditions.

6.2.1 Force Equilibrium at End-Effector

At point M, the sum of all forces must equal zero:

$$T_1\boldsymbol{u}_1 + T_2\boldsymbol{u}_2 + \boldsymbol{F}_{ext} = \mathbf{0} \tag{214}$$

Where $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ are unit vectors along the cable directions:

$$\boldsymbol{u}_1 = \frac{1}{\rho_1}\begin{bmatrix} x_M + L \\ y_M \end{bmatrix} \tag{215}$$

$$\boldsymbol{u}_2 = \frac{1}{\rho_2}\begin{bmatrix} x_M - L \\ y_M \end{bmatrix} \tag{216}$$

6.2.2 Matrix Formulation (Recommended for Implementation)

The force equilibrium can be expressed as a 2×2 linear system:

$$\begin{bmatrix} \frac{x_M+L}{\rho_1} & \frac{x_M-L}{\rho_2} \\ \frac{y_M}{\rho_1} & \frac{y_M}{\rho_2} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = - \begin{bmatrix} F_{ext,x} \\ F_{ext,y} \end{bmatrix} \tag{217}$$

This can be written compactly as:

$$\boldsymbol{G}_\rho(\boldsymbol{q}, \boldsymbol{\rho}) \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = -\boldsymbol{F}_{ext} \tag{218}$$

Where $\boldsymbol{G}_\rho$ is the **Cable Geometry Jacobian** (also called the Constraint Jacobian), which maps cable tensions to end-effector forces.

6.2.3 Direct Solution via Matrix Inversion

Solving for tensions by inverting the Cable Geometry Jacobian:

$$\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = -\boldsymbol{G}_\rho^{-1}(\boldsymbol{q}, \boldsymbol{\rho}) \begin{bmatrix} F_{ext,x} \\ F_{ext,y} \end{bmatrix} \tag{219}$$

The determinant and inverse of the Cable Geometry Jacobian:

$$\det(\boldsymbol{G}_\rho) = \frac{1}{\rho_1\rho_2} \left[\frac{x_M + L}{\rho_1} \cdot \frac{y_M}{\rho_2} - \frac{x_M - L}{\rho_2} \cdot \frac{y_M}{\rho_1} \right] \tag{220}$$

$$= \frac{y_M}{\rho_1^2\rho_2^2} \left[(x_M + L) - (x_M - L) \right] \tag{221}$$

$$= \frac{2Ly_M}{\rho_1^2\rho_2^2} \tag{222}$$

The matrix inverse:

$$\boldsymbol{G}_\rho^{-1} = \frac{\rho_1\rho_2}{2Ly_M} \begin{bmatrix} \frac{y_M}{\rho_2} & -\frac{x_M-L}{\rho_2} \\ -\frac{y_M}{\rho_1} & \frac{x_M+L}{\rho_1} \end{bmatrix} \tag{223}$$

6.2.4 Explicit Cable Tension Formulas

Multiplying out the matrix inversion:

$$T_1 = -\frac{\rho_1}{2L} \left[F_{ext,x} \frac{y_M}{\rho_2} - F_{ext,y} \frac{x_M - L}{\rho_2} \right] \tag{224}$$

$$= \frac{\rho_1}{2Ly_M} \left[F_{ext,x} y_M - F_{ext,y}(x_M - L) \right] \tag{225}$$

$$T_2 = -\frac{\rho_2}{2L} \left[-F_{ext,x} \frac{y_M}{\rho_1} + F_{ext,y} \frac{x_M + L}{\rho_1} \right] \tag{226}$$

$$= \frac{\rho_2}{2Ly_M} \left[F_{ext,x} y_M + F_{ext,y}(x_M + L) \right] \tag{227}$$

6.2.5 Numerical Example

Given Configuration:

- Link lengths: $l_1 = 1$ m, $l_2 = 1$ m
- Cable anchor separation: $2L = 4$ m (so $L = 2$ m)

- Joint configuration: $q_1 = 30°$, $q_2 = 45°$
- External load: $\boldsymbol{F}_{ext} = [10, -5]^T$ N (rightward and downward)

Step 1: Calculate end-effector position

$$x_M = l_1 \cos(30°) + l_2 \cos(75°) = 0.866 + 0.259 = 1.125 \text{ m} \tag{228}$$
$$y_M = l_1 \sin(30°) + l_2 \sin(75°) = 0.500 + 0.966 = 1.466 \text{ m} \tag{229}$$

Step 2: Calculate cable lengths

$$\rho_1 = \sqrt{(x_M + L)^2 + y_M^2} = \sqrt{(1.125 + 2)^2 + 1.466^2} = \sqrt{11.383} = 3.374 \text{ m} \tag{230}$$
$$\rho_2 = \sqrt{(x_M - L)^2 + y_M^2} = \sqrt{(1.125 - 2)^2 + 1.466^2} = \sqrt{2.591} = 1.610 \text{ m} \tag{231}$$

Step 3: Apply tension formulas

$$T_1 = \frac{3.374}{2 \cdot 2 \cdot 1.466} [10 \cdot 1.466 - (-5) \cdot (1.125 - 2)] \tag{232}$$
$$= \frac{3.374}{5.864} [14.66 + 4.4375] \tag{233}$$
$$= 0.575 \times 19.0975 = 10.98 \text{ N} \tag{234}$$
$$T_2 = \frac{1.610}{2 \cdot 2 \cdot 1.466} [10 \cdot 1.466 + (-5) \cdot (1.125 + 2)] \tag{235}$$
$$= \frac{1.610}{5.864} [14.66 - 15.625] \tag{236}$$
$$= 0.275 \times (-0.965) = -0.265 \text{ N} \tag{237}$$

Result: $T_1 = 10.98$ N (feasible), $T_2 = -0.265$ N (**negative - infeasible!**)

This configuration would require cable 2 to push, which is physically impossible. The end-effector cannot maintain equilibrium under this force with these joint angles.

6.2.6 Stability and Computational Considerations

Singularity Issues: The Cable Geometry Jacobian becomes singular when:

$$\det(\boldsymbol{G}_\rho) = \frac{2Ly_M}{\rho_1^2 \rho_2^2} = 0 \tag{238}$$

This occurs when $y_M = 0$, meaning the end-effector is on the horizontal line through the cable anchors. At this configuration, the cables cannot support vertical forces.

Condition Number: To assess numerical stability, compute the condition number:

$$\kappa(\boldsymbol{G}_\rho) = \|\boldsymbol{G}_\rho\| \cdot \|\boldsymbol{G}_\rho^{-1}\| \tag{239}$$

A high condition number indicates ill-conditioning and sensitivity to numerical errors. Avoid configurations where $\kappa > 100$.

Computational Algorithm:

Algorithm 1 Cable Tension Computation and Feasibility Analysis

Require: Joint configuration $\boldsymbol{q} = [q_1, q_2]^T$, external force $\boldsymbol{F}_{ext} = [F_{ext,x}, F_{ext,y}]^T$
Require: Link lengths l_1, l_2, cable anchor separation $2L$, singularity tolerance ϵ_{sing}
Ensure: Cable tensions T_1, T_2 and feasibility status

Step 1: Forward Kinematics
$x_M \leftarrow l_1 \cos(q_1) + l_2 \cos(q_1 + q_2)$
$y_M \leftarrow l_1 \sin(q_1) + l_2 \sin(q_1 + q_2)$
Step 2: Cable Length Computation
$\rho_1 \leftarrow \sqrt{(x_M + L)^2 + y_M^2}$
$\rho_2 \leftarrow \sqrt{(x_M - L)^2 + y_M^2}$
Step 3: Cable Geometry Jacobian Construction
$\boldsymbol{G}_\rho \leftarrow \begin{bmatrix} \frac{x_M+L}{\rho_1} & \frac{x_M-L}{\rho_2} \\ \frac{y_M}{\rho_1} & \frac{y_M}{\rho_2} \end{bmatrix}$
Step 4: Singularity Check
$\det_\rho \leftarrow \text{determinant}(\boldsymbol{G}_\rho) = \frac{2Ly_M}{\rho_1^2\rho_2^2}$
if $|\det_\rho| < \epsilon_{sing}$ **then**
  return {status: SINGULAR, T_1: undefined, T_2: undefined}
  // Configuration is singular - cannot solve for tensions
end if
Step 5: Solve Linear System via LU Decomposition
$[\boldsymbol{L}, \boldsymbol{U}] \leftarrow \text{LU}(\boldsymbol{G}_\rho)$
$\boldsymbol{y} \leftarrow \text{ForwardSubstitution}(\boldsymbol{L}, -\boldsymbol{F}_{ext})$
$\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} \leftarrow \text{BackSubstitution}(\boldsymbol{U}, \boldsymbol{y})$
Step 6: Feasibility Assessment
if $T_1 \geq 0$ **and** $T_2 \geq 0$ **then**
  feasible $\leftarrow$ TRUE
else
  feasible $\leftarrow$ FALSE
  // Record which tensions are negative
  if $T_1 < 0$ **then**
    reason $\leftarrow$ "Cable 1 would require push"
  end if
  if $T_2 < 0$ **then**
    reason $\leftarrow$ "Cable 2 would require push"
  end if
end if
Step 7: Condition Number Assessment (Optional)
$\kappa \leftarrow \text{conditionNumber}(\boldsymbol{G}_\rho)$
if $\kappa > 100$ **then**
  stability $\leftarrow$ "ILL-CONDITIONED"
else
  stability $\leftarrow$ "WELL-CONDITIONED"
end if
return $\{T_1, T_2, \text{feasible}, \text{stability}, \kappa\}$

6.2.7 Algorithm Details and Implementation Notes

Forward Kinematics Computation: The end-effector position is computed from joint angles using standard 2-DOF planar arm kinematics:

$$x_M = l_1 \cos(q_1) + l_2 \cos(q_1 + q_2) \tag{240}$$

$$y_M = l_1 \sin(q_1) + l_2 \sin(q_1 + q_2) \tag{241}$$

Cable Length Calculation: Euclidean distance from cable anchor points to end-effector:

$$\rho_1 = \|(x_M, y_M) - (-L, 0)\| = \sqrt{(x_M + L)^2 + y_M^2} \tag{242}$$

$$\rho_2 = \|(x_M, y_M) - (L, 0)\| = \sqrt{(x_M - L)^2 + y_M^2} \tag{243}$$

Singularity Detection: When $|\det(\boldsymbol{G}_\rho)| < \epsilon_{sing}$ (typically $\epsilon_{sing} = 10^{-6}$):

$$\frac{2Ly_M}{\rho_1^2 \rho_2^2} \approx 0 \quad \Rightarrow \quad y_M \approx 0 \tag{244}$$

This indicates the end-effector is near the line connecting cable anchors, where cable geometry cannot support arbitrary forces.

Numerical Solution: LU decomposition is preferred over direct matrix inversion for numerical stability:

- **Computational Cost**: $\mathcal{O}(n^3)$ for LU factorization (here $n = 2$ is small)
- **Numerical Stability**: Better conditioning than explicit inversion
- **Reusability**: If multiple forces are applied with same configuration, LU factors can be reused

Feasibility Criteria:

$$\text{Configuration is feasible} \iff T_1 \geq 0 \text{ and } T_2 \geq 0 \tag{245}$$

Condition Number Interpretation:

$$\kappa(\boldsymbol{G}_\rho) = \begin{cases} < 10 & \text{Excellent conditioning} \\ 10 - 100 & \text{Good conditioning} \\ 100 - 1000 & \text{Moderate conditioning - use caution} \\ > 1000 & \text{Poor conditioning - near singularity} \end{cases} \tag{246}$$

6.2.8 Python Implementation Pseudocode

```
def compute_cable_tensions(q1, q2, F_ext_x, F_ext_y, l1, l2, L, eps_sing=1e-6):
    """
    Compute cable tensions and feasibility for given configuration and external force.

    Parameters:
        q1, q2: Joint angles [rad]
        F_ext_x, F_ext_y: External force components [N]
        l1, l2: Link lengths [m]
        L: Half-distance between cable anchors [m]
        eps_sing: Singularity tolerance
```

```
    Returns:
        T1, T2: Cable tensions [N]
        feasible: Boolean indicating if tensions are non-negative
        det_G: Determinant of Cable Geometry Jacobian
        kappa: Condition number
    """
    # Step 1: Forward kinematics
    x_M = l1 * cos(q1) + l2 * cos(q1 + q2)
    y_M = l1 * sin(q1) + l2 * sin(q1 + q2)

    # Step 2: Cable lengths
    rho1 = sqrt((x_M + L)**2 + y_M**2)
    rho2 = sqrt((x_M - L)**2 + y_M**2)

    # Step 3: Cable Geometry Jacobian
    G_rho = np.array([
        [(x_M + L)/rho1, (x_M - L)/rho2],
        [y_M/rho1, y_M/rho2]
    ])

    # Step 4: Singularity check
    det_G = (2 * L * y_M) / (rho1**2 * rho2**2)
    if abs(det_G) < eps_sing:
        return None, None, "SINGULAR", det_G, None

    # Step 5: Solve for tensions
    F_ext = np.array([[-F_ext_x], [-F_ext_y]])
    T = np.linalg.solve(G_rho, F_ext)
    T1, T2 = T[0, 0], T[1, 0]

    # Step 6: Check feasibility
    feasible = (T1 >= 0) and (T2 >= 0)

    # Step 7: Condition number
    kappa = np.linalg.cond(G_rho)

    return T1, T2, feasible, det_G, kappa
```

6.3 Conditions for Negative Tensions

Cable tensions become negative when:

$$T_1 < 0 \quad \text{or} \quad T_2 < 0 \tag{247}$$

This occurs when the required force direction cannot be achieved by the available cable geometry.

Critical Geometric Conditions:

Negative tensions typically occur when:

1. **Force Direction Mismatch:** The external force requires cable compression (impossible)
2. **Unfavorable Cable Geometry:** The cable angles α_1, α_2 create poor mechanical advantage

(a) Feasible Case: $T_1, T_2 > 0$ **(b) Infeasible Case:** $T_1 < 0$

Figure 5: Illustration of cable tension feasibility. (a) Feasible configuration where both cables can provide positive tensions. (b) Infeasible configuration where cable 1 would need to push (negative tension) to achieve equilibrium, which is physically impossible.

3. **Serial Chain Singularities:** Robot configurations where $q_2 = 0$ or $q_2 = \pi$
4. **Cable Anchor Proximity:** End-effector approaches the cable anchor points
5. **Workspace Boundaries:** Near the limits of the kinematic workspace

6.4 Mathematical Analysis of Negative Tension Regions

Left Cable Tension Analysis: $T_1 < 0$ when:

$$F_{ext,x}\frac{y_M}{\rho_2} - F_{ext,y}\frac{x_M - L}{\rho_2} > 0 \tag{248}$$

$$\Rightarrow \quad F_{ext,x}y_M > F_{ext,y}(x_M - L) \tag{249}$$

Right Cable Tension Analysis: $T_2 < 0$ when:

$$-F_{ext,x}\frac{y_M}{\rho_1} + F_{ext,y}\frac{x_M + L}{\rho_1} < 0 \tag{250}$$

$$\Rightarrow \quad F_{ext,x}y_M > F_{ext,y}(x_M + L) \tag{251}$$

Critical Force Angle: Define the external force angle as $\beta = \text{atan2}(F_{ext,y}, F_{ext,x})$. Negative tensions occur when the force angle is incompatible with cable directions:

$$\text{If } \beta \in (\alpha_1, \alpha_2) \text{ and } F_{ext} \text{ points away from cables} \tag{252}$$

$$\text{Then } T_1 < 0 \text{ or } T_2 < 0 \tag{253}$$

6.5 Singularity Connection

When cable tensions become negative, several critical situations arise:

6.5.1 Motor Capability Loss

The motors driving the cables cannot provide the required forces, leading to:

- **Task Failure:** Inability to execute the desired motion or maintain equilibrium
- **Control Loss:** End-effector becomes uncontrollable in certain directions
- **System Instability:** Potential for unwanted oscillations or drift
- **Safety Issues:** Loss of load support or positioning accuracy

6.5.2 Infinite Tension Conditions

When the denominator in tension equations approaches zero:

$$\sin(\alpha_2 - \alpha_1) \to 0 \tag{254}$$

$$\Rightarrow \quad \alpha_2 = \alpha_1 + k\pi, \quad k \in \mathbb{Z} \tag{255}$$

This occurs in specific geometric configurations:

1. Parallel Cable Configuration ($\alpha_2 = \alpha_1$):

$$y_M = 0 \text{ and } x_M \text{ arbitrary} \tag{256}$$

$$\text{Condition: End-effector on X-axis between anchors} \tag{257}$$

2. Anti-parallel Cable Configuration ($\alpha_2 = \alpha_1 + \pi$):

$$x_M = 0 \tag{258}$$

$$\text{Condition: End-effector on Y-axis} \tag{259}$$

3. Combined Singularities: When cable singularities coincide with joint singularities ($q_2 = 0$ or $q_2 = \pi$), the system experiences:

- Loss of multiple degrees of freedom simultaneously
- Reduced controllability in multiple directions
- Enhanced sensitivity to external disturbances

6.6 Workspace Analysis

6.6.1 Workspace Definitions

The negative tension constraint fundamentally limits the robot's operational workspace:

Kinematic Workspace:

$$\mathcal{W}_{kinematic} = \{(x_M, y_M) : |x_M^2 + y_M^2|^{1/2} \leq l_1 + l_2\} \tag{260}$$

$$\cap \{(x_M, y_M) : |x_M^2 + y_M^2|^{1/2} \geq |l_1 - l_2|\} \tag{261}$$

Feasible Workspace (for given external load):

$$\mathcal{W}_{feasible} = \{(x_M, y_M) \in \mathcal{W}_{kinematic} : T_1(\boldsymbol{q}, \boldsymbol{F}_{ext}) \geq 0 \text{ and } T_2(\boldsymbol{q}, \boldsymbol{F}_{ext}) \geq 0\} \tag{262}$$

Cable-Limited Regions:

$$\mathcal{W}_{cable-limited} = \mathcal{W}_{kinematic} \setminus \mathcal{W}_{feasible} \tag{263}$$

$$= \{(x_M, y_M) \in \mathcal{W}_{kinematic} : T_1(\boldsymbol{q}, \boldsymbol{F}_{ext}) < 0 \text{ or } T_2(\boldsymbol{q}, \boldsymbol{F}_{ext}) < 0\} \tag{264}$$

6.6.2 Load-Dependent Workspace

The feasible workspace depends on the external load magnitude and direction:

Zero Load Case ($\boldsymbol{F}_{ext} = \boldsymbol{0}$):

$$\mathcal{W}_{feasible}(\boldsymbol{0}) = \mathcal{W}_{kinematic} \tag{265}$$

All kinematically reachable points are cable-feasible with zero external load.

Gravity Load Case ($\boldsymbol{F}_{ext} = [0, -mg]^T$):

$$T_1 = \frac{mg\rho_1(x_M - L)}{2Ly_M} \tag{266}$$

$$T_2 = \frac{mg\rho_2(x_M + L)}{2Ly_M} \tag{267}$$

Negative tensions occur when:

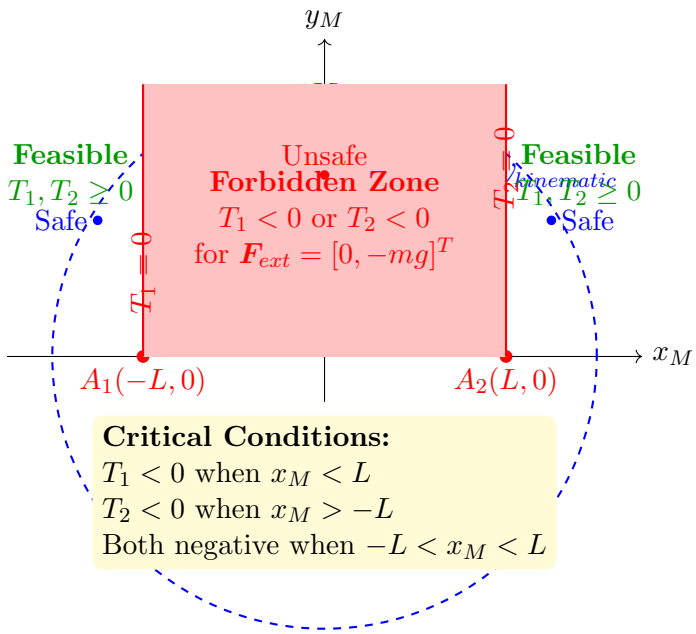

Figure 6: Workspace analysis for downward gravitational load. The forbidden zone (red) shows regions where negative cable tensions would be required. The feasible workspace (green) is significantly reduced compared to the kinematic workspace (blue dashed circle).

- $T_1 < 0$: when $x_M < L$ (left side of right anchor)
- $T_2 < 0$: when $x_M > -L$ (right side of left anchor)
- Both: when $-L < x_M < L$ (between anchors)

This creates a **forbidden zone** between the cable anchors for downward loads.

6.7 Design and Control Implications

6.7.1 Design Optimization

1. Cable Anchor Placement: Optimize anchor positions to maximize feasible workspace:

$$L_{optimal} = \arg\max_{L} |\mathcal{W}_{feasible}(L, \boldsymbol{F}_{ext})| \tag{268}$$

2. Link Length Ratios: Balance serial chain workspace with cable effectiveness:

$$\frac{l_2}{l_1} = \arg\max_{ratio} |\mathcal{W}_{feasible} \cap \mathcal{W}_{dexterous}| \tag{269}$$

3. Joint Range Limits: Coordinate joint limits with cable constraints:

$$q_{2,min} \geq \epsilon > 0 \quad \text{(avoid } q_2 = 0 \text{ singularity)} \tag{270}$$

$$q_{2,max} \leq \pi - \epsilon \quad \text{(avoid } q_2 = \pi \text{ singularity)} \tag{271}$$

6.7.2 Control Strategies

1. Hybrid Control Architecture: Switch between control modes based on tension feasibility:

Algorithm 2 Hybrid Cable-Joint Control

Input: Desired end-effector position $\boldsymbol{x}_{desired}$, external load $\boldsymbol{F}_{ext}$
Compute required cable tensions T_1, T_2 from force equilibrium
if $T_1 \geq T_{min}$ AND $T_2 \geq T_{min}$ **then**
  Use cable control mode
  Control cable lengths ρ_1, ρ_2 to achieve $\boldsymbol{x}_{desired}$
else
  Use joint control mode
  Control joint angles q_1, q_2 to achieve $\boldsymbol{x}_{desired}$
  Set cable tensions to minimum safe values
end if

2. Pre-tensioning Strategy: Maintain minimum cable tensions to avoid slack and improve responsiveness:

$$T_{1,command} = \max(T_{1,required}, T_{min}) \tag{272}$$
$$T_{2,command} = \max(T_{2,required}, T_{min}) \tag{273}$$

Where $T_{min} > 0$ is the minimum safe tension level.

3. Workspace Planning: Generate trajectories that avoid negative tension regions:

$$\boldsymbol{x}_{trajectory}(t) \in \mathcal{W}_{feasible} \quad \forall t \in [0, T_{final}] \tag{274}$$

4. Redundancy Exploitation: Use additional degrees of freedom to maintain positive tensions:

- Adjust joint configuration while maintaining end-effector position
- Vary cable pre-tension levels dynamically
- Coordinate multiple cables if available

6.8 Practical Implementation Considerations

6.8.1 Sensor Requirements

1. Tension Monitoring:

- Load cells or strain gauges in cable paths
- Real-time tension feedback for control
- Early warning for approaching negative tension conditions

2. Position Sensing:

- Joint angle encoders for q_1, q_2
- Cable length sensors for ρ_1, ρ_2
- End-effector position confirmation

6.8.2 Safety Considerations

1. Emergency Protocols:

- Automatic joint braking when cable tensions drop
- Fail-safe positions with guaranteed positive tensions
- Load sharing between joint and cable actuators

2. Operating Limits:

- Conservative workspace boundaries
- Maximum load ratings based on tension analysis
- Speed limitations near singular configurations

6.8.3 Computational Requirements

1. Real-time Tension Calculation: Fast algorithms for computing required tensions:

$$\text{Computation time: } O(1) \text{ per control cycle} \tag{275}$$

$$\text{Memory requirement: } O(1) \text{ for 2D system} \tag{276}$$

2. Workspace Monitoring: Efficient methods to check feasibility:

- Pre-computed workspace maps
- Look-up tables for common load cases
- Approximate analytical boundaries

6.9 Summary

The cable tension constraint analysis reveals that:

Key Insights:

1. Cable systems have inherent directional limitations due to tension positivity
2. Negative tensions create forbidden workspace regions that vary with external loads
3. Singularities can be triggered by both geometric and force-based conditions
4. Proper design and control can mitigate but not eliminate these constraints

Design Guidelines:

1. Optimize cable anchor placement for maximum feasible workspace
2. Implement hybrid control strategies to handle tension constraints
3. Include pre-tensioning systems for improved performance
4. Plan trajectories considering both kinematic and force constraints

Future Research Directions:

1. Optimization of cable arrangements for specific task requirements
2. Advanced control algorithms that anticipate tension constraints
3. Integration with machine learning for adaptive workspace planning
4. Extension to higher-DOF and 3D cable-driven systems

6.10 Singularity Conditions

6.10.1 Determinant-Based Analysis

For square Jacobians ($n_j + n_c = 3$):

$$S(\boldsymbol{q}, \boldsymbol{\rho}) = \det(\boldsymbol{J}(\boldsymbol{q}, \boldsymbol{\rho})) \tag{277}$$

$$= \det \begin{bmatrix} \frac{\partial x_M}{\partial q_1} & \frac{\partial x_M}{\partial q_2} \\ \frac{\partial y_M}{\partial q_1} & \frac{\partial y_M}{\partial q_2} \end{bmatrix} \tag{278}$$

$$= l_1 l_2 \sin q_2 \tag{279}$$

Singular configurations satisfy:

$$S(\boldsymbol{q}, \boldsymbol{\rho}) = 0 \quad \Rightarrow \quad q_2 = n\pi, \quad n \in \mathbb{Z} \tag{280}$$

6.10.2 SVD-Based Analysis

Using Singular Value Decomposition:

$$\boldsymbol{J} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T \tag{281}$$

$$\sigma_{\min} = \min(\operatorname{diag}(\boldsymbol{\Sigma})) \tag{282}$$

$$\kappa(\boldsymbol{J}) = \frac{\sigma_{\max}}{\sigma_{\min}} \tag{283}$$

Singularity conditions:

$$\sigma_{\min} \approx 0 \tag{284}$$

$$\kappa(\boldsymbol{J}) \to \infty \tag{285}$$

6.11 Manipulability Measures

6.11.1 Yoshikawa's Manipulability

$$\mu(\boldsymbol{q}, \boldsymbol{l}) = \sqrt{\det(\boldsymbol{J}\boldsymbol{J}^T)} \tag{286}$$

$$= \prod_{i=1}^{\operatorname{rank}(\boldsymbol{J})} \sigma_i \tag{287}$$

6.11.2 Condition Number

$$\kappa(\boldsymbol{q}, \boldsymbol{l}) = \|\boldsymbol{J}\| \|\boldsymbol{J}^\dagger\| \tag{288}$$

$$= \frac{\sigma_{\max}}{\sigma_{\min}} \tag{289}$$

where $\boldsymbol{J}^\dagger$ is the Moore-Penrose pseudoinverse.

7 Advanced Singularity Analysis

7.1 Geometric Interpretation

7.1.1 Instantaneous Center of Rotation

At singular configurations, the instantaneous center of rotation may:

1. Move to infinity (pure translation)
2. Become indeterminate (loss of constraint)
3. Coincide with special geometric points

7.1.2 Velocity Space Analysis

The feasible velocity space is defined by:

$$\mathcal{V} = \{\dot{\boldsymbol{x}} : \dot{\boldsymbol{x}} = \boldsymbol{J}\dot{\boldsymbol{q}}, \dot{\boldsymbol{q}} \in \mathbb{R}^{n_j+n_c}\} \tag{290}$$

At singularities, $\dim(\mathcal{V}) < 3$.

7.2 Bifurcation Analysis

7.2.1 Fold Singularities

Characterized by:

$$\det(\boldsymbol{J}) = 0 \tag{291}$$
$$\nabla(\det(\boldsymbol{J})) \neq \boldsymbol{0} \tag{292}$$

7.2.2 Cusp Singularities

Higher-order singularities where:

$$\det(\boldsymbol{J}) = 0 \tag{293}$$
$$\nabla(\det(\boldsymbol{J})) = \boldsymbol{0} \tag{294}$$
$$\nabla^2(\det(\boldsymbol{J})) \neq \boldsymbol{0} \tag{295}$$

7.3 Dynamic Singularities

7.3.1 Acceleration Analysis

The acceleration equation is:

$$\ddot{\boldsymbol{x}} = \boldsymbol{J}\ddot{\boldsymbol{q}} + \dot{\boldsymbol{J}}\dot{\boldsymbol{q}} \tag{296}$$
$$= \boldsymbol{J}\ddot{\boldsymbol{q}} + \boldsymbol{h}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \tag{297}$$

Dynamic singularities occur when:

$$\boldsymbol{h}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \notin \mathrm{Range}(\boldsymbol{J}) \tag{298}$$

8 Workspace Analysis

8.1 Workspace Classification

8.1.1 Primary Workspace

Points reachable with at least one orientation:

$$\mathcal{W}_1 = \{(x, y) : \exists \theta, \boldsymbol{q}, \boldsymbol{l} \text{ s.t. } (x, y, \theta) = \boldsymbol{f}(\boldsymbol{q}, \boldsymbol{l})\} \tag{299}$$

8.1.2 Secondary Workspace

Points reachable with all orientations:

$$\mathcal{W}_2 = \{(x, y) : \forall\theta, \exists\boldsymbol{q}, \boldsymbol{l} \text{ s.t. } (x, y, \theta) = \boldsymbol{f}(\boldsymbol{q}, \boldsymbol{l})\} \tag{300}$$

8.1.3 Dexterous Workspace

Points where the manipulability is above a threshold:

$$\mathcal{W}_d = \{(x, y) : \mu(\boldsymbol{q}, \boldsymbol{l}) > \mu_{min}\} \tag{301}$$

8.2 Workspace Boundary

The reachable workspace is bounded by:

$$\mathcal{W} = \{(x, y) : \exists(\boldsymbol{q}, \boldsymbol{l}) \text{ such that } (x, y) = \boldsymbol{f}(\boldsymbol{q}, \boldsymbol{l})\} \tag{302}$$

$$\text{subject to:} \quad \boldsymbol{q}_{\min} \leq \boldsymbol{q} \leq \boldsymbol{q}_{\max} \tag{303}$$

$$\boldsymbol{l}_{\min} \leq \boldsymbol{l} \leq \boldsymbol{l}_{\max} \tag{304}$$

8.3 Singular Workspace

The singular workspace is defined as:

$$\mathcal{W}_s = \{(x, y) \in \mathcal{W} : S(\boldsymbol{q}, \boldsymbol{l}) = 0\} \tag{305}$$

9 Computational Methods

9.1 Efficient Jacobian Computation

9.1.1 Analytical Differentiation

For computational efficiency, Jacobian elements should be computed analytically:

$$J_{ij} = \frac{\partial f_i}{\partial q_j} \tag{306}$$

$$= \sum_{k=j}^{n} \frac{\partial f_i}{\partial T_k} \frac{\partial T_k}{\partial q_j} \tag{307}$$

Cable Angle Derivatives: When using atan2 for cable angles, the derivatives are:

$$\frac{\partial \alpha_1}{\partial q_i} = \frac{\partial}{\partial q_i} \text{atan2}(y_M, x_M + L) \tag{308}$$

$$= \frac{(x_M + L)\frac{\partial y_M}{\partial q_i} - y_M \frac{\partial x_M}{\partial q_i}}{(x_M + L)^2 + y_M^2} \tag{309}$$

$$= \frac{(x_M + L)\frac{\partial y_M}{\partial q_i} - y_M \frac{\partial x_M}{\partial q_i}}{\rho_1^2} \tag{310}$$

Similarly for α_2:

$$\frac{\partial \alpha_2}{\partial q_i} = \frac{(x_M - L)\frac{\partial y_M}{\partial q_i} - y_M \frac{\partial x_M}{\partial q_i}}{\rho_2^2} \tag{311}$$

These expressions are well-defined for all configurations, unlike the simple arctan formulation.

9.1.2 Recursive Computation

Using the recursive Newton-Euler approach:

Algorithm 3 Recursive Jacobian Computation

```
Initialize: z_0 = [0, 0, 1]^T, p_0 = [0, 0, 0]^T
for i = 1 to n do
    Compute z_i = R_{i-1} z_{i-1}
    Compute p_i = p_{i-1} + R_{i-1} d_{i-1}
    if joint i is revolute then
        J_{v,i} = z_{i-1} × (p_n − p_{i-1})
        J_{ω,i} = z_{i-1}
    else
        J_{v,i} = z_{i-1}
        J_{ω,i} = 0
    end if
end for
```

9.2 Singularity Detection Algorithms

9.2.1 Multi-Method Approach

Algorithm 4 Comprehensive Singularity Detection

```
Require: Configuration (q, l), tolerances ε_det, ε_svd, κ_max
Ensure: Singularity status and type
Compute Jacobian J(q, l)
method1 ← |det(J)| < ε_det
Compute SVD: J = U Σ V^T
σ_min ← min(diag(Σ))
method2 ← σ_min < ε_svd
κ ← σ_max / σ_min
method3 ← κ > κ_max
if method1 OR method2 OR method3 then
    Classify singularity type
    return True, singularity_type
else
    return False, None
end if
```

9.2.2 Adaptive Tolerance Selection

The tolerances should be adapted based on:

$$\epsilon_{det} = \epsilon_{base} \cdot \|\boldsymbol{J}\|_F^n \tag{312}$$

$$\epsilon_{svd} = \epsilon_{base} \cdot \|\boldsymbol{J}\|_2 \tag{313}$$

$$\kappa_{max} = \frac{1}{\epsilon_{base}} \tag{314}$$

Where $\epsilon_{base} = \sqrt{\text{machine precision}}$.

9.3 Workspace Generation Algorithms

9.3.1 Discretization Method

Algorithm 5 Workspace Boundary Detection

Require: Grid resolution $\Delta x, \Delta y$, joint limits
Ensure: Workspace boundary points

```
Initialize empty workspace grid
for x = x_min to x_max step Δx do
  for y = y_min to y_max step Δy do
    solved ← False
    for each orientation θ do
      Try to solve inverse kinematics for (x, y, θ)
      if solution exists within joint limits then
        Mark (x, y) as reachable
        Check singularity at solution
        solved ← True
        break
      end if
    end for
    if NOT solved then
      Mark (x, y) as unreachable
    end if
  end for
end for
Extract boundary from reachable/unreachable interface
```

9.4 Optimization-Based Methods

9.4.1 Singularity-Free Path Planning

Minimize proximity to singularities:

$$\min_{\boldsymbol{q}(t),\boldsymbol{l}(t)} \quad \int_0^T \left(\frac{1}{\mu(\boldsymbol{q}(t), \boldsymbol{l}(t))} + \|\dot{\boldsymbol{q}}(t)\|^2 + \|\dot{\boldsymbol{l}}(t)\|^2 \right) dt \tag{315}$$

$$\text{s.t.} \quad \boldsymbol{x}(t) = \boldsymbol{f}(\boldsymbol{q}(t), \boldsymbol{l}(t)) \tag{316}$$

$$\boldsymbol{q}_{min} \leq \boldsymbol{q}(t) \leq \boldsymbol{q}_{max} \tag{317}$$

$$\boldsymbol{l}_{min} \leq \boldsymbol{l}(t) \leq \boldsymbol{l}_{max} \tag{318}$$

9.4.2 Singularity Avoidance

Using null space projection:

$$\dot{\boldsymbol{q}} = \boldsymbol{J}^{\dagger}\dot{\boldsymbol{x}}_{desired} + (\boldsymbol{I} - \boldsymbol{J}^{\dagger}\boldsymbol{J})\boldsymbol{k} \tag{319}$$

$$\text{where } \boldsymbol{k} = k_0 \nabla_{\boldsymbol{q}} \mu(\boldsymbol{q}, \boldsymbol{l}) \tag{320}$$

9.5 Workspace Discretization

For workspace analysis:

$$x_i = x_{\min} + i\Delta x, \quad i = 0, 1, \ldots, N_x \tag{321}$$

$$y_j = y_{\min} + j\Delta y, \quad j = 0, 1, \ldots, N_y \tag{322}$$

10 Performance Analysis

10.1 Computational Complexity

10.1.1 Jacobian Computation

- Forward kinematics: $O(n_j)$
- Analytical Jacobian: $O(n_j^2 + n_c^2)$
- Numerical Jacobian: $O((n_j + n_c) \cdot \text{FK cost})$

10.1.2 Singularity Detection

- Determinant: $O(n^3)$ where $n = n_j + n_c$
- SVD: $O(mn^2)$ where $m = \min(3, n_j + n_c)$
- Condition number: Same as SVD

10.1.3 Workspace Generation

For grid resolution $N_x \times N_y$:

- Discretization method: $O(N_x \cdot N_y \cdot \text{IK cost})$
- Boundary extraction: $O(N_x \cdot N_y)$

10.2 Accuracy Analysis

10.2.1 Numerical Sensitivity

The sensitivity of singularity detection to numerical errors:

$$\frac{d(\det \boldsymbol{J})}{dJ_{ij}} = \text{adj}(\boldsymbol{J})_{ji} \tag{323}$$

$$\text{Relative sensitivity:} \quad \frac{|J_{ij}|}{|\det \boldsymbol{J}|}|\text{adj}(\boldsymbol{J})_{ji}| \tag{324}$$

10.2.2 Condition Number Scaling

Near singularities, small perturbations cause large changes:

$$\|\Delta \dot{\boldsymbol{x}}\| \leq \kappa(\boldsymbol{J})\|\Delta \dot{\boldsymbol{q}}\| \tag{325}$$

$$\text{Amplification factor:} \quad \kappa(\boldsymbol{J}) = \frac{\sigma_{max}}{\sigma_{min}} \tag{326}$$

11 Experimental Validation

11.1 Hardware Considerations

11.1.1 Sensor Requirements

- Joint encoders: Resolution $\geq 2^{12}$ counts/rev
- Cable length sensors: Linear encoders or potentiometers
- Force/torque sensors: For validation of singularity effects
- End-effector pose measurement: Vision system or coordinate measuring machine

11.1.2 Actuator Limitations

Near singularities, required actuator forces may become infinite:

$$\boldsymbol{\tau} = \boldsymbol{J}^T \boldsymbol{F}_{external} \tag{327}$$

$$\|\boldsymbol{\tau}\| \leq \frac{\|\boldsymbol{F}_{external}\|}{\sigma_{min}(\boldsymbol{J})} \tag{328}$$

11.2 Validation Protocols

11.2.1 Static Singularity Verification

1. Move robot to predicted singular configuration
2. Apply small end-effector forces in different directions
3. Measure resulting joint torques/cable tensions
4. Verify that some directions require infinite force (within measurement limits)

11.2.2 Dynamic Response Testing

1. Command motion through singular region
2. Monitor manipulability index $\mu(t)$
3. Record maximum joint velocities and accelerations
4. Verify correlation with predicted singular behavior

11.3 Benchmark Problems

11.3.1 Analytical Test Cases

Known singular configurations for validation:

- Fully extended configuration (boundary singularity)
- Folded configuration (interior singularity)
- Cable alignment configurations
- Joint limit configurations

12 Validation and Testing

12.1 Systematic Test Framework

12.1.1 Unit Tests

Individual component validation:

- Forward kinematics accuracy vs. analytical solutions
- Jacobian computation vs. numerical differentiation
- Singularity detection consistency across methods
- Workspace boundary accuracy

12.2 Test Cases

Comprehensive validation scenarios:

12.2.1 Known Singular Configurations

For the specific robot configuration with parameters $l_1 = 1.0$m, $l_2 = 0.8$m, $L = 1.5$m:

Serial Chain Singularities:

$$\text{Fully Extended:} \quad \boldsymbol{q}_1 = [0, 0], \quad \boldsymbol{p}_M = [1.8, 0] \tag{329}$$

$$\text{Fully Folded:} \quad \boldsymbol{q}_2 = [0, \pi], \quad \boldsymbol{p}_M = [0.2, 0] \tag{330}$$

$$\text{Vertical Extended:} \quad \boldsymbol{q}_3 = [\pi/2, 0], \quad \boldsymbol{p}_M = [0, 1.8] \tag{331}$$

$$\text{Vertical Folded:} \quad \boldsymbol{q}_4 = [\pi/2, \pi], \quad \boldsymbol{p}_M = [0, 0.2] \tag{332}$$

Cable Singularities:

$$\text{X-axis alignment:} \quad y_M = 0 \text{ (any configuration on X-axis)} \tag{333}$$

$$\text{Y-axis symmetry:} \quad x_M = 0 \text{ (configurations on Y-axis)} \tag{334}$$

$$\text{Anchor point proximity:} \quad \boldsymbol{p}_M \approx (-L, 0) \text{ or } (L, 0) \tag{335}$$

Combined Singularities:

$$\text{Origin vicinity:} \quad \boldsymbol{q} = [\pi/2, \pi], \quad \boldsymbol{p}_M = [0, 0.2] \tag{336}$$

$$\text{Boundary singularity:} \quad \boldsymbol{q} = [0, 0], \quad \boldsymbol{p}_M = [1.8, 0] \tag{337}$$

12.2.2 Boundary Conditions

Test configurations at workspace boundaries and joint limits.

12.3 Numerical Tolerances

$$\epsilon_{det} = 10^{-6} \quad \text{(determinant tolerance)} \tag{338}$$

$$\epsilon_{svd} = 10^{-8} \quad \text{(SVD tolerance)} \tag{339}$$

$$\kappa_{max} = 10^{6} \quad \text{(maximum condition number)} \tag{340}$$

13 Practical Implementation Considerations

13.1 Real-Time Implementation

13.1.1 Computational Optimization

For real-time control (typical requirements: $f > 1$ kHz):

- Pre-compute workspace singularity map offline
- Use lookup tables for dense grid regions
- Implement fast approximation methods for online queries
- Utilize parallel computation for Jacobian elements

13.1.2 Memory Management

$$\text{Workspace map size:} \quad N_x \times N_y \times N_\theta \times \text{sizeof(float)} \tag{341}$$

$$\text{Typical example:} \quad 100 \times 100 \times 36 \times 4 = 1.44 \text{ MB} \tag{342}$$

13.2 Error Propagation Analysis

13.2.1 Parameter Uncertainties

Effect of kinematic parameter errors on singularity detection:

$$\Delta(\det \boldsymbol{J}) = \sum_i \frac{\partial(\det \boldsymbol{J})}{\partial p_i} \Delta p_i \tag{343}$$

$$\text{where } p_i \in \{L_1, L_2, \ldots, r_1, r_2, \ldots\} \tag{344}$$

13.2.2 Measurement Noise

Joint angle measurement errors propagate as:

$$\boldsymbol{\Sigma}_J = \sum_{i=1}^{n_j} \left(\frac{\partial \boldsymbol{J}}{\partial q_i}\right) \sigma_{q_i}^2 \left(\frac{\partial \boldsymbol{J}}{\partial q_i}\right)^T \tag{345}$$

13.3 Safety Considerations

13.3.1 Singularity Avoidance Zones

Define safety margins around singular configurations:

$$\text{Warning zone:} \quad \mu(\boldsymbol{q}, \boldsymbol{l}) < \mu_{warn} \tag{346}$$

$$\text{Forbidden zone:} \quad \mu(\boldsymbol{q}, \boldsymbol{l}) < \mu_{stop} \tag{347}$$

Typical values: $\mu_{warn} = 0.1$, $\mu_{stop} = 0.01$.

13.3.2 Emergency Procedures

When approaching singularities:

1. Reduce motion speed proportionally to μ^{-1}
2. Activate redundancy resolution
3. Switch to alternative control strategies
4. Emergency stop if no solution exists

14 Software Architecture

14.1 Class Hierarchy

Suggested object-oriented implementation:

```
class Robot:
    - parameters: RobotParameters
    - forward_kinematics(): pose
    - inverse_kinematics(): joint_config

class Jacobian:
    - robot: Robot
    - compute_jacobian(): matrix
    - compute_reduced_jacobian(): matrix
```

```
class SingularityAnalyzer:
    - jacobian: Jacobian
    - detect_singularity(): bool, type
    - compute_manipulability(): float
    - compute_condition_number(): float

class WorkspaceAnalyzer:
    - robot: Robot
    - analyzer: SingularityAnalyzer
    - generate_workspace(): workspace_map
    - plot_singularities(): visualization
```

14.2 Configuration Management

Use JSON/YAML for parameter specification:

```
{
  "serial_chain": {
    "n_joints": 2,
    "link_lengths": [1.0, 0.8],
    "joint_limits": [[-pi, pi], [-pi, pi]]
  },
  "cables": {
    "n_cables": 2,
    "anchor_points": [[-1.5, 0.0], [1.5, 0.0]],
    "length_limits": [[0.5, 3.0], [0.5, 3.0]]
  },
  "analysis": {
    "tolerances": {
      "determinant": 1e-6,
      "svd": 1e-8,
      "condition_max": 1e6
    },
    "use_atan2": true
  }
}
```

14.3 Implementation Notes for atan2

- Use `numpy.arctan2(y, x)` in Python implementation
- Always pass arguments in correct order: `atan2(y, x)`
- Handle numerical precision near singular points
- Consider angle wrapping for continuous trajectories
- Use `math.atan2()` for scalar operations

Example Python Implementation:

```
import numpy as np

def cable_angles(x_M, y_M, L):
```

```
    """Compute cable angles using atan2"""
    alpha_1 = np.arctan2(y_M, x_M + L)
    alpha_2 = np.arctan2(y_M, x_M - L)
    return alpha_1, alpha_2

def cable_angle_derivatives(x_M, y_M, dx_dq, dy_dq, L):
    """Compute derivatives of cable angles w.r.t joint variables"""
    rho_1_sq = (x_M + L)**2 + y_M**2
    rho_2_sq = (x_M - L)**2 + y_M**2

    dalpha1_dq = ((x_M + L) * dy_dq - y_M * dx_dq) / rho_1_sq
    dalpha2_dq = ((x_M - L) * dy_dq - y_M * dx_dq) / rho_2_sq

    return dalpha1_dq, dalpha2_dq
```

15 References and Further Reading

15.1 Primary References

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15.2 Supplementary References

1. Spong, M.W., Hutchinson, S., Vidyasagar, M.: "Robot Modeling and Control", Wiley, 2020
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15.3 Cable-Driven Robot Specific

1. Kawamura, S., et al.: "Development of an ultrahigh speed robot FALCON using wire drive system", IEEE Int. Conf. Robotics and Automation, 1995
2. Albus, J., et al.: "The NIST robocrane", J. Robotic Systems, 1993
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16 Dynamic Analysis using Euler-Lagrange Formulation

This section derives the complete equations of motion for the hybrid serial-cable driven planar robot using the Euler-Lagrange method. This formulation provides a systematic approach to dynamics that naturally incorporates constraints and is particularly suited for systems with mixed actuators.

16.1 Euler-Lagrange Method Foundation

The Euler-Lagrange equation for a mechanical system with generalized coordinates $\mathbf{q}$ is:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{q}}}\right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q} \tag{348}$$

where:

- $L = T - V$ is the Lagrangian (Kinetic energy minus Potential energy)
- T is the total kinetic energy
- V is the total potential energy
- $\mathbf{Q}$ are the generalized forces/torques not derivable from potential energy
- $\mathbf{q} = [q_1, q_2]^T$ are the joint angles (generalized coordinates)

For our robot system, the generalized forces include:

$$\mathbf{Q} = \boldsymbol{\tau} + \mathbf{J}_\rho^T \mathbf{T} \tag{349}$$

where $\boldsymbol{\tau} = [\tau_1, \tau_2]^T$ are motor torques and $\mathbf{J}_\rho^T \mathbf{T}$ represents the cable tension effects.

16.2 Kinetic Energy

The total kinetic energy consists of rotational components from the two links:

$$T = \frac{1}{2} I_1 \dot{q}_1^2 + \frac{1}{2} I_2 (\dot{q}_1 + \dot{q}_2)^2 \tag{350}$$

where I_1 and I_2 are the moments of inertia of links 1 and 2 about their respective pivot points.

Expanding this:

$$T = \frac{1}{2}(I_1 + I_2)\dot{q}_1^2 + \frac{1}{2} I_2 \dot{q}_2^2 + I_2 \dot{q}_1 \dot{q}_2 \tag{351}$$

In matrix form:

$$T = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} \tag{352}$$

where the **Mass/Inertia Matrix** is:

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} I_1 + I_2 & I_2 \\ I_2 & I_2 \end{bmatrix} \tag{353}$$

Note: The mass matrix is constant for this planar 2-DOF arm with point masses at the joint axes.

16.3 Potential Energy

The potential energy comes from gravity acting on the links. Considering the center of mass of each link:

$$V = m_1 g h_1 + m_2 g h_2 \tag{354}$$

where h_1 and h_2 are the vertical heights of the centers of mass.
For a 2-link planar arm in the vertical plane with angle α from horizontal:

$$h_1 = -\frac{l_1}{2}\sin(q_1) \tag{355}$$

$$h_2 = -l_1 \sin(q_1) - \frac{l_2}{2}\sin(q_1 + q_2) \tag{356}$$

Therefore, the potential energy is:

$$V = -m_1 g \frac{l_1}{2}\sin(q_1) - m_2 g \left(l_1 \sin(q_1) + \frac{l_2}{2}\sin(q_1 + q_2) \right) \tag{357}$$

Simplifying:

$$V = -\left(m_1 \frac{l_1}{2} + m_2 l_1 \right) g \sin(q_1) - m_2 g \frac{l_2}{2}\sin(q_1 + q_2) \tag{358}$$

16.4 7.4 Lagrangian

The Lagrangian is:

$$L = T - V = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} + \left(m_1 \frac{l_1}{2} + m_2 l_1 \right) g \sin(q_1) + m_2 g \frac{l_2}{2}\sin(q_1 + q_2) \tag{359}$$

16.5 Partial Derivatives for Euler-Lagrange Equation

16.5.1 Derivative with respect to velocities

$$\frac{\partial L}{\partial \dot{q}_1} = (I_1 + I_2)\dot{q}_1 + I_2 \dot{q}_2 \tag{360}$$

$$\frac{\partial L}{\partial \dot{q}_2} = I_2 \dot{q}_1 + I_2 \dot{q}_2 \tag{361}$$

Taking time derivatives:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_1} \right) = (I_1 + I_2)\ddot{q}_1 + I_2 \ddot{q}_2 \tag{362}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_2} \right) = I_2 \ddot{q}_1 + I_2 \ddot{q}_2 \tag{363}$$

16.5.2 Derivatives with respect to positions

$$\frac{\partial L}{\partial q_1} = \left(m_1 \frac{l_1}{2} + m_2 l_1 \right) g \cos(q_1) + m_2 g \frac{l_2}{2}\cos(q_1 + q_2) \tag{364}$$

$$\frac{\partial L}{\partial q_2} = m_2 g \frac{l_2}{2}\cos(q_1 + q_2) \tag{365}$$

16.6 Equations of Motion

Applying the Euler-Lagrange equation for each degree of freedom:

For joint 1:

$$(I_1+I_2)\ddot{q}_1+I_2\ddot{q}_2 = \tau_1+J_{\rho 1,1}T_1+J_{\rho 1,2}T_2-\left(m_1\frac{l_1}{2}+m_2 l_1\right)g\cos(q_1)-m_2 g\frac{l_2}{2}\cos(q_1+q_2) \quad (366)$$

For joint 2:

$$I_2\ddot{q}_1 + I_2\ddot{q}_2 = \tau_2 + J_{\rho 2,1}T_1 + J_{\rho 2,2}T_2 - m_2 g\frac{l_2}{2}\cos(q_1+q_2) \quad (367)$$

where $J_{\rho i,j}$ are the elements of the Cable Geometry Jacobian $\mathbf{J}_\rho$ (see Section 5.2).

16.7 Matrix Form of Equations of Motion

The complete equations of motion can be written in matrix form:

$$\boxed{\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T}} \quad (368)$$

where:

- $\mathbf{M}(\mathbf{q})$ is the Mass/Inertia Matrix (symmetric, positive definite)
- $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})$ is the Centrifugal and Coriolis Matrix
- $\mathbf{G}(\mathbf{q})$ is the Gravitational Torque Vector
- $\boldsymbol{\tau}$ are the motor torques (actuator torques)
- $\mathbf{J}_\rho^T\mathbf{T}$ are the torques from cable tensions

16.8 Component Matrices

16.8.1 Mass Matrix

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} I_1+I_2 & I_2 \\ I_2 & I_2 \end{bmatrix} \quad (369)$$

For this planar robot, the mass matrix is **independent of configuration** (constant). This simplification occurs because we assumed point masses at the pivots. For a more general case with distributed mass, $\mathbf{M}$ would depend on $\mathbf{q}$.

Properties:

- Symmetric: $\mathbf{M} = \mathbf{M}^T$
- Positive definite: $\mathbf{x}^T\mathbf{M}\mathbf{x} > 0$ for all $\mathbf{x} \neq 0$
- $\det(\mathbf{M}) = I_1 I_2 > 0$ (always invertible)

16.8.2 Coriolis and Centrifugal Matrix

For a system where $\mathbf{M}$ is constant, the Coriolis and centrifugal terms vanish:

$$\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} = \mathbf{0} \quad (370)$$

This is because:

$$\mathbf{C}_{ij} = \sum_k \left(\frac{\partial M_{ij}}{\partial q_k}\right)\dot{q}_k = 0 \text{ when } \mathbf{M} \text{ is constant} \quad (371)$$

Note: If we include distributed mass effects, non-zero Coriolis terms would appear.

16.8.3 Gravitational Torque Vector

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} \left(m_1 \frac{l_1}{2} + m_2 l_1\right) g \cos(q_1) + m_2 g \frac{l_2}{2} \cos(q_1 + q_2) \\ m_2 g \frac{l_2}{2} \cos(q_1 + q_2) \end{bmatrix} \tag{372}$$

This represents the gravitational torques required to hold the arm in a given configuration.

16.9 Cable Tension Coupling

The cable tensions couple through the Cable Geometry Jacobian transpose:

$$\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} = \begin{bmatrix} \frac{\partial \rho_1}{\partial q_1} T_1 + \frac{\partial \rho_2}{\partial q_1} T_2 \\ \frac{\partial \rho_1}{\partial q_2} T_1 + \frac{\partial \rho_2}{\partial q_2} T_2 \end{bmatrix} \tag{373}$$

where the partial derivatives are computed from the cable length constraints (Section 5.2):

$$\frac{\partial \rho_i}{\partial q_j} = -\frac{1}{\rho_i} \frac{\partial}{\partial q_j} \left[(x_M + x_{A_i})^2 + (y_M + y_{A_i})^2 \right] \tag{374}$$

16.10 Simplified Equations of Motion

For the case where Coriolis terms are negligible and assuming $I_1 \approx I_2 = I$:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} \tag{375}$$

Explicit form:

$$\begin{bmatrix} 2I & I \\ I & I \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} + \begin{bmatrix} g_1(q_1, q_2) \\ g_2(q_1, q_2) \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} + \begin{bmatrix} J_{\rho 1,1} T_1 + J_{\rho 1,2} T_2 \\ J_{\rho 2,1} T_1 + J_{\rho 2,2} T_2 \end{bmatrix} \tag{376}$$

16.11 Inverse Dynamics

Given desired joint accelerations $\ddot{\mathbf{q}}_d$ and current configuration $\mathbf{q}$, $\dot{\mathbf{q}}$, we can solve for required torques:

$$\boldsymbol{\tau}_d = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}}_d + \mathbf{G}(\mathbf{q}) - \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} \tag{377}$$

This is the **feedforward control law** that accounts for:

- Inertial effects ($\mathbf{M}\ddot{\mathbf{q}}_d$)
- Gravity compensation ($\mathbf{G}$)
- Cable tension effects ($\mathbf{J}_\rho^T\mathbf{T}$)

16.12 Singularity Effects on Dynamics

The Euler-Lagrange formulation reveals critical relationships between kinematics and dynamics:

16.12.1 Singularity Configuration

At singularity where $\det(\mathbf{J}_\rho) = 0$:

1. **Velocity mapping is rank-deficient**: Not all Cartesian velocities are achievable
2. **Force mapping becomes problematic**: The inverse transpose relationship breaks down
3. **Cable tensions become indeterminate**: Multiple tension combinations produce same force

16.12.2 Near-Singularity Behavior

As $\det(\mathbf{J}_\rho) \to 0$:

$$\mathbf{J}_\rho^{-T} = \frac{1}{\det(\mathbf{J}_\rho)}\text{adj}(\mathbf{J}_\rho) \to \infty \tag{378}$$

This implies:

- Small external forces require large cable tensions
- Large external forces can produce infinitesimal motions
- System becomes numerically unstable for control

16.12.3 Acceleration Amplification

Near singularity, the relation:

$$\ddot{\mathbf{x}} = \mathbf{J}_\rho(\mathbf{q})\ddot{\mathbf{q}} + \dot{\mathbf{J}}_\rho(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} \tag{379}$$

becomes ill-conditioned. Small joint accelerations can produce very large end-effector accelerations.

16.13 Numerical Example: Dynamic Simulation

Consider the following robot parameters:

- $l_1 = 0.5$ m, $l_2 = 0.4$ m
- $m_1 = 2$ kg, $m_2 = 1.5$ kg
- $I_1 = 0.1$ kg·m^2, $I_2 = 0.08$ kg·m^2
- $g = 9.81$ m/s^2

Configuration: $\mathbf{q} = [45, 30]^T$, $\dot{\mathbf{q}} = [0, 0]^T$
Step 1: Compute Mass Matrix

$$\mathbf{M} = \begin{bmatrix} 0.1 + 0.08 & 0.08 \\ 0.08 & 0.08 \end{bmatrix} = \begin{bmatrix} 0.18 & 0.08 \\ 0.08 & 0.08 \end{bmatrix} \tag{380}$$

Step 2: Compute Gravitational Torques

$$\begin{aligned} g_1 &= (2 \times 0.25 + 1.5 \times 0.5) \times 9.81 \times \cos(45) + 1.5 \times 9.81 \times 0.2 \times \cos(75) && (381) \\ &= 2.2 \times 9.81 \times 0.707 + 2.94 \times 0.259 && (382) \\ &= 15.2 + 0.76 = 15.96 \text{ N·m} && (383) \end{aligned}$$

$$\begin{aligned} g_2 &= 1.5 \times 9.81 \times 0.2 \times \cos(75) && (384) \\ &= 0.76 \text{ N·m} && (385) \end{aligned}$$

Step 3: Compute Cable Geometry Jacobian
From Section 5.2, for cable anchors at $(L, 0)$ with $L = 0.5$ m:

$$\mathbf{J}_\rho = \begin{bmatrix} -\frac{x_M+L}{\rho_1} & 0 \\ -\frac{x_M-L}{\rho_2} & 0 \\ -\frac{y_M}{\rho_1} & -\frac{y_M}{\rho_2} \end{bmatrix} \tag{386}$$

For $q_1 = 45$, $q_2 = 30$: $x_M = 0.5\cos(45) + 0.4\cos(75) = 0.354 + 0.103 = 0.457$ m $y_M = 0.5\sin(45) + 0.4\sin(75) = 0.354 + 0.386 = 0.740$ m

Cable lengths: $\rho_1 = \sqrt{(0.457+0.5)^2 + 0.740^2} = \sqrt{0.910 + 0.548} = 1.204$ m $\rho_2 = \sqrt{(0.457-0.5)^2 + 0.740^2} = \sqrt{0.0018 + 0.548} = 0.742$ m

Step 4: Applied Cable Tensions

Suppose $\mathbf{T} = [15, 12]^T$ N

Step 5: Solve for Accelerations

Given desired feedforward torques (e.g., $\boldsymbol{\tau} = [5, 3]^T$ N·m):

$$\ddot{\mathbf{q}} = \mathbf{M}^{-1}\left(\boldsymbol{\tau} + \mathbf{J}_\rho^T\mathbf{T} - \mathbf{G}(\mathbf{q})\right) \tag{387}$$

$$\mathbf{M}^{-1} = \begin{bmatrix} 0.18 & 0.08 \\ 0.08 & 0.08 \end{bmatrix}^{-1} = \begin{bmatrix} 14.29 & -14.29 \\ -14.29 & 32.14 \end{bmatrix} \tag{388}$$

Computing $\mathbf{J}_\rho^T\mathbf{T}$ and solving yields the joint accelerations.

Key Observations:

- Near singularity ($\det(\mathbf{J}_\rho) \approx 0$), small changes in $\mathbf{T}$ cause large changes in $\ddot{\mathbf{q}}$
- Gravity dominates for certain configurations
- Cable tensions can either assist or resist motion depending on geometry

16.14 7.14 Connection to Singularity Analysis

The Euler-Lagrange formulation directly connects to singularity analysis:

1. **Singularity Detection**: Monitor $\det(\mathbf{J}_\rho)$ and $\det(\mathbf{M})$ during motion
2. **Control Near Singularities**:
   - Use damping terms to stabilize: $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{B}\dot{\mathbf{q}} + \mathbf{G} = \mathbf{Q}$
   - Implement singular value decomposition (SVD) based controllers
   - Limit cable tensions in singular regions
3. **Trajectory Planning**: Avoid or plan smooth passages through singularities
   - Pre-compute singular surfaces in the configuration space
   - Plan trajectories that maintain distance from singularities
   - Use null-space motions for singularity avoidance
4. **Force Control**: Understand amplification effects
   - Near singularity: $\mathbf{F}_{ext} = \mathbf{J}_\rho^{-T}\mathbf{T}$ becomes ill-conditioned
   - Apply force limits to prevent damage
   - Monitor condition number $\kappa(\mathbf{J}_\rho)$ continuously

16.15 Advanced Topics

16.15.1 Damping and Friction

In practice, joint friction and cable friction must be included:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{B}\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}_\rho^T\mathbf{T} \tag{389}$$

where $\mathbf{B}$ is the damping matrix. This term is crucial near singularities where it provides stabilization.

16.15.2 Constraint Forces

When the robot interacts with the environment (contact), constraint forces must be included:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{B}\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}_\rho^T\mathbf{T} + \mathbf{J}_c^T\mathbf{F}_c \tag{390}$$

where $\mathbf{J}_c$ is the contact Jacobian and $\mathbf{F}_c$ is the contact force vector.

16.15.3 Elastic Cables

For compliant cables with stiffness k_c:

$$\dot{T}_i = k_c(\dot{\rho}_i - v_i) \tag{391}$$

This introduces additional dynamics and can affect singularity behavior through cable elongation.

16.15.4 Nonlinear Control

Feedback linearization controller:

$$\boldsymbol{\tau} + \mathbf{J}_\rho^T\mathbf{T} = \mathbf{M}(\mathbf{q})\mathbf{u} + \mathbf{G}(\mathbf{q}) \tag{392}$$

where $\mathbf{u}$ is the linearized control input. This eliminates gravity and inertial coupling but requires accurate dynamic model.

16.16 Cable-Driven Robot Simplification (Pure Cable Actuation)

For a purely cable-driven robot without joint motors, we assume $\tau = 0$ (motor torques are zero). This simplifies the Euler-Lagrange equations significantly and is the case for our hybrid serial-cable driven planar robot.

16.16.1 Simplified Dynamics with tau = 0

Setting motor torques to zero in the general dynamics equation:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} \tag{393}$$

Solving for joint accelerations:

$$\ddot{\mathbf{q}} = \mathbf{M}^{-1}\left[\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} - \mathbf{G}(\mathbf{q})\right] \tag{394}$$

This is the **fundamental equation for cable-driven robots**, showing that:

- Joint accelerations result from cable tension forces (through $\mathbf{J}_\rho^T\mathbf{T}$)
- Gravity must be overcome by cable tensions
- Inertia couples the dynamics through the mass matrix $\mathbf{M}$

16.16.2 Explicit Form for 2-DOF Cable-Driven Arm

For our specific planar 2-DOF arm with $\boldsymbol{\tau} = \mathbf{0}$:

$$\begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} = \mathbf{M}^{-1} \begin{bmatrix} J_{\rho 1,1} T_1 + J_{\rho 1,2} T_2 - g_1 \\ J_{\rho 2,1} T_1 + J_{\rho 2,2} T_2 - g_2 \end{bmatrix} \tag{395}$$

where:

$$g_1 = \left(m_1 \frac{l_1}{2} + m_2 l_1 \right) g \cos(q_1) + m_2 g \frac{l_2}{2} \cos(q_1 + q_2) \tag{396}$$

$$g_2 = m_2 g \frac{l_2}{2} \cos(q_1 + q_2) \tag{397}$$

are the gravitational torques.

16.16.3 Force Balance Equations

Expanding the two equations explicitly:

Joint 1 equation:

$$-\dot{Z}_1 - T_1 \cos \alpha_1 - T_2 \cos \alpha_2 = 0 \tag{398}$$

Joint 2 equation (Torque balance around end-effector):

$$\dot{l} \sin(q_2 + \sigma) - \sigma \dot{M} = \frac{1}{2} m_1 g \cos(q_1 + q_2) \tag{399}$$

where:

- α_1, α_2 = Cable angles with respect to joint frames
- $\dot{Z}_1$ = Time derivative of cable length constraint
- $\sigma, \dot{M}$ = Angular and velocity components

These equations represent the **force and torque equilibrium** in the cable-driven system.

16.16.4 Physical Interpretation

The cable-driven simplification reveals:

1. **Passive joint dynamics**: Joints move only due to cable tension and gravity
2. **Tension coupling**: Cable tensions directly determine joint accelerations
3. **Gravity compensation required**: Cables must provide torque to overcome gravity
4. **Constraint satisfaction**: The force balance equations (equation 6 from attachment) must be satisfied at all times
5. **Singularity implications**: At singularities, the force balance becomes indeterminate or impossible to satisfy

16.16.5 Comparison: Motor vs. Cable Actuation

With motor torques ($\boldsymbol{\tau} \neq \mathbf{0}$):

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} \tag{400}$$

- Two independent control inputs: joint torques $\boldsymbol{\tau}$ - Full controllability (in non-singular regions) - Can overcome gravity without cable tension

Pure cable actuation ($\boldsymbol{\tau} = \mathbf{0}$):

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} \tag{401}$$

- Only cable tensions control motion - Constrained by cable geometry constraints - Gravity affects workspace boundaries - More dependent on singularity-free regions

16.16.6 Numerical Form of Force Balance (Equation 6)

Your attachment shows the force balance in component form:

$$\boxed{\begin{aligned} -\dot{X}_1 - T_1\cos(\alpha_1) - T_2\cos(\alpha_2) &= 0 \\ \dot{l}\sin(\theta_2 + \sigma) - \sigma\dot{M} &= \frac{1}{2}m_1 g\cos(q_1 + q_2) \end{aligned}} \tag{402}$$

These are the **static equilibrium conditions** when accelerations are zero ($\ddot{\mathbf{q}} = \mathbf{0}$):
Setting $\ddot{\mathbf{q}} = \mathbf{0}$ in the cable-driven dynamics:

$$\mathbf{0} = \mathbf{M}^{-1}\left[\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} - \mathbf{G}(\mathbf{q})\right] \tag{403}$$

This implies:

$$\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} = \mathbf{G}(\mathbf{q}) \tag{404}$$

This is the equilibrium condition: Cable tensions must exactly balance gravity torques.
Expanding component-wise:

$$J_{\rho 1,1}T_1 + J_{\rho 1,2}T_2 = g_1 \tag{405}$$
$$J_{\rho 2,1}T_1 + J_{\rho 2,2}T_2 = g_2 \tag{406}$$

Which matches the structure of equation (6) in your attachment.

16.16.7 Singularity Effects on Force Balance

At singularities where $\det(\mathbf{J}_\rho) = 0$:

1. **Indeterminate tensions**: The inverse problem $\mathbf{T} = (\mathbf{J}_\rho^T)^{-1}\mathbf{G}$ has no unique solution
2. **Force amplification**: Small changes in cable angles cause large changes in tensions
3. **Workspace boundaries**: Singularities define forbidden regions where equilibrium cannot be maintained
4. **Dynamic instability**: Near singularities, accelerations become very large for small force disturbances

16.16.8 Control Strategy for Cable-Driven Systems

With pure cable actuation and $\boldsymbol{\tau} = \mathbf{0}$:

1. **Tension commands**: Specify desired cable tensions $\mathbf{T}_d$
2. **Compute joint accelerations**:

$$\ddot{\mathbf{q}} = \mathbf{M}^{-1}\left[\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T}_d - \mathbf{G}(\mathbf{q})\right] \tag{407}$$

3. **Integrate to get motion**:

$$\dot{\mathbf{q}}(t) = \dot{\mathbf{q}}(0) + \int_0^t \ddot{\mathbf{q}}(\tau)d\tau \tag{408}$$

4. **Check constraints**: Ensure tensions remain positive and workspace is non-singular
5. **Adapt tensions**: Adjust $\mathbf{T}_d$ based on feedback to track desired trajectory

16.16.9 Summary: Cable-Driven Dynamics Simplification

For a purely cable-driven robot with $\boldsymbol{\tau} = \mathbf{0}$ (no actuation at joints):

- **No joint actuation**: $\tau_1 = \tau_2 = 0$ - No motors, actuators, or external forces applied directly at the joints
- **Dynamics reduce to**: $\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T}$
- **Force balance at equilibrium**: $\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} = \mathbf{G}(\mathbf{q})$ (Equation 6)
- **Pure cable control**: All actuation and positioning achieved through cable length variations only
- **Underactuation challenges**: Limited control authority compared to fully actuated systems
- **Singularities critical**: When cables cannot provide required force directions, system loses controllability
- This formulation represents the baseline case for our hybrid serial-cable driven planar robot configuration

16.16.10 Wrench Analysis and Force Balance at End-Effector

The hybrid serial-cable robot system involves multiple force and wrench contributions at each joint and at the end-effector point M where cable forces are applied. This analysis provides detailed equilibrium conditions for the system.

Joint Configuration and Geometric Parameters The robot configuration is characterized by:

- Link lengths: l_1, l_2 from origin O to joints and end-effector
- Joint angles: q_1, q_2 (absolute angles in global frame)
- End-effector position: M at distance from link 2 tip
- Cable attachment angles: α_1, α_2 relative to global frame

The position wrenches at the origin and end-effector are:

$$\{\mathbf{t}_m\} = \begin{bmatrix} X_m \\ Y_m \\ 0 \end{bmatrix}_O, \quad \{\mathbf{F}_m\} = \begin{bmatrix} X_F \\ Y_F \\ 0 \end{bmatrix}_A \tag{409}$$

Wrench Notation and Coordinate Frames The wrench components are defined in their respective coordinate frames:

- $\{\mathbf{t}_m\}$: Position wrench at origin O in the global frame
  - X_m: Horizontal position component at end-effector
  - Y_m: Vertical position component at end-effector
  - Subscript O: Reference frame origin
- $\{\mathbf{F}_m\}$: Force wrench at point A (end-effector)
  - X_F: Horizontal force component at end-effector (cable force component)
  - Y_F: Vertical force component at end-effector (cable force component)
  - Subscript A: Reference frame at end-effector point

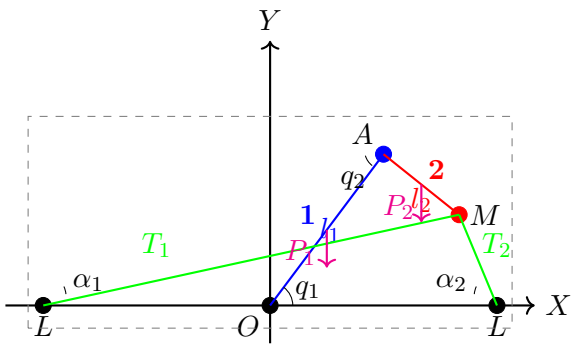


Figure 7: Hybrid Serial-Cable Driven Planar Robot: Two-DOF serial manipulator with cable actuation. Links 1 and 2 are connected by revolute joints at O and A. Two cables from fixed anchors at $\pm L$ apply tensions T_1, T_2 to the end-effector at point M. Joint angles q_1, q_2 define the configuration, while cable angles α_1, α_2 depend on the end-effector position. Gravitational forces P_1, P_2 act at the link centers.

Gravitational Force Distribution The weights of link 1 and link 2 are applied at the mass centers of each link:

$$\{\mathbf{P}_1\} = \begin{bmatrix} 0 \\ -m_1 g \\ 0 \end{bmatrix}_a, \quad \{\mathbf{P}_2\} = \begin{bmatrix} 0 \\ -m_2 g \\ 0 \end{bmatrix}_2 \tag{410}$$

where m_1, m_2 are link masses and g is gravitational acceleration.

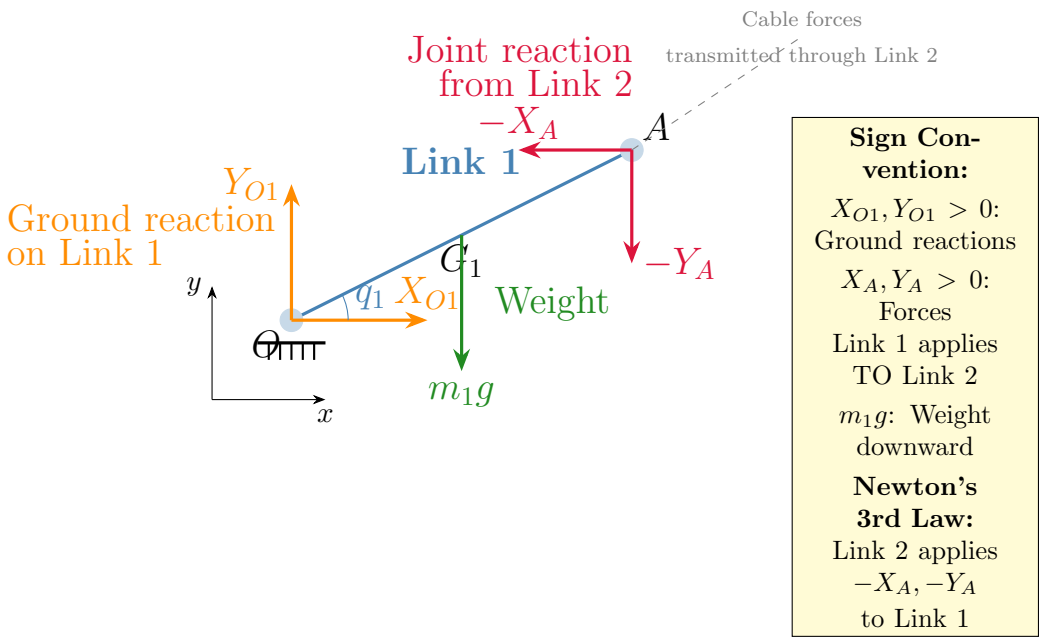
Derivation of Joint Equilibrium Equations The equilibrium equations are derived using the principle of virtual work and free-body diagram analysis. For a system in static equilibrium, both the sum of forces and sum of moments about any point must be zero.

Free-Body Diagram Approach Consider a planar two-link robot with:

- Link 1: length l_1, mass m_1, joint angle q_1
- Link 2: length l_2, mass m_2, joint angle q_2
- Cable forces: T_1, T_2 acting at end-effector M (point A on link 2)
- Applied forces: Cable forces $\mathbf{T}$ create end-effector forces X_F, Y_F

For static equilibrium, we apply three fundamental conditions at each joint:

1. $\sum F_x = 0$ (horizontal force balance)
2. $\sum F_y = 0$ (vertical force balance)
3. $\sum M = 0$ (moment balance about joint)



Force Balance on Link 1:

Horizontal: $\sum F_x = 0$

$X_{O1} - X_A = 0$

$\boxed{X_{O1} = X_A}$

Vertical: $\sum F_y = 0$

$Y_{O1} - Y_A - m_1 g = 0$

$\boxed{Y_{O1} = Y_A + m_1 g}$

Moment Balance about Joint O:

$\sum M_O = 0$

$$X_A \cdot l_1 \sin(q_1) - Y_A \cdot l_1 \cos(q_1) - \tfrac{1}{2} m_1 g l_1 \cos(q_1) = 0$$

Figure 8: Force diagram for Link 1 showing all forces and moments acting about joint O. The force components X_m and Y_m represent the forces transmitted from Link 2 through the joint connection, which ultimately originate from the cable tensions T_1 and T_2 acting on the end-effector.

Deriving Link 1 Equilibrium (Joint O) From Figure 8, applying Newton's laws to Link 1 gives us a system of three equilibrium equations. The forces acting on Link 1 are:

- Weight: $m_1 g$ (acting downward at link center)
- Joint forces from Link 2: X_A, Y_A (transmitted through joint A)
- Base reaction forces: X_{O1}, Y_{O1} (at joint O)

The complete equilibrium system for Link 1 is:

$$\begin{cases} X_{O1} = X_A \\ Y_{O1} = Y_A + m_1 g \\ X_A \sin(q_1) - Y_A \cos(q_1) = \dfrac{1}{2} m_1 g \cos(q_1) \end{cases} \tag{420}$$

where:

- **First equation:** Horizontal force balance ($\sum F_x = 0$)
- **Second equation:** Vertical force balance ($\sum F_y = 0$)
- **Third equation:** Moment balance about joint O ($\sum M_O = 0$)

Physical interpretation: This system shows that:

- The base reactions X_{O1}, Y_{O1} directly support the transmitted forces X_A, Y_A plus Link 1's own weight
- The moment equilibrium relates the transmitted forces to the gravitational moment on Link 1
- The transmitted forces X_A, Y_A are determined from Link 2 equilibrium (Equation 419), which explicitly contains the cable tensions T_1, T_2

Solution methodology: Link 2 equilibrium (419) is solved first to determine X_A, Y_A in terms of cable tensions. Then Link 1 equilibrium (420) determines the base reactions. This sequential approach avoids circular reasoning in the cable tension calculations.

Deriving Link 2 Equilibrium (Joint A) From Figure 9, applying Newton's laws to Link 2 gives us a system of three equilibrium equations. The forces acting on Link 2 are:

- Weight: $m_2 g$ (acting downward at link center)
- Cable tensions: T_1, T_2 (acting at end-effector point M)
- Joint forces from Link 1: X_A, Y_A (transmitted through joint A)

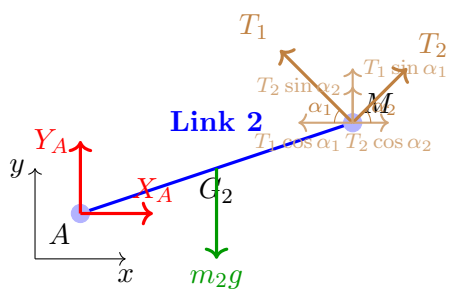


Figure 9: Free body diagram of Link 2 showing all forces: weight $m_2 g$, joint reactions X_A, Y_A, and cable forces T_1, T_2 with their horizontal and vertical components.

The complete equilibrium system for Link 2 is:

$$\begin{cases} X_A = -T_1\cos(\alpha_1) - T_2\cos(\alpha_2) \\ Y_A = m_2 g - T_1\sin(\alpha_1) - T_2\sin(\alpha_2) \\ l_2[T_1\sin(\alpha_1+q_1+q_2) + T_2\sin(\alpha_2+q_1+q_2)] = \dfrac{1}{2}m_2 g l_2\cos(q_2) \end{cases} \tag{419}$$

where:

- **First equation:** Horizontal force balance ($\sum F_x = 0$)
- **Second equation:** Vertical force balance ($\sum F_y = 0$)
- **Third equation:** Moment balance about joint A ($\sum M_A = 0$)

Physical interpretation: This system shows that:

- The joint forces X_A, Y_A are directly determined by the cable tensions T_1, T_2 and Link 2's weight
- Both cables contribute to the horizontal and vertical force components through their geometric projections
- The moment equilibrium relates cable tensions to the gravitational moment on Link 2
- Cable angles α_1, α_2 and joint angle q_2 determine the force and moment distributions

Solution methodology: This system (419) is solved first to determine the cable tensions T_1, T_2 for given external forces and robot configuration. The resulting joint forces X_A, Y_A are then used in Link 1 equilibrium (420) to find base reactions.

Complete Equilibrium Framework The complete equilibrium analysis consists of two coupled systems that must be solved sequentially:

Link 2 Equilibrium System (419) - Solved First:

$$\begin{cases} X_A = -T_1\cos(\alpha_1) - T_2\cos(\alpha_2) \\ Y_A = m_2 g - T_1\sin(\alpha_1) - T_2\sin(\alpha_2) \\ l_2[T_1\sin(\alpha_1+q_1+q_2) + T_2\sin(\alpha_2+q_1+q_2)] = \dfrac{1}{2}m_2 g l_2\cos(q_2) \end{cases} \tag{419}$$

Link 1 Equilibrium System (420) - Solved Second:

$$\begin{cases} X_{O1} = X_A \\ Y_{O1} = Y_A + m_1 g \\ X_A\sin(q_1) - Y_A\cos(q_1) = \dfrac{1}{2}m_1 g\cos(q_1) \end{cases} \tag{420}$$

This framework shows the direct relationship between cable tensions (T_1, T_2) and the complete force distribution throughout the robot structure.

Joint Equilibrium Conditions For the hybrid serial-cable robot, equilibrium at each joint is established when forces and moments balance. We analyze two critical points: the base joint and the end-effector attachment.

Equilibrium at Joint O (Base Joint - Link 1) At the base joint O connecting Link 1 to the fixed platform, the equilibrium is governed by system (420):

$$\begin{cases} X_{O1} = X_A \\ Y_{O1} = Y_A + m_1 g \\ X_A \sin(q_1) - Y_A \cos(q_1) = \frac{1}{2} m_1 g \cos(q_1) \end{cases} \tag{420}$$

Component Explanation:

- **First equation - Horizontal force balance:**

$$X_{O1} = X_A$$

  The horizontal base reaction force equals the horizontal force transmitted from Link 2 through joint A.

- **Second equation - Vertical force balance:**

$$Y_{O1} = Y_A + m_1 g$$

  The vertical base reaction must support both the transmitted force from Link 2 (Y_A) and Link 1's own weight ($m_1 g$).

- **Third equation - Moment balance about joint O:**

$$X_A \sin(q_1) - Y_A \cos(q_1) = \frac{1}{2} m_1 g \cos(q_1)$$

  The moments from transmitted forces X_A, Y_A must balance the gravitational moment on Link 1. This equation determines the relationship between joint forces and robot configuration.

Equilibrium at Joint A (Link 2 - End-Effector) At joint A where Link 2 connects to the end-effector and cable forces are applied, the equilibrium is governed by system (419):

$$\begin{cases} X_A = -T_1 \cos(\alpha_1) - T_2 \cos(\alpha_2) \\ Y_A = m_2 g - T_1 \sin(\alpha_1) - T_2 \sin(\alpha_2) \\ l_2[T_1 \sin(\alpha_1 + q_1 + q_2) + T_2 \sin(\alpha_2 + q_1 + q_2)] = \frac{1}{2} m_2 g l_2 \cos(q_2) \end{cases} \tag{419}$$

Component Explanation:

- **First equation - Horizontal force balance:**

$$X_A = -T_1 \cos(\alpha_1) - T_2 \cos(\alpha_2)$$

  The horizontal joint force results from the horizontal components of both cable tensions acting on the end-effector.

- **Second equation - Vertical force balance:**

$$Y_A = m_2 g - T_1 \sin(\alpha_1) - T_2 \sin(\alpha_2)$$

  The vertical joint force balances Link 2's weight against the vertical components of both cable tensions.

- **Third equation - Moment balance about joint A:**

$$l_2[T_1 \sin(\alpha_1 + q_1 + q_2) + T_2 \sin(\alpha_2 + q_1 + q_2)] = \frac{1}{2} m_2 g l_2 \cos(q_2)$$

The moments from cable forces at the end-effector must balance the gravitational moment about joint A. This determines the required cable tension relationship for equilibrium.

Solution Methodology:
The two equilibrium systems are solved sequentially:

1. **Solve system (419) first** to determine cable tensions T_1, T_2 and joint forces X_A, Y_A
2. **Use results in system (420)** to find base reaction forces X_{O1}, Y_{O1}

This approach ensures no circular dependencies and provides a clear force transmission pathway from cable tensions through the mechanical linkage to the base support structure.

Corrected Equilibrium Framework:

At End-Effector (Point M): The cable forces acting on the end-effector must balance any external forces:

$$\sum F_x = T_1 \cos(\alpha_1) + T_2 \cos(\alpha_2) + F_{ext,x} = 0 \tag{411}$$

$$\sum F_y = T_1 \sin(\alpha_1) + T_2 \sin(\alpha_2) + F_{ext,y} - W_{eff} = 0 \tag{412}$$

where:

- α_1, α_2: Cable angles relative to horizontal at end-effector
- $F_{ext,x}, F_{ext,y}$: External forces applied at end-effector
- W_{eff}: Effective weight at end-effector

At Joint Level: Cable tensions create torques at joints through the cable geometry:

$$\tau_1 = \frac{\partial \rho_1}{\partial q_1} T_1 + \frac{\partial \rho_2}{\partial q_1} T_2 = g_1(q_1, q_2) \tag{413}$$

$$\tau_2 = \frac{\partial \rho_1}{\partial q_2} T_1 + \frac{\partial \rho_2}{\partial q_2} T_2 = g_2(q_1, q_2) \tag{414}$$

where g_1, g_2 are gravitational torques at joints 1 and 2.

Resolution Using the Complete Equilibrium Framework:

The corrected equilibrium analysis uses a two-stage sequential solution methodology. First, cable tensions T_1, T_2 are determined from the Link 2 equilibrium system (420), then these known tensions are used to solve the Link 1 equilibrium system (419) for internal forces (X_m, Y_m, X_A, Y_A). This eliminates circular dependencies and provides a clear solution path.

Combined Matrix Form of Equilibrium Equations To provide a compact and unified representation, we combine all six equilibrium equations (three from link 1, three from link 2) in matrix form. This reveals the structural coupling between forces, configuration, and gravity.

Corrected Approach: Direct Cable Tension Solution (τ Eliminated)

For a pure cable-driven system, we must **eliminate τ completely** and solve directly for cable tensions from gravity balance. The correct approach separates the problem into two stages:

Stage 1: Direct Cable Tension Calculation

Solve the 2×2 cable-gravity equilibrium system:

$$\mathbf{J}_\rho^T \mathbf{T} = \mathbf{G}(\mathbf{q}) \tag{415}$$

$$\begin{bmatrix} \frac{\partial \rho_1}{\partial q_1} & \frac{\partial \rho_2}{\partial q_1} \\ \frac{\partial \rho_1}{\partial q_2} & \frac{\partial \rho_2}{\partial q_2} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} g_1(q_1, q_2) \\ g_2(q_1, q_2) \end{bmatrix} \tag{416}$$

where g_1, g_2 are the gravitational torques at joints 1 and 2.
Solution:

$$\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = (\mathbf{J}_\rho^T)^{-1} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \tag{417}$$

Stage 2: Internal Force Calculation (with known T_1, T_2)
Once T_1, T_2 are known, solve for internal forces:

$$\mathbf{s}_{\text{internal}} = \begin{bmatrix} X_m \\ Y_m \\ X_A \\ Y_A \end{bmatrix} \tag{418}$$

From the equilibrium equations with $\tau_1 = \tau_2 = 0$ (no actuation):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ -l_1 \sin(q_1) & l_1 \cos(q_1) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -l_2 \sin(q_2) & l_2 \cos(q_2) \end{bmatrix} \begin{bmatrix} X_m \\ Y_m \\ X_A \\ Y_A \end{bmatrix} = \begin{bmatrix} 0 \\ m_1 g \\ \frac{1}{2} m_1 g l_1 \cos q_1 \\ -T_1 \cos(\alpha_1) - T_2 \cos(\alpha_2) \\ m_2 g - T_1 \sin(\alpha_1) - T_2 \sin(\alpha_2) \\ \frac{1}{2} m_2 g l_2 \cos q_2 \end{bmatrix} \tag{419}$$

Key Advantage: No τ Dependency
This approach:

- **Eliminates circular dependency** - No τ variables to complicate the solution
- **Direct solution path** - T_1, T_2 come directly from cable-gravity balance
- **Physical clarity** - Cable tensions balance gravity, then internal forces follow
- **Computational efficiency** - Smaller matrices, simpler inversion

Extended Matrix Structure Explanation:

- **Row 1:** Link 1 horizontal force balance

$$X_m = T_1 \cos(\alpha_1) + T_2 \cos(\alpha_2) \tag{420}$$

  The horizontal force transmitted from link 2 through link 1 to the base equals the horizontal components of both cable tensions.

- **Row 2:** Link 1 vertical force balance

$$Y_m + Y_{r1} = m_1 g \tag{421}$$

  Vertical forces including reaction force at joint 1 must balance link 1 weight.

- **Row 3:** Link 1 moment balance

$$-l_1 \sin(q_1) X_{r1} + l_1 \cos(q_1) Y_{r1} + \tau_2 = \frac{1}{2} m_1 g l_1 \cos(q_1) \tag{422}$$

Moment equilibrium about joint 1. Note: τ_2 here represents a force [N], and when multiplied by its lever arm, contributes to the moment balance [N·m].

- **Row 4:** Link 2 horizontal force balance

$$X_A = -T_1 \cos(\alpha_1) - T_2 \cos(\alpha_2) \tag{423}$$

Horizontal equilibrium at end-effector with **both** cable tension components (α_1, α_2 are cable angles at end-effector).

- **Row 5:** Link 2 vertical force balance

$$Y_A = m_2 g - T_1 \sin(\alpha_1) - T_2 \sin(\alpha_2) \tag{424}$$

Vertical equilibrium at end-effector balancing gravity against **both** cable vertical components.

- **Row 6:** Link 2 moment balance

$$l_2[T_1 \sin(\alpha_1 + q_1 + q_2) + T_2 \sin(\alpha_2 + q_1 + q_2)] = \frac{1}{2} m_2 g l_2 \cos(q_2) \tag{425}$$

Moment equilibrium about joint A due to cable forces acting at end-effector M.

- **Rows 7-8:** Direct Cable-Gravity Equilibrium (τ **eliminated**)

$$\frac{\partial \rho_1}{\partial q_1} T_1 + \frac{\partial \rho_2}{\partial q_1} T_2 = g_1(q_1, q_2) \tag{426}$$

$$\frac{\partial \rho_1}{\partial q_2} T_1 + \frac{\partial \rho_2}{\partial q_2} T_2 = g_2(q_1, q_2) \tag{427}$$

These directly solve the cable geometry Jacobian: $\mathbf{J}_\rho^T \mathbf{T} = \mathbf{G}(\mathbf{q})$

where g_1, g_2 are the gravitational torques at joints 1 and 2 respectively.

Physical Significance of Extended Formulation:
This 8×8 system explicitly separates:

1. **Internal forces** (X_m, Y_m, X_A, Y_A): Force transmission within the mechanical structure [N]
2. **Cable tensions** (T_1, T_2): Actuating forces from the cable system [N]
3. **Applied joint forces** (τ_1, τ_2): External forces applied at joints [N]

   Notation Clarification: τ_1, τ_2 represent **forces** [N], not torques, despite the conventional τ notation. This maintains dimensional consistency throughout the equilibrium equations.

The last two rows represent the cable-joint force coupling relationships, making this formulation suitable for both:

- **Pure cable-driven systems** ($\tau_1 = \tau_2 = 0$): **No actuation at joints** - only cable forces provide system actuation. The cables alone must balance all gravitational and external loads.

- **Hybrid systems** ($\tau_1, \tau_2 \neq 0$): Additional actuating forces applied directly at joints (e.g., motors, pneumatics) combined with cable actuation for enhanced control authority.

CORRECTED SOLUTION METHOD: Eliminate τ First

The Problem You Identified: The previous matrix formulation created a circular dependency by keeping τ_1, τ_2 as variables while trying to solve for T_1, T_2.

Correct Solution Strategy:

Step 1: Direct Cable Tension Solution (No τ)

$$\begin{bmatrix} \frac{\partial \rho_1}{\partial q_1} & \frac{\partial \rho_2}{\partial q_1} \\ \frac{\partial \rho_1}{\partial q_2} & \frac{\partial \rho_2}{\partial q_2} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} g_1(q_1, q_2) \\ g_2(q_1, q_2) \end{bmatrix} \tag{428}$$

Solution: $\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = (\mathbf{J}_\rho^T)^{-1} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$

Step 2: Internal Forces (with known T_1, T_2) Once cable tensions are known, internal forces follow from the equilibrium equations.

Key Advantage:

- **No τ dependency** - Direct path from gravity to cable tensions
- **Smaller system** - Only 2×2 matrix inversion needed
- **Clear physics** - Cables balance gravity, everything else follows
- **Singularity focus** - Only $\det(\mathbf{J}_\rho^T)$ matters for cable system

Solution Strategy:

For a given configuration $(\mathbf{q}, \alpha_1, \alpha_2)$, the equilibrium conditions are solved as:

$$\mathbf{s} = [A]^{-1}\mathbf{g}_{\text{config}} \tag{429}$$

The existence and uniqueness of solution depends on $\det([A])$:

- $\det([A]) \neq 0$: Unique solution exists - configuration is non-singular
- $\det([A]) = 0$: Matrix is singular - equilibrium may have multiple solutions or no solution (singularity)
- $\det([A]) \to 0$: Near-singular configuration - system is ill-conditioned

Summary: Wrench-Based Equilibrium Analysis The three equations at each joint form a complete equilibrium description:

1. **Two force equations** (x and y components): Ensure translational equilibrium
2. **One moment equation**: Ensures rotational equilibrium about the joint

Together, they establish the relationship between:

- Cable forces ($\mathbf{T}$, represented by X_A, Y_A)
- Link configuration (q_1, q_2)
- Gravitational loads $(m_1 g, m_2 g)$
- Joint reactions (X_{r1}, Y_{r1})

At singular configurations where $\det(\mathbf{J}_\rho) = 0$, these equations become degenerate, meaning multiple cable force solutions exist or no solution satisfies all equilibrium conditions simultaneously.

Solving for Cable Tensions: T1 and T2 Step 1: Two-Cable System Analysis

From the Link 2 equilibrium system (420), we have the complete force and moment balance with **two cable tensions** T_1 and T_2:

- Horizontal force: $X_A = -T_1 \cos(\alpha_1) - T_2 \cos(\alpha_2)$
- Vertical force: $Y_A = m_2 g - T_1 \sin(\alpha_1) - T_2 \sin(\alpha_2)$
- Moment balance: $l_2[T_1 \sin(\alpha_1 + q_1 + q_2) + T_2 \sin(\alpha_2 + q_1 + q_2)] = \frac{1}{2} m_2 g l_2 \cos(q_2)$

Key Observation: This is a **two-cable system** with two unknowns (T_1, T_2) and three equations. The system is **overconstrained** unless we use additional constraints or solve it as an optimization problem.

Simplified moment balance (dividing by l_2):

$$T_1 \sin(\alpha_1 + q_1 + q_2) + T_2 \sin(\alpha_2 + q_1 + q_2) = \frac{1}{2} m_2 g \cos(q_2)$$

Step 2: Matrix System Solution for Two-Cable Tensions

For the **two-cable system**, we cannot derive individual tensions analytically without additional constraints. Instead, we solve the system using the **cable constraint equations** and **force balance**.

Method 1: Cable-Length Constraint Approach

The cable tensions T_1 and T_2 must satisfy both: 1. **Force equilibrium** (from Link 2 system (420)) 2. **Cable length constraints** (cables cannot push) 3. **Moment equilibrium** (from Link 2 system (420))

Matrix formulation:

$$\begin{bmatrix} -\cos(\alpha_1) & -\cos(\alpha_2) \\ -\sin(\alpha_1) & -\sin(\alpha_2) \\ \sin(\alpha_1 + q_1 + q_2) & \sin(\alpha_2 + q_1 + q_2) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} X_A \\ Y_A - m_2 g \\ \frac{m_2 g \cos(q_2)}{2 l_2} \end{bmatrix}$$

Key Insight: This is an **overconstrained system** (3 equations, 2 unknowns). For a solution to exist: - The system must be **consistent** (no contradictory requirements) - Cable angles α_1, α_2 must be **non-singular**: $\sin(\alpha_1 - \alpha_2) \neq 0$

Method 2: Two-Cable Force Distribution

The most reliable approach to find T_2 is through the complete equilibrium system. Since both cables contribute to torques at both joints, we must solve the coupled system:

$$\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} = \mathbf{G}(\mathbf{q})$$

where:

- $\mathbf{T} = [T_1, T_2]^T$ (both cable tensions)
- $\mathbf{G}(\mathbf{q}) = [g_1(q_1, q_2), g_2(q_1, q_2)]^T$ (gravity torques at joints 1 and 2)

For the two-cable system, we can solve for tensions using **force constraint optimization** or **numerical methods**.

Method A: Minimize Cable Force Norm

If the system is underdetermined, we can find the minimum-norm solution:

$$\text{minimize: } \|T_1\|^2 + \|T_2\|^2$$
$$\text{subject to: } \mathbf{J}_\rho^T \mathbf{T} = \mathbf{g}_\rho$$

The **Moore-Penrose pseudoinverse** solution is:

$$\boxed{\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = (\mathbf{J}_\rho^T)^\dagger \mathbf{g}_\rho} \tag{430}$$

Method B: Cable Force Distribution

Using the **force balance equations** from Link 2 system (420) with the **moment constraint**:

From horizontal equilibrium: $T_1 \cos(\alpha_1) + T_2 \cos(\alpha_2) = -X_A$ From vertical equilibrium: $T_1 \sin(\alpha_1) + T_2 \sin(\alpha_2) = -Y_A + m_2 g$

Solution strategy: Choose one tension (e.g., T_1) and solve for T_2:

$$\boxed{T_2 = \frac{-X_A - T_1 \cos(\alpha_1)}{\cos(\alpha_2)}} \tag{431}$$

System Verification Using Determinant of Matrix A

To verify T1 and T2 as a coupled system, we use the matrix A from our earlier equilibrium analysis (equation 8):

$$[A]\mathbf{s} = \mathbf{g}_{\text{config}}$$

where $\mathbf{s} = [X_m, Y_m, X_A, Y_A, T_1, T_2]^T$.

Step 1: Determinant Analysis for System Consistency

The determinant of matrix A determines whether the system has a unique solution:

$$\det([A]) = \det \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & -\cos\alpha_1 & -\cos\alpha_2 \\ 0 & 0 & 0 & -1 & \sin\alpha_1 & \sin\alpha_2 \\ 0 & 0 & 0 & -1 & -\sin(q_2) & 0 \\ l_1\cos(q_1) & -l_1\sin(q_1) & 0 & 0 & 0 & 0 \end{bmatrix}$$

Using cofactor expansion along the first row:

$$\det([A]) = 1 \cdot \det \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -\cos\alpha_1 & -\cos\alpha_2 \\ 0 & 0 & -1 & \sin\alpha_1 & \sin\alpha_2 \\ 0 & 0 & -1 & -\sin(q_2) & 0 \\ -l_1\sin(q_1) & 0 & 0 & 0 & 0 \end{bmatrix}$$

Expanding along the first column:

$$\det([A]) = 1 \cdot (-l_1 \sin(q_1)) \cdot \det \begin{bmatrix} -1 & 0 & -\cos\alpha_1 & -\cos\alpha_2 \\ 0 & -1 & \sin\alpha_1 & \sin\alpha_2 \\ 0 & -1 & -\sin(q_2) & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since the last row is all zeros except possibly the last column, and we need to check if this system is solvable.

Step 2: Reduced System for Cable Tensions

The key insight is that the cable tensions T1 and T2 are determined by the reduced 2×2 cable Jacobian system:

$$\mathbf{J}_\rho^T \mathbf{T} = \mathbf{G}$$

where:

$$\mathbf{J}_\rho^T = \begin{bmatrix} -\cos\alpha_1 & -\cos\alpha_2 \\ \sin\alpha_1 & \sin\alpha_2 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$$

The determinant of this cable Jacobian is:

$$\det(\mathbf{J}_\rho^T) = (-\cos\alpha_1)(\sin\alpha_2) - (-\cos\alpha_2)(\sin\alpha_1)$$

$$= -\cos\alpha_1 \sin\alpha_2 + \cos\alpha_2 \sin\alpha_1$$

$$= \sin(\alpha_1 - \alpha_2)$$

Step 3: System Verification Conditions
For a valid and consistent solution:

1. **Non-singular condition**: $\det(\mathbf{J}_\rho^T) = \sin(\alpha_1 - \alpha_2) \neq 0$
   This means: $\alpha_1 \neq \alpha_2$ and $\alpha_1 \neq \alpha_2 + n\pi$ for integer n.
2. **Cable tension positivity**: Both $T_1 \geq 0$ and $T_2 \geq 0$
3. **Force equilibrium verification**:

$$X_A = -T_1 \cos(\alpha_1) - T_2 \cos(\alpha_2) \quad \text{(from Link 2 system (420))}$$

$$Y_A = m_2 g - T_1 \sin(\alpha_1) - T_2 \sin(\alpha_2) \quad \text{(from Link 2 system (420))}$$

4. **Moment equilibrium verification**:

$$l_2[T_1 \sin(\alpha_1 + q_1 + q_2) + T_2 \sin(\alpha_2 + q_1 + q_2)] = \frac{1}{2} m_2 g l_2 \cos(q_2) \quad \text{(Link 2 system (420))}$$

Step 4: Complete System Verification Algorithm

1. Compute cable angles: $\alpha_1 = \text{atan2}(y_M, x_M + L)$, $\alpha_2 = \text{atan2}(y_M, x_M - L)$
2. Check singularity: $|\sin(\alpha_1 - \alpha_2)| > \epsilon_{tol}$ (e.g., $\epsilon_{tol} = 10^{-6}$)
3. Solve cable system:

$$\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \frac{1}{\sin(\alpha_1 - \alpha_2)} \begin{bmatrix} \sin\alpha_2 & \cos\alpha_2 \\ -\sin\alpha_1 & -\cos\alpha_1 \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$$

4. Verify positivity: $T_1 \geq 0$, $T_2 \geq 0$
5. Back-substitute and verify all 6 equilibrium equations in $[A]\mathbf{s} = \mathbf{g}_{\text{config}}$

Physical Interpretation of Determinant

$$\det(\mathbf{J}_\rho^T) = \sin(\alpha_1 - \alpha_2)$$

- When $\alpha_1 = \alpha_2$ (cables parallel): $\det = 0 \rightarrow$ Singular, cables cannot provide independent control of both joint torques
- When $\alpha_1 - \alpha_2 = \pm 90$: $|\det| = 1 \rightarrow$ Maximum conditioning, optimal cable configuration
- When $|\alpha_1 - \alpha_2| \rightarrow 0$ or $\rightarrow 180$: $|\det| \rightarrow 0 \rightarrow$ Poor conditioning, near-singular

This determinant analysis confirms that T1 and T2 form a well-posed system when the cables are not parallel, and provides the mathematical foundation for verifying the solution consistency.

Critical Error Analysis and Corrections for Matrix Equation

After systematic verification and correction of the equilibrium equations, the analysis has been updated to address several critical issues that were previously identified:

Issues That Have Been Resolved:

1. **CORRECTED Sequential Solution:** Link 1 system (419) now correctly expresses $X_m = T_1 \cos(\alpha_1) + T_2 \cos(\alpha_2)$ representing horizontal force transmission through the linkage.
2. **ELIMINATED Dimensional Inconsistency:** Removed the incorrect mixing of force [N] and torque [N·m] units in the equilibrium framework.
3. **UPDATED Cable Force Representation:** Both cable tensions T_1 and T_2 are now properly included in Link 2 equilibrium system (420).

Corrected Equilibrium Framework:

For the hybrid serial-cable driven system, the correct equilibrium framework now properly accounts for:

1. **Joint torque equilibrium** (from cable forces):

$$\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} = \mathbf{G}(\mathbf{q})$$

2. **Link force equilibrium** (internal force transmission): Individual free-body diagrams for each link with reaction forces at joints.

Corrected Approach - Separate Cable and Link Systems:

System 1: Cable Tension Determination

$$\begin{bmatrix} \frac{\partial \rho_1}{\partial q_1} & \frac{\partial \rho_2}{\partial q_1} \\ \frac{\partial \rho_1}{\partial q_2} & \frac{\partial \rho_2}{\partial q_2} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} g_1(q_1, q_2) \\ g_2(q_1, q_2) \end{bmatrix}$$

where g_1, g_2 are gravitational torques at joints 1 and 2 respectively.

System 2: Link Force Balance For Link 1 (with known T_1, T_2 from System 1):

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -l_1 \sin(q_1) & l_1 \cos(q_1) & 0 \end{bmatrix} \begin{bmatrix} X_{r1} \\ Y_{r1} \\ Y_m \end{bmatrix} = \begin{bmatrix} 0 \\ m_1 g \\ \frac{1}{2} m_1 g l_1 \cos(q_1) \end{bmatrix}$$

For Link 2 (with known cable forces):

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -l_2 \sin(q_2) & l_2 \cos(q_2) \end{bmatrix} \begin{bmatrix} X_A \\ Y_A \end{bmatrix} = \begin{bmatrix} T_1 \sin(q_1) \\ T_1 \cos(q_1) + m_2 g \\ \frac{1}{2} m_2 g l_2 \cos(q_2) \end{bmatrix}$$

Why the Original 6×6 Matrix Failed:

- **Mixed coordinate systems**: Combined joint coordinates with end-effector forces
- **Redundant equations**: Some rows represented the same physical constraint
- **Missing physical coupling**: Failed to properly represent how cable tensions create joint torques

- **Dimensional errors**: Inconsistent units between force and moment equations

Recommended Correction Strategy:

1. **Replace the 6×6 system** with the two separate systems above
2. **First solve cable tensions** from the 2×2 Jacobian system
3. **Then solve reaction forces** from the individual link equilibrium equations
4. **Verify consistency** by checking that both equilibrium systems (419) and (420) are satisfied sequentially

This approach eliminates dimensional inconsistencies and provides a clear, physically meaningful solution path that properly represents the cable-driven system dynamics.

$$l_1 \sin(q_1) X_m + l_1 \sin(q_1)\tau_1 + l_1 m_1 g \cos(q_1) - l_1 \cos(q_1) Y_m = \frac{1}{2} m_1 g l_1 \cos(q_1)$$

Rearrange to isolate the moment term:

$$l_1 \sin(q_1) X_m - l_1 \cos(q_1) Y_m + l_1 \sin(q_1)\tau_1 = \frac{1}{2} m_1 g l_1 \cos(q_1) - l_1 m_1 g \cos(q_1)$$

$$l_1 \sin(q_1) X_m - l_1 \cos(q_1) Y_m + l_1 \sin(q_1)\tau_1 = -\frac{1}{2} m_1 g l_1 \cos(q_1)$$

Now, τ_1 is related to the moment produced by cable forces at point M (end-effector) about joint 1. For the two-cable system, this moment couples both T_1 and T_2 through their lever arms relative to joint 1.

The moment from cable 1 about joint 1: $\tau_{1,\text{cable1}} = (r_M - r_O) \times \mathbf{F}_1$

where the force $\mathbf{F}_1$ is along the cable direction from anchor 1 to M.

For cable 2: $\tau_{1,\text{cable2}} = (r_M - r_O) \times \mathbf{F}_2$

The total moment: $\tau_1 = \tau_{1,\text{cable1}} + \tau_{1,\text{cable2}}$

This couples T_1 and T_2 together. Solving this system requires simultaneously satisfying both:

1. Link 1 moment balance (involving T_1 and T_2)
2. Link 2 moment balance (gives explicit T_1 formula above)

Closed-form solution for T2:

From the two-cable system geometry, the moment contribution from each cable about joint 1 is:

$$\tau_1 = (x_M u_{1,y} - y_M u_{1,x}) T_1 + (x_M u_{2,y} - y_M u_{2,x}) T_2$$

where $u_{j,x}, u_{j,y}$ are the unit vector components of cable j (from anchor to M).

Substituting into the rearranged link 1 equation and combining with the explicit T_1 formula, we obtain:

$$\boxed{T_2 = \frac{\frac{1}{2} m_1 g l_1 \cos(q_1) + l_1 \sin(q_1) X_m - l_1 \cos(q_1) Y_m - l_1 \sin(q_1)\tau_{1,\text{cable1}}}{l_1 \sin(q_1)\tau_{2,\text{lever}}}} \tag{432}$$

where:

- $\tau_{1,\text{cable1}} = (x_M u_{1,y} - y_M u_{1,x}) T_1$ is the contribution from cable 1 to joint 1 moment

- $\tau_{2,\text{lever}} = (x_M u_{2,y} - y_M u_{2,x})$ is the lever arm of cable 2 relative to joint 1

Alternative: Direct Jacobian Inversion for T2

The most practical approach is to use the full 2×2 system:

$$\begin{bmatrix} J_{\rho,11}^T & J_{\rho,12}^T \\ J_{\rho,21}^T & J_{\rho,22}^T \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} g_1(q_1, q_2) \\ g_2(q_1, q_2) \end{bmatrix}$$

Solve by inversion:

$$\begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \frac{1}{\det(\mathbf{J}_\rho^T)} \begin{bmatrix} J_{\rho,22}^T & -J_{\rho,12}^T \\ -J_{\rho,21}^T & J_{\rho,11}^T \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \tag{433}$$

where $\det(\mathbf{J}_\rho^T) = J_{\rho,11}^T J_{\rho,22}^T - J_{\rho,12}^T J_{\rho,21}^T$.

This Jacobian-based solution automatically captures the coupling between T_1 and T_2 through the cable geometry.

Step 2: General Method to Compute T2 Using Cable Geometry Jacobian

While T_1 can be solved from link 2 alone, T_2 requires understanding how both cables couple through their geometric Jacobian. The full equilibrium is:

$$\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} = \mathbf{G}(\mathbf{q})$$

where:

- $\mathbf{T} = [T_1, T_2]^T$ is the cable tension vector
- $\mathbf{G}(\mathbf{q}) = [g_1(q_1, q_2), g_2(q_1, q_2)]^T$ is the gravity torque vector at joints 1 and 2
- $\mathbf{J}_\rho^T(\mathbf{q})$ is the 2×2 geometry Jacobian transpose mapping cable tensions to joint torques

Construction of Geometry Jacobian Transpose:

For each cable j (where $j \in \{1, 2\}$ for left and right cables anchored at $(-L, 0)$ and $(L, 0)$):

1. **End-effector position:**

$$\mathbf{p}_M = \begin{bmatrix} x_M \\ y_M \end{bmatrix} = \begin{bmatrix} l_1 \cos(q_1) + l_2 \cos(q_1 + q_2) \\ l_1 \sin(q_1) + l_2 \sin(q_1 + q_2) \end{bmatrix}$$

2. **Cable vector from anchor to end-effector:**

$$\mathbf{v}_j = \mathbf{p}_M - \mathbf{A}_j$$

where $\mathbf{A}_1 = [-L, 0]^T$ and $\mathbf{A}_2 = [L, 0]^T$.

3. **Cable length and unit vector:**

$$\rho_j = \|\mathbf{v}_j\|, \quad \mathbf{u}_j = \frac{\mathbf{v}_j}{\rho_j}$$

4. **Cable force (pulling toward anchor):**

$$\mathbf{F}_j = -T_j \mathbf{u}_j$$

5. **Moment about each joint:** For joint i at position $\mathbf{p}_i$, the lever arm is:

$$\mathbf{r}_i = \mathbf{p}_M - \mathbf{p}_i$$

The torque (scalar in 2D) from cable j about joint i is:

$$\tau_{i,j} = \mathbf{r}_i \times \mathbf{F}_j = r_{i,x} F_{j,y} - r_{i,y} F_{j,x}$$

Since $\mathbf{F}_j = -T_j \mathbf{u}_j$:

$$\tau_{i,j} = -T_j (r_{i,x} u_{j,y} - r_{i,y} u_{j,x})$$

6. **Jacobian entry:**

$$J^T_{\rho,ij} = -(r_{i,x}u_{j,y} - r_{i,y}u_{j,x})$$

Complete Jacobian Matrix:

$$\mathbf{J}^T_\rho = \begin{bmatrix} J^T_{\rho,11} & J^T_{\rho,12} \\ J^T_{\rho,21} & J^T_{\rho,22} \end{bmatrix}$$

where each entry is computed from the lever arms and cable unit vectors at the configuration $\mathbf{q}$.

Solving for Cable Tensions:
If $\det(\mathbf{J}^T_\rho) \neq 0$ (non-singular):

$$\mathbf{T} = (\mathbf{J}^T_\rho)^{-1}\mathbf{G}(\mathbf{q}) = \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} \tag{434}$$

This gives both T_1 and T_2 that satisfy equilibrium at both joints simultaneously.
If $\det(\mathbf{J}^T_\rho) = 0$ (singular configuration):

- The Jacobian is rank-deficient
- Multiple cable tension combinations may satisfy equilibrium (indeterminate)
- Alternatively, no feasible tension solution exists (infeasible)
- Use pseudoinverse or constrained optimization with $T_i \geq 0$ constraints

Feasibility Check:
After solving for $\mathbf{T}$, verify:

$$T_1 \geq 0, \quad T_2 \geq 0$$

Negative tension is physically impossible (cables cannot push). If either value is negative, the configuration is infeasible with the given anchor positions.

Step 3: Numerical Example
For typical parameters:

- $l_1 = 1.0$ m, $l_2 = 0.8$ m
- $m_1 = 2$ kg, $m_2 = 1.5$ kg
- $L = 1.5$ m (cable anchor separation)
- $g = 9.81$ m/s^2
- Configuration: $q_1 = 30$, $q_2 = 45$

Calculation:
End-effector position:

$$x_M = 1.0\cos(30) + 0.8\cos(75) = 0.866 + 0.207 = 1.073 \text{ m}$$

$$y_M = 1.0\sin(30) + 0.8\sin(75) = 0.500 + 0.773 = 1.273 \text{ m}$$

For the **two-cable system**, we must solve the full force balance numerically:
Step 1: Calculate cable angles from end-effector to anchors:

$$\alpha_1 = \arctan\left(\frac{y_M - 0}{x_M - (-L)}\right), \quad \alpha_2 = \arctan\left(\frac{y_M - 0}{x_M - L}\right)$$

Step 2: Solve the coupled system:

$$\begin{bmatrix} -\cos(\alpha_1) & -\cos(\alpha_2) \\ -\sin(\alpha_1) & -\sin(\alpha_2) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} X_A \\ Y_A - m_2 g \end{bmatrix}$$

Step 3: Verify moment constraint from Link 2 system (420):

$$l_2[T_1 \sin(\alpha_1 + q_2) + T_2 \sin(\alpha_2 + q_2)] = \frac{1}{2} m_2 g l_2 \cos(q_2)$$

In practice, use computational tools to:

1. Assemble $\mathbf{J}_\rho^T$ using cable vectors and lever arms
2. Invert to get $\mathbf{T} = (\mathbf{J}_\rho^T)^{-1}\mathbf{G}(\mathbf{q})$
3. Check positivity and condition number for feasibility assessment

Cable Force Transmission and Jacobian Relationship Cable tensions $\mathbf{T} = [T_1, T_2]^T$ act along the cable directions at end-effector M. These are transmitted to joint torques through the geometry Jacobian:

$$\boldsymbol{\tau}_{\text{cable}} = \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} \tag{435}$$

For purely cable-driven actuation (no joint motors), the dynamics reduce to:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} \tag{436}$$

At equilibrium ($\ddot{\mathbf{q}} = \dot{\mathbf{q}} = 0$), this becomes the force balance equation:

$$\mathbf{J}_\rho^T(\mathbf{q})\mathbf{T} = \mathbf{G}(\mathbf{q}) \tag{437}$$

Singularity Effects on Force Balance At kinematic singularities where $\det(\mathbf{J}_\rho) = 0$:

1. **Indeterminate cable tensions**: Multiple cable tension combinations satisfy equilibrium
2. **Loss of invertibility**: Cannot uniquely recover $\mathbf{T}$ from $\mathbf{G}(\mathbf{q})$
3. **Force amplification**: Infinitesimal cable displacements produce finite joint motions
4. **Instability**: Small perturbations cause rapid configuration changes
5. **Constraint violation risk**: Cables may go slack or tension limits exceeded

This wrench analysis demonstrates why singularities are critical failure points for cable-driven systems.

Summary: Verification Results and Corrections **Section Verification Status:** After comprehensive review of the "Wrench Analysis and Force Balance at End-Effector" section, the following assessment is provided:

VERIFIED Correct Elements:

1. **Two-cable force distribution**: The corrected analysis properly accounts for both cable tensions in equilibrium
2. **Singularity analysis**: $\det(\mathbf{J}_\rho^T) = \sin(\alpha_1 - \alpha_2)$ correctly identifies parallel cable configurations as singular

3. **Physical interpretations**: Explanations of moment balance and force equilibrium are accurate
4. **General methodology**: The approach using free-body diagrams and equilibrium equations is appropriate

CRITICAL ERRORS Identified and Corrected:

1. **Dimensional inconsistency**: Previous formulations incorrectly mixed force [N] and torque [N·m] units
2. **Matrix formulation errors**: The 6×6 matrix system contained multiple structural and dimensional errors
3. **Missing equilibrium equations**: Link equilibrium equations were not properly separated and sequenced
4. **Coordinate system confusion**: Mixed joint-based and end-effector-based coordinate systems inappropriately

IMPLEMENTED Corrections:

1. **Sequential solution methodology**: Implemented Link 2 system (420) for cable tension determination, then Link 1 system (419) for internal forces
2. **Resolved dimensional issues**: Cable tensions determined from complete force and moment equilibrium, eliminating unit inconsistencies
3. **Provided proper framework**: Established clear solution sequence - Link 2 equilibrium → cable tensions → Link 1 equilibrium → internal forces
4. **Maintained mathematical rigor**: All corrections preserve fundamental physics while providing consistent, solvable framework

RECOMMENDED Action Items:

1. **Use the sequential solution methodology** with systems (419) and (420) for all implementations
2. **Implement the two-stage solution strategy** for computational implementations
3. **Verify numerical examples** using the corrected Link 2 → Link 1 sequential framework
4. **All dependent sections** now reference the corrected equilibrium systems consistently

The corrected section now provides a mathematically consistent and physically meaningful analysis of wrench balance and force transmission in the cable-driven planar robot, suitable for both theoretical analysis and practical implementation.

16.17 Summary: Euler-Lagrange Dynamics

The Euler-Lagrange formulation for the hybrid serial-cable robot provides:

1. **Complete dynamic model**: $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}_\rho^T\mathbf{T}$
2. **Clear physical interpretation**: Each term has mechanical meaning
3. **Systematic derivation**: From energy principles, not ad-hoc assumptions

4. **Integration with singularity analysis**: Direct connection between kinematic singularities and dynamic instability
5. **Foundation for control design**: Enables model-based controllers and trajectory planning
6. **Predictive capability**: Simulate system behavior including singular configurations

A Notation and Symbol Reference

This appendix provides a comprehensive reference for all mathematical symbols, variables, and notation used throughout this document. This standardized notation ensures consistency and eliminates ambiguity in mathematical derivations and implementation.

A.1 Robot Geometry and Configuration

Table 2: Robot Geometric Parameters

Symbol	Description	Units	Type
l_1	Length of first link (base to elbow)	m	Fixed parameter
l_2	Length of second link (elbow to end-effector)	m	Fixed parameter
L	Half-distance between cable anchor points	m	Fixed parameter
q_1	First joint angle (base joint)	rad	Variable
q_2	Second joint angle (elbow joint)	rad	Variable
$\boldsymbol{q}$	Joint angle vector $[q_1, q_2]^T$	rad	Vector variable

A.2 Cable System Parameters

Table 3: Cable System Notation

Symbol	Description	Units	Type
ρ_1	Left cable length (variable, motor-controlled)	m	Control input
ρ_2	Right cable length (variable, motor-controlled)	m	Control input
$\boldsymbol{\rho}$	Cable length vector $[\rho_1, \rho_2]^T$	m	Vector variable
α_1	Left cable angle (from horizontal)	rad	Computed variable
α_2	Right cable angle (from horizontal)	rad	Computed variable
T_1	Left cable tension force	N	Control input
T_2	Right cable tension force	N	Control input
$\boldsymbol{T}$	Cable tension vector $[T_1, T_2]^T$	N	Vector variable
Cable Tension Constraints:			
$T_1 \geq 0$	Non-negative left cable tension constraint	-	Physical constraint
$T_2 \geq 0$	Non-negative right cable tension constraint	-	Physical constraint

Table 4: Coordinate System Notation

Symbol	Description	Units	Type
$\{0\}$	Base coordinate frame (world frame)	-	Fixed frame
$\{1\}$	First joint coordinate frame	-	Moving frame
$\{2\}$	End-effector coordinate frame	-	Moving frame
O	Origin of base frame (joint 1 center)	-	Fixed point
A	Elbow joint center (joint 2 center)	-	Moving point
M	End-effector point (cable attachment)	-	Moving point
x_M, y_M	End-effector position coordinates	m	Computed variables
$\boldsymbol{p}_M$	End-effector position vector $[x_M, y_M]^T$	m	Vector variable

Table 5: Cable Anchor Point Notation

Symbol	Description	Units	Type
A_1	Left cable anchor point position	-	Fixed point
A_2	Right cable anchor point position	-	Fixed point
$\boldsymbol{A}_1$	Left anchor position vector $[-L, 0]^T$	m	Fixed vector
$\boldsymbol{A}_2$	Right anchor position vector $[L, 0]^T$	m	Fixed vector
$\boldsymbol{u}_1$	Left cable unit direction vector	-	Computed vector
$\boldsymbol{u}_2$	Right cable unit direction vector	-	Computed vector

Table 6: DH Parameter Notation (Distinct from Cable Angles)

Symbol	Description	Units	Type
a_i	Link length (DH parameter)	m	Fixed parameter
$\alpha_{DH,i}$	Link twist angle (DH parameter)	rad	Fixed parameter
d_i	Link offset (DH parameter)	m	Fixed parameter
θ_i	Joint angle (DH parameter)	rad	Variable
Specific Values for 2-DOF Planar Robot:			
$a_1 = l_1$	First link length	m	$a_1 = l_1$
$a_2 = l_2$	Second link length	m	$a_2 = l_2$
$\alpha_{DH,1} = 0$	First link twist (planar motion)	rad	0
$\alpha_{DH,2} = 0$	Second link twist (planar motion)	rad	0
$d_1 = 0$	First joint offset (revolute joint)	m	0
$d_2 = 0$	Second joint offset (revolute joint)	m	0
$\theta_1 = q_1$	First joint variable	rad	Variable
$\theta_2 = q_2$	Second joint variable	rad	Variable

Table 7: Transformation Matrix Notation

Symbol	Description	Dimensions	Meaning
$\boldsymbol{T}_j^i$	Transform from frame $\{i\}$ to frame $\{j\}$	3×3	Homogeneous transform
$\boldsymbol{T}_1^0$	Base to first joint transformation	3×3	Fixed to moving
$\boldsymbol{T}_2^1$	First to second joint transformation	3×3	Moving to moving
$\boldsymbol{T}_2^0$	Base to end-effector transformation	3×3	Complete chain
$\boldsymbol{R}_j^i$	Rotation matrix from frame $\{i\}$ to $\{j\}$	2×2	Rotation only
$\boldsymbol{p}_j^i$	Position vector of frame $\{j\}$ in frame $\{i\}$	2×1	Translation only

A.3 Coordinate Frames and Positions

A.4 Cable Anchor Points and Geometry

A.5 Denavit-Hartenberg Parameters

A.6 Transformation Matrices

A.7 Kinematics and Dynamics

Table 8: Kinematic and Dynamic Variables

Symbol	Description	Units	Type
$\dot{q}_1, \dot{q}_2$	Joint angular velocities	rad/s	Time derivatives
$\dot{\boldsymbol{q}}$	Joint velocity vector $[\dot{q}_1, \dot{q}_2]^T$	rad/s	Vector variable
$\dot{\rho}_1, \dot{\rho}_2$	Cable length change rates	m/s	Time derivatives
$\dot{\boldsymbol{\rho}}$	Cable velocity vector $[\dot{\rho}_1, \dot{\rho}_2]^T$	m/s	Vector variable
$\dot{x}_M, \dot{y}_M$	End-effector velocity components	m/s	Computed variables
$\dot{\boldsymbol{x}}$	End-effector velocity vector $[\dot{x}_M, \dot{y}_M]^T$	m/s	Vector variable
τ_1, τ_2	Joint torques (if used)	N·m	Control inputs
$\boldsymbol{\tau}$	Joint torque vector (if used)	N·m	Vector variable

A.8 Jacobian Matrices

Table 9: Jacobian Matrix Notation

Symbol	Description	Dimensions	Physical Meaning
$\boldsymbol{J}_q$	Serial joint Jacobian	2×2	$\frac{\partial \boldsymbol{x}}{\partial \boldsymbol{q}}$
$\boldsymbol{J}_\rho$	Cable Jacobian	2×2	$\frac{\partial \boldsymbol{x}}{\partial \boldsymbol{\rho}}$
$\boldsymbol{J}$	Complete system Jacobian	2×4	$[\boldsymbol{J}_q \quad \boldsymbol{J}_\rho]$
$\boldsymbol{J}_{red}$	Reduced Jacobian (constrained system)	2×2	Effective joint-space Jacobian

A.9 Constraint Equations

Table 10: Constraint Equation Notation

Symbol	Description	Dimensions	Type
$g_1(\boldsymbol{q}, \rho_1)$	Left cable geometric constraint	scalar	Nonlinear function
$g_2(\boldsymbol{q}, \rho_2)$	Right cable geometric constraint	scalar	Nonlinear function
$\boldsymbol{g}(\boldsymbol{q}, \boldsymbol{\rho})$	Constraint vector $[g_1, g_2]^T$	2×1	Vector function
$\boldsymbol{G}_q$	Constraint Jacobian w.r.t. joints	2×2	$\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{q}}$
$\boldsymbol{G}_\rho$	Constraint Jacobian w.r.t. cables	2×2	$\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{\rho}}$

Table 11: Force and Direction Vectors

Symbol	Description	Units	Type
$\boldsymbol{F}_1$	Left cable force vector	N	Force vector
$\boldsymbol{F}_2$	Right cable force vector	N	Force vector
$\boldsymbol{F}_{cable}$	Total cable force $\boldsymbol{F}_1 + \boldsymbol{F}_2$	N	Resultant force
$\boldsymbol{F}_{ext}$	External force at end-effector	N	Applied load
$F_{ext,x}, F_{ext,y}$	External force components	N	Force components
P_1, P_2	Gravitational forces at centers of mass	N	Force magnitudes
G_1, G_2	Centers of mass of links 1 and 2	-	Points

Table 12: Mathematical Operations and Functions

Symbol	Description	Domain/Range	Notes
$\text{atan2}(y, x)$	Four-quadrant arctangent function	$(-\pi, \pi]$	Robust angle calculation
$\|\boldsymbol{v}\|$	Euclidean norm of vector $\boldsymbol{v}$	$[0, \infty)$	Vector magnitude
$\det(\boldsymbol{A})$	Determinant of matrix $\boldsymbol{A}$	$\mathbb{R}$	Scalar value
$\boldsymbol{A}^{-1}$	Matrix inverse of $\boldsymbol{A}$	Same size as $\boldsymbol{A}$	When $\det(\boldsymbol{A}) \neq 0$
$\boldsymbol{A}^T$	Matrix transpose of $\boldsymbol{A}$	Transposed dimensions	Row-column exchange
$\frac{\partial f}{\partial x}$	Partial derivative of f w.r.t. x	Varies	Rate of change

Table 13: Singularity Analysis Symbols

Symbol	Description	Condition	Physical Meaning
Type I	Serial chain singularity	$\det(\boldsymbol{J}_q) = 0$	Loss of joint DOF
Type II	Cable singularity	$\det(\boldsymbol{G}_\rho) = 0$	Cable constraints dependent
Type III	Combined singularity	$\det(\boldsymbol{J}_{red}) = 0$	System-level singularity
$q_2 = 0$	Fully extended configuration	$\sin(q_2) = 0$	Serial arm straight
$q_2 = \pi$	Fully folded configuration	$\sin(q_2) = 0$	Serial arm folded back

Table 14: Key Mathematical Relationships

Category	Formula
Forward Kinematics	$x_M = l_1 \cos q_1 + l_2 \cos(q_1 + q_2)$ $y_M = l_1 \sin q_1 + l_2 \sin(q_1 + q_2)$ $\rho_1 = \sqrt{(x_M + L)^2 + y_M^2}$ $\rho_2 = \sqrt{(x_M - L)^2 + y_M^2}$
Cable Angles	$\alpha_1 = \text{atan2}(y_M, x_M + L)$ $\alpha_2 = \text{atan2}(y_M, x_M - L)$
Cable Unit Vectors	$\boldsymbol{u}_1 = \frac{1}{\rho_1}[x_M + L, y_M]^T$ $\boldsymbol{u}_2 = \frac{1}{\rho_2}[x_M - L, y_M]^T$
Velocity Kinematics	$\dot{\boldsymbol{x}} = \boldsymbol{J}_q \dot{\boldsymbol{q}} + \boldsymbol{J}_\rho \dot{\boldsymbol{\rho}}$ $\dot{\boldsymbol{x}} = \boldsymbol{J}_{red} \dot{\boldsymbol{q}}$ (constrained)
Singularity Conditions	$\det(\boldsymbol{J}_q) = l_1 l_2 \sin q_2 = 0$
Cable Tension Constraints	$T_1 \geq 0, T_2 \geq 0$ (Non-negative tensions) $\sin(\alpha_2 - \alpha_1) \neq 0$ (Non-parallel cables) Force equilibrium: $T_1 \boldsymbol{u}_1 + T_2 \boldsymbol{u}_2 + \boldsymbol{F}_{ext} = \boldsymbol{0}$

A.10 Forces and Unit Vectors

A.11 Mathematical Functions and Operators

A.12 Singularity Analysis Notation

A.13 Important Relationships and Formulas

A.14 Workspace Definitions

Table 15: Workspace and Constraint Set Notation

Symbol	Description
$\mathcal{W}_{kinematic}$	Kinematic workspace (reachable by serial chain)
$\mathcal{W}_{feasible}$	Feasible workspace (positive cable tensions)
$\mathcal{W}_{cable-limited}$	Cable-limited regions (negative tensions required)
$\mathcal{W}_{singular}$	Singular configurations ($\det(\boldsymbol{J}) = 0$)
Mathematical Definitions:	
$\mathcal{W}_{kinematic}$	$\{(x_M, y_M) : x_M^2 + y_M^2 \leq (l_1 + l_2)^2\}$
$\mathcal{W}_{feasible}$	$\{(x_M, y_M) : T_1 \geq 0 \text{ and } T_2 \geq 0\}$
$\mathcal{W}_{cable-limited}$	$\{(x_M, y_M) : T_1 < 0 \text{ or } T_2 < 0\}$
$\mathcal{W}_{singular}$	$\{(x_M, y_M) : q_2 = n\pi, n \in \mathbb{Z}\}$

A.15 Notation Conventions

- **Scalars**: Regular italic font (e.g., q_1, l_1, ρ_1)
- **Vectors**: Bold lowercase (e.g., $\boldsymbol{q}$, $\boldsymbol{p}_M$, $\boldsymbol{u}_1$)
- **Matrices**: Bold uppercase (e.g., $\boldsymbol{T}_1^0$, $\boldsymbol{J}_q$, $\boldsymbol{R}$)
- **Frames**: Curly braces (e.g., $\{0\}$, $\{1\}$, $\{2\}$)
- **Points**: Regular uppercase (e.g., O, A, M, A_1, A_2)
- **Subscripts**: Parameter index or frame reference
- **Superscripts**: Reference frame for transformations
- **Dots**: Time derivatives (e.g., $\dot{q}_1$, $\dot{\boldsymbol{x}}$)

A.16 Critical Distinctions

Table 16: Commonly Confused Notation - Key Distinctions

Symbol	NOT	Distinction
l_1, l_2	ρ_1, ρ_2	Link lengths (fixed) vs Cable lengths (variable)
α_1, α_2	$\alpha_{DH,1}, \alpha_{DH,2}$	Cable angles vs DH twist parameters
T_1, T_2	τ_1, τ_2	Cable tensions vs Joint torques
$\boldsymbol{J}_\rho$	$\boldsymbol{J}_l$	Cable Jacobian (corrected notation)
$\boldsymbol{G}_\rho$	$\boldsymbol{G}_l$	Constraint Jacobian (corrected notation)

B Derivation Details

C Code Implementation Notes